



Department of Materials Science & Engineering

MM-101 Materials and Nanotechnology (Spring 2025)

Post Midterm Examination (Week 9-15)

Course Instructors: Dr. Hamza Mohsin & Dr. Shanza Rehan

Email: hamza.mohsin@giki.edu.pk / shanza.rehan@giki.edu.pk

Office: G-09 (Faculty Lobby, FMCE) / G-10 (Faculty Lobby, FMCE)

Topics covered/To be covered



Pre-Mid

Topic	Chapter (Callister 7 th Edition)	Chapter (Callister 10 th /11 th Edition)
Introduction	1	1
Atomic Structure and Interatomic Bonding	2	2
Structure of Crystalline Solids - Metals	3, 4	3
Structure of Crystalline Solids - Ceramics	4	12
Imperfections	4	5, 12

Post-Mid

Topic	Chapter (Callister 7 th Edition)	Chapter (Callister 10 th /11 th Edition)
Mechanical Properties of Metals	9	6
Diffusion	6	5
Composites	15	16
Nanomaterials and Nanotechnology		
Electrical, Magnetic, Thermal and Optical Properties	17, 18, 19, 20	18, 19, 20, 21

Mechanical Properties



Materials have a set of mechanical properties:

- Elasticity
- Plasticity
- Strength
- Hardness
- Ductility
- Brittleness
- Toughness
- Stiffness
- Resilience





Materials have a set of mechanical properties:

- **Elasticity**
- Plasticity
- Strength
- Hardness
- Ductility
- Brittleness
- Toughness
- Stiffness
- Resilience



Elasticity in materials refers to the ability of a material to return to its original shape and size after being deformed (stretched, compressed, or twisted) by an applied force, once the force is removed.



Mechanical Properties

Materials have a set of mechanical properties:

- Elasticity
- **Plasticity**
- Strength
- Hardness
- Ductility
- Brittleness
- Toughness
- Stiffness
- Resilience



Plasticity in materials refers to the ability of a material to undergo permanent deformation (change in shape or size) when subjected to an external force, without breaking.



Mechanical Properties

Materials have a set of mechanical properties:

- Elasticity
- Plasticity
- **Strength**
- Hardness
- Ductility
- Brittleness
- Toughness
- Stiffness
- Resilience



Strength in materials refers to the ability of a material to withstand an applied force without failing or breaking. It is a measure of how much load a material can bear before it deforms permanently or fractures.



Materials have a set of mechanical properties:

- Elasticity
- Plasticity
- Strength
- **Hardness**
- **Ductility**
- Brittleness
- Toughness
- Stiffness
- Resilience



Hardness in materials refers to a material's resistance to localized plastic deformation.

Ductility in materials refers to the ability of a material to undergo significant plastic deformation (stretching, bending, or elongating) before it fractures or breaks.



Materials have a set of mechanical properties:

- Elasticity
- Plasticity
- Strength
- Hardness
- Ductility
- **Brittleness**
- Toughness
- Stiffness
- Resilience



Brittleness in materials refers to the tendency of a material to fracture or break with little to no plastic deformation when subjected to stress.



Materials have a set of mechanical properties:

- Elasticity
- Plasticity
- Strength
- Hardness
- Ductility
- Brittleness
- **Toughness**
- Stiffness
- Resilience



Toughness in materials refers to the ability of a material to absorb energy and withstand fracture when subjected to load/stress/force before it fractures. Essentially, toughness describes how much energy a material can absorb till fracture.



Materials have a set of mechanical properties:

- Elasticity
- Plasticity
- Strength
- Hardness
- Ductility
- Brittleness
- Toughness
- **Stiffness**
- Resilience



Stiffness in materials is a measure of how much a material resists changes in shape when subjected to stress, particularly when experiencing elastic deformation. The stiffer a material is, the less it deforms under a given load.



Materials have a set of mechanical properties:

- Elasticity
- Plasticity
- Strength
- Hardness
- Ductility
- Brittleness
- Toughness
- Stiffness
- **Resilience**



Resilience in materials refers to the ability of a material to absorb energy when it is deformed elastically and to release that energy when the stress is removed.

Mechanical Properties

Why are mechanical properties so important?





Mechanical Properties

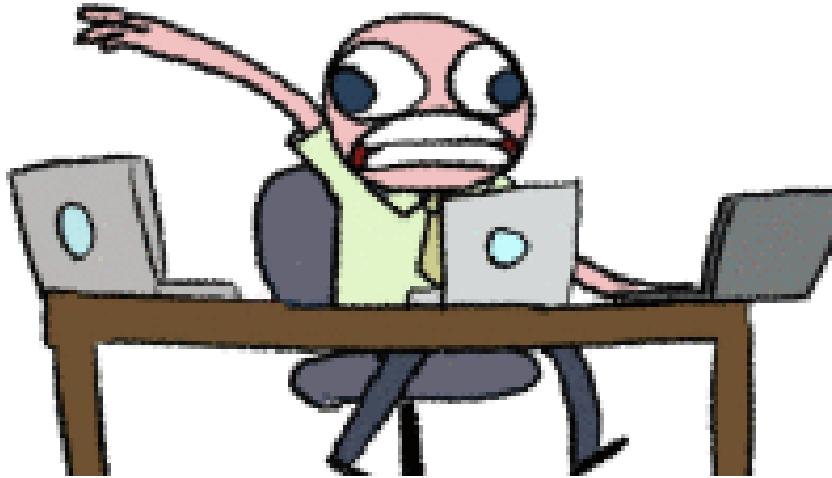
Why are mechanical properties so important??

- **Since properties influenced by:**
 - type of material/composition etc.
 - nature of the applied load (stress i.e. force / area) during service
 - duration of load (constant or fluctuating in time)
 - environmental conditions (temperature, corrosive)
- **Mechanical properties are of great concern to:**
 - Producers, Consumers, Government agencies, Research organizations
- **Professional bodies are involved in defining / measuring mechanical properties such as**

“American Society for Testing and Materials (ASTM)”

Mechanical Properties

Stress



Vacuum Implosion

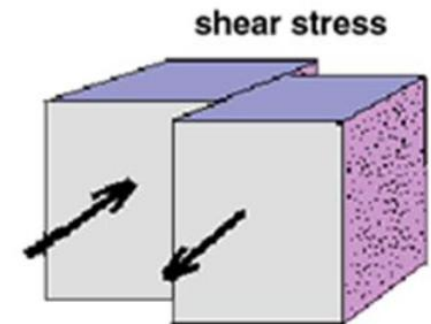
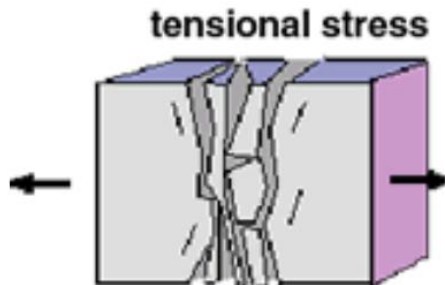
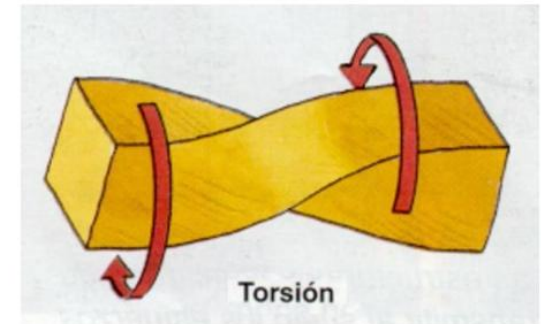


Mechanical Properties

Stress

Stress is the measure of what the material feels from externally applied forces. It is simply a ratio of the external forces to the cross- sectional area of the material.

$$\text{stress} = \frac{\text{applied force}}{\text{c.s.a. of a material}} = \frac{F}{A_0}$$



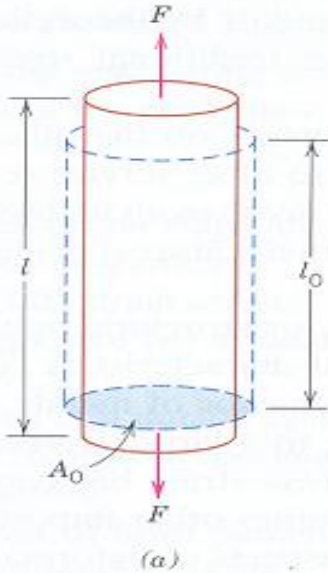
Mechanical Properties



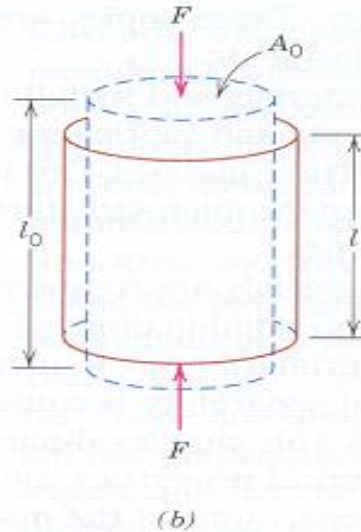
Types of Stress

Dotted lines represent pre-stressed shape

Tensile stress



Compressive Stress

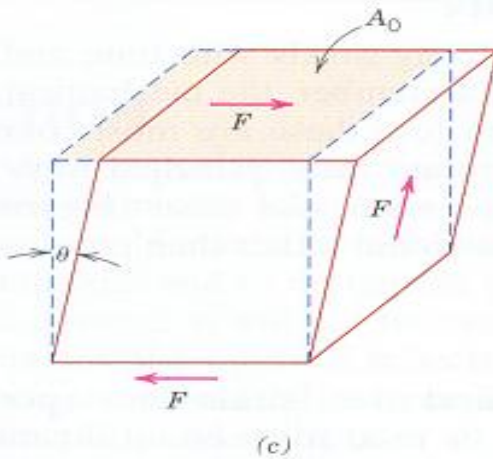


Stress

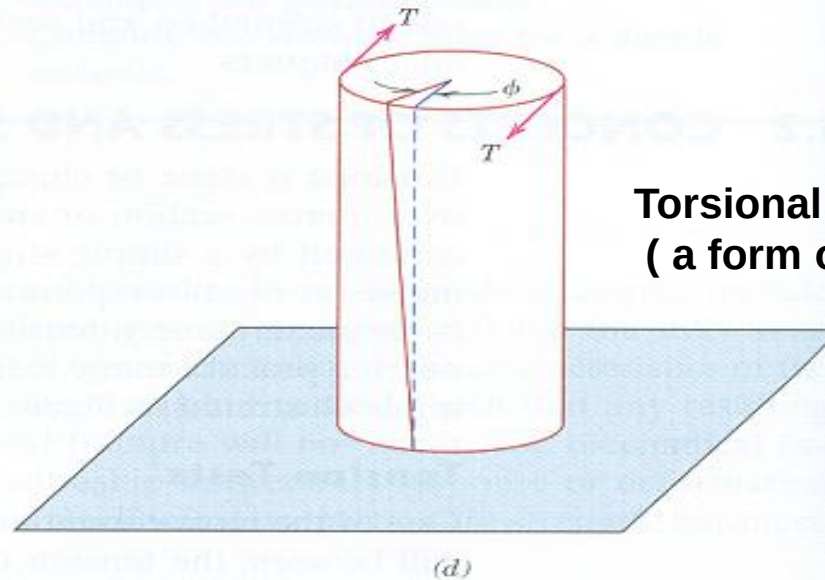
$$\sigma = F / A_0$$

Units = Pascal = N / m²
(generally, MPa)

Shear stress



Torsional stress
(a form of shear)

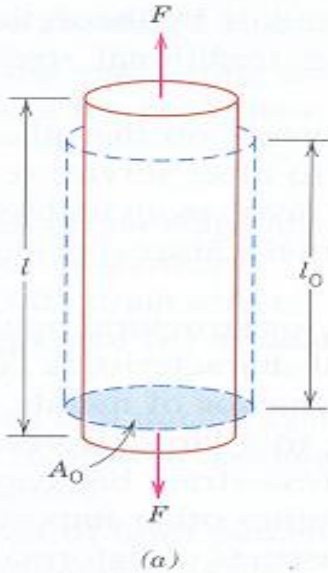


Mechanical Properties

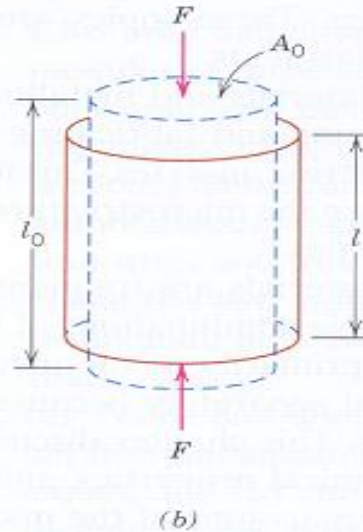
Types of Stress

Dotted lines represent pre-stressed shape

Tensile stress



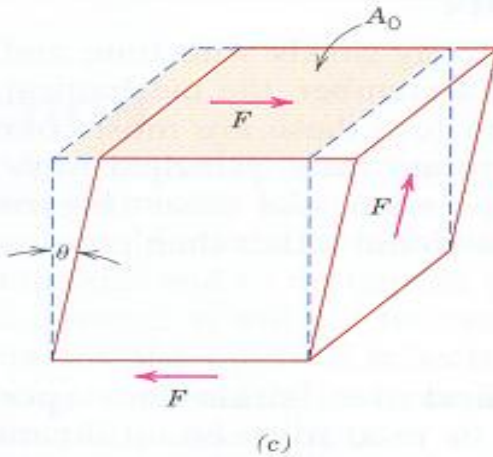
Compressive Stress



Strain

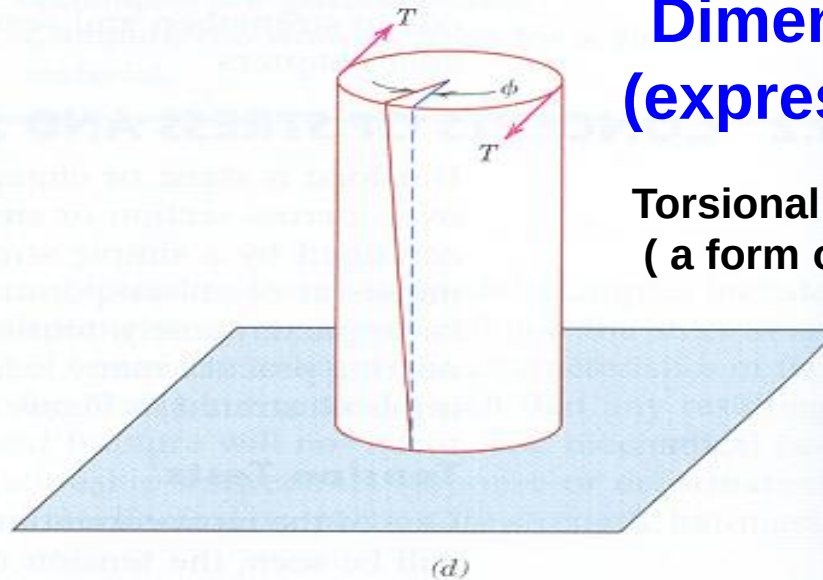
$$\epsilon = \frac{l_i - l_0}{l_0} = \frac{\Delta l}{l_0}$$

Shear stress



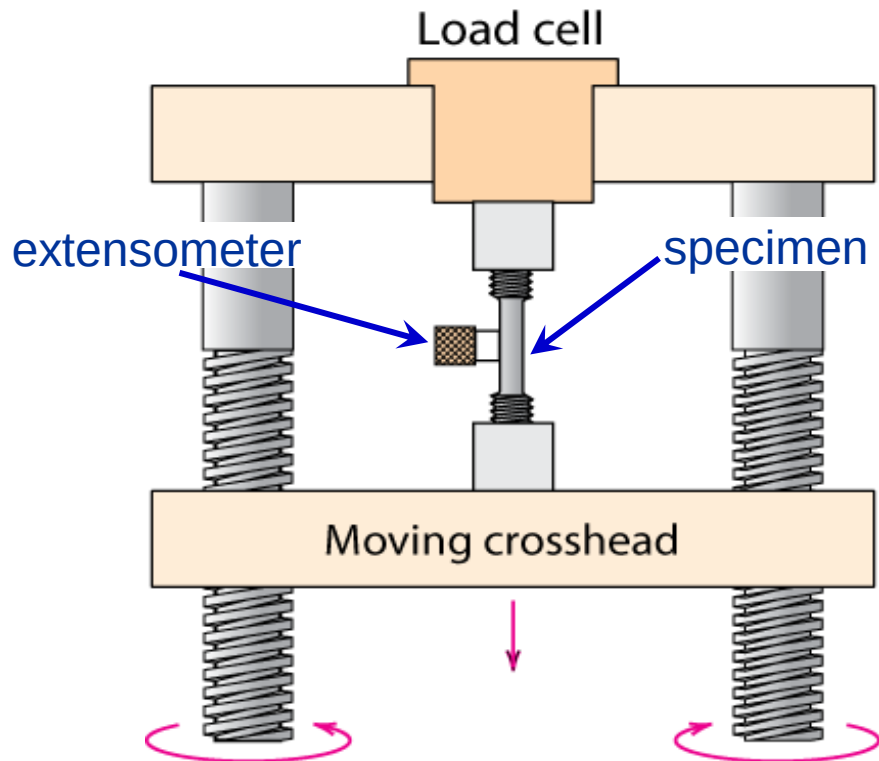
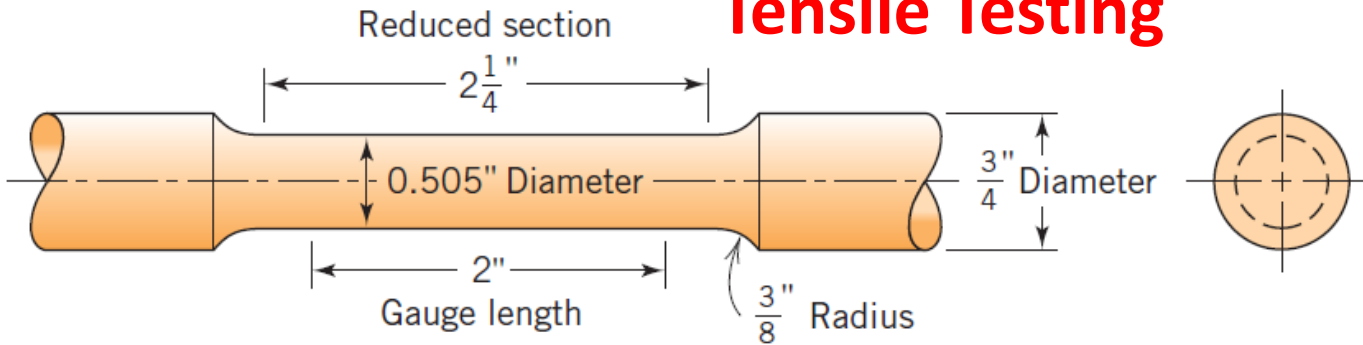
Dimensionless
(expressed in %)

Torsional stress
(a form of shear)

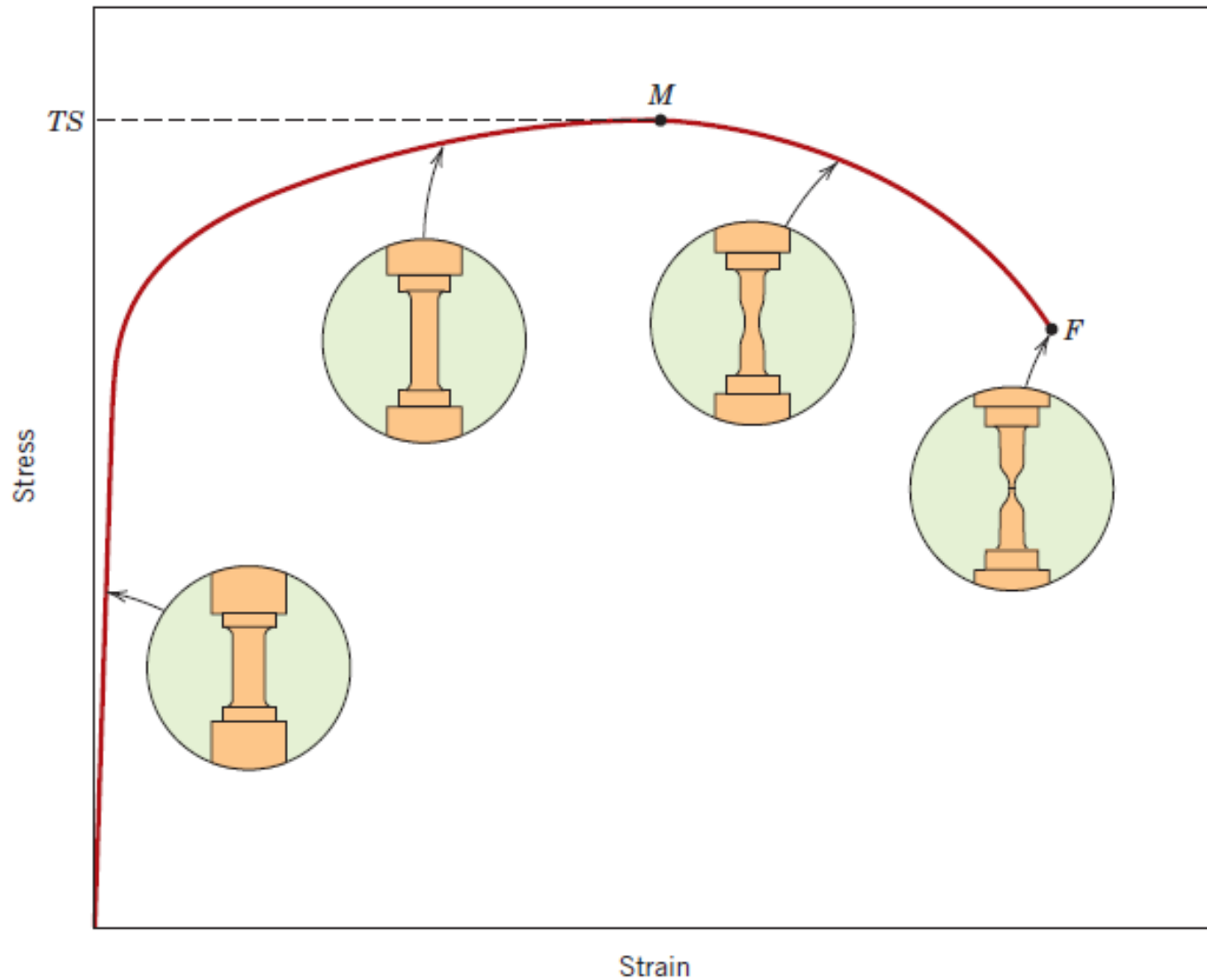


Mechanical Properties

Tensile Testing



Tensile Testing



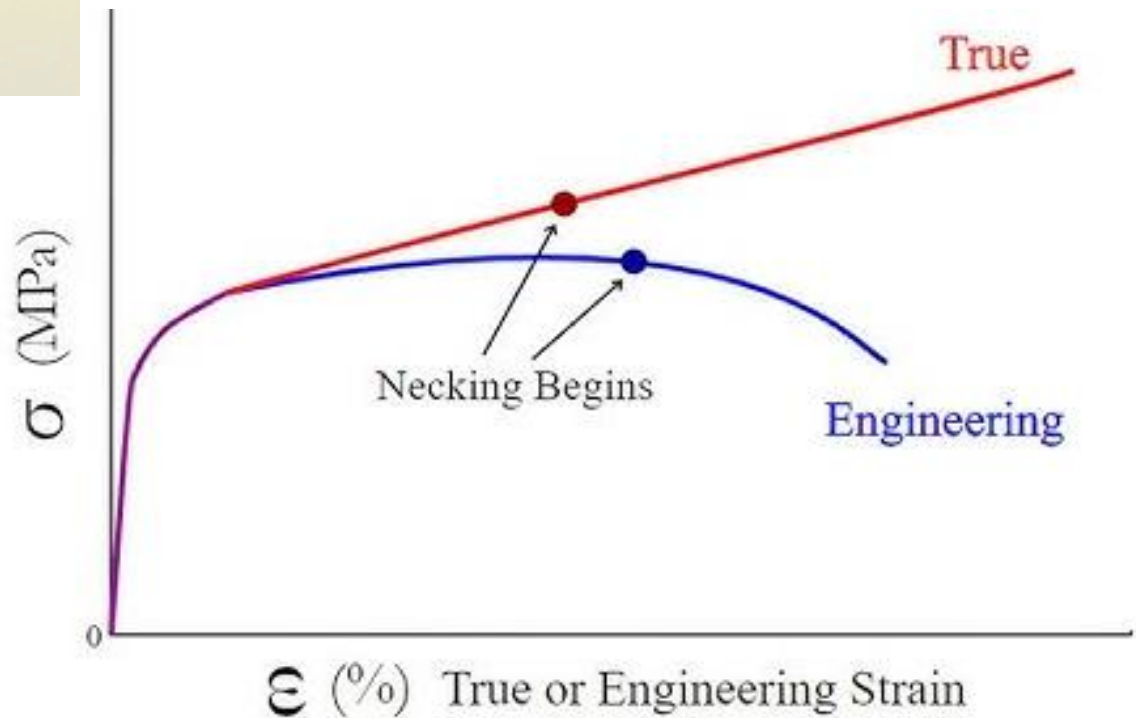
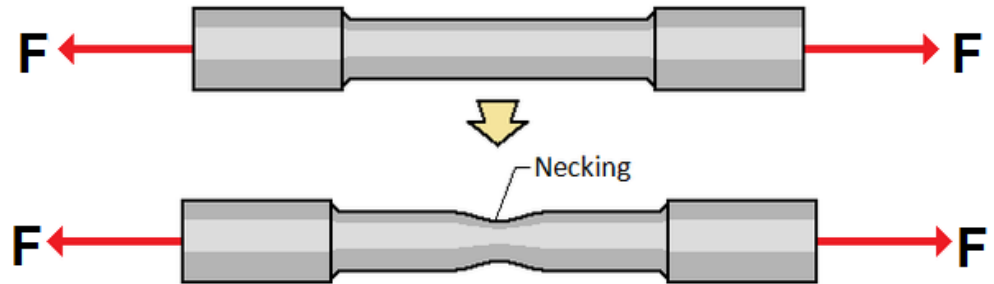
Mechanical Properties

Tensile Testing



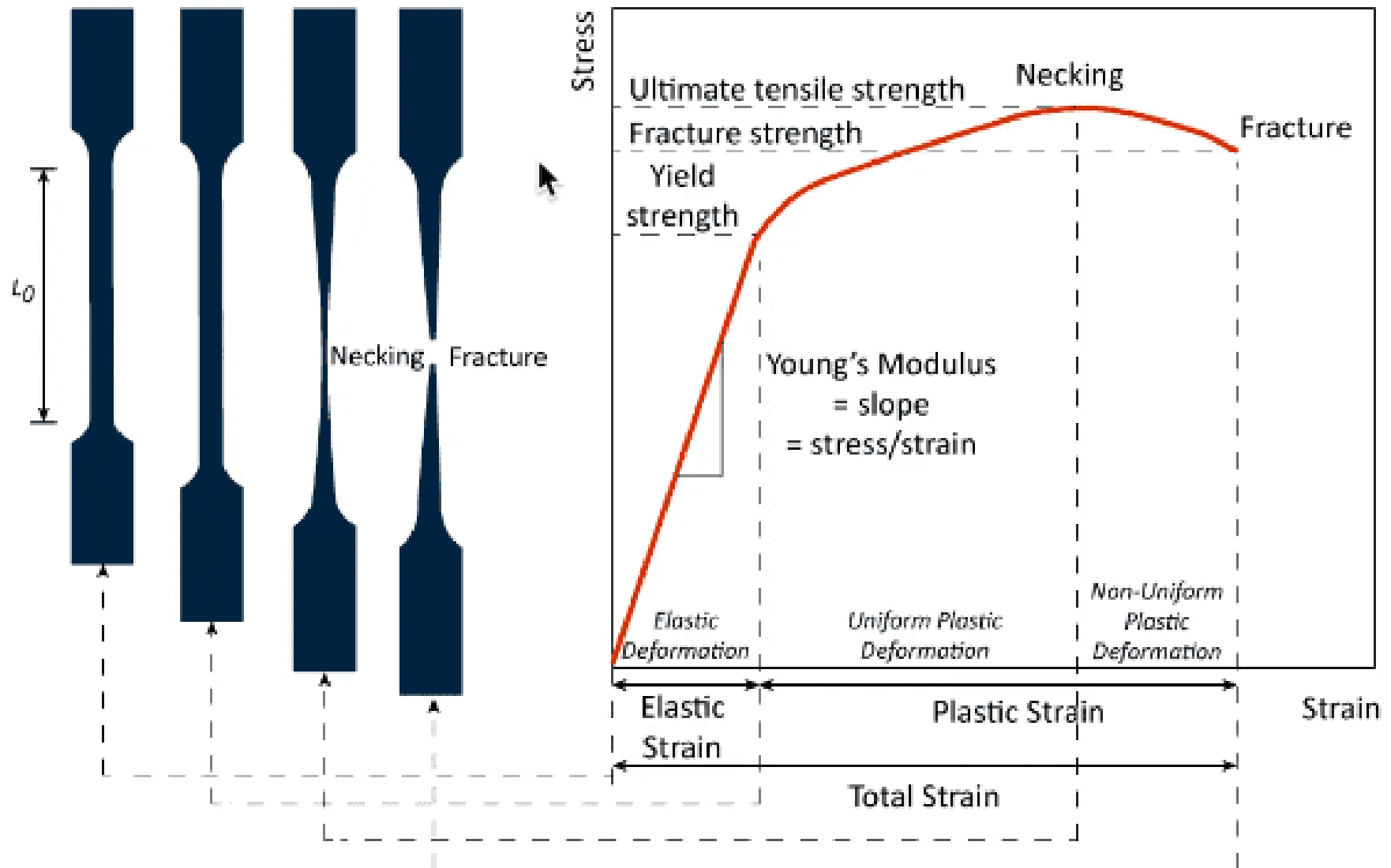
Engineering Stress

Engineering Strain



Mechanical Properties

Tensile Testing





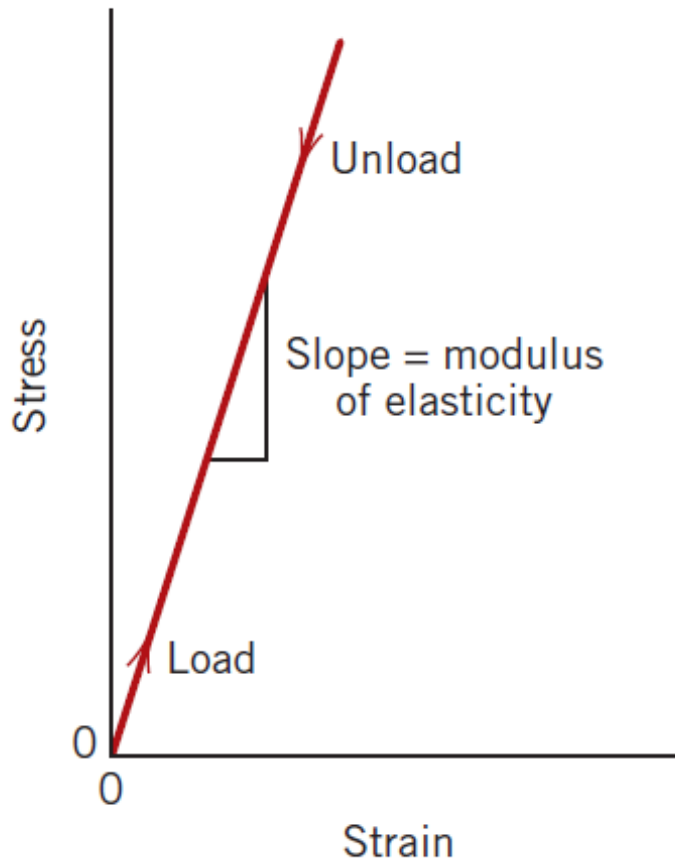
Mechanical Properties

Elastic Deformation

(reversible and Linear)

Hooke's Law:

stress and strain are proportional to each other through the relationship



$$\sigma = E\varepsilon$$

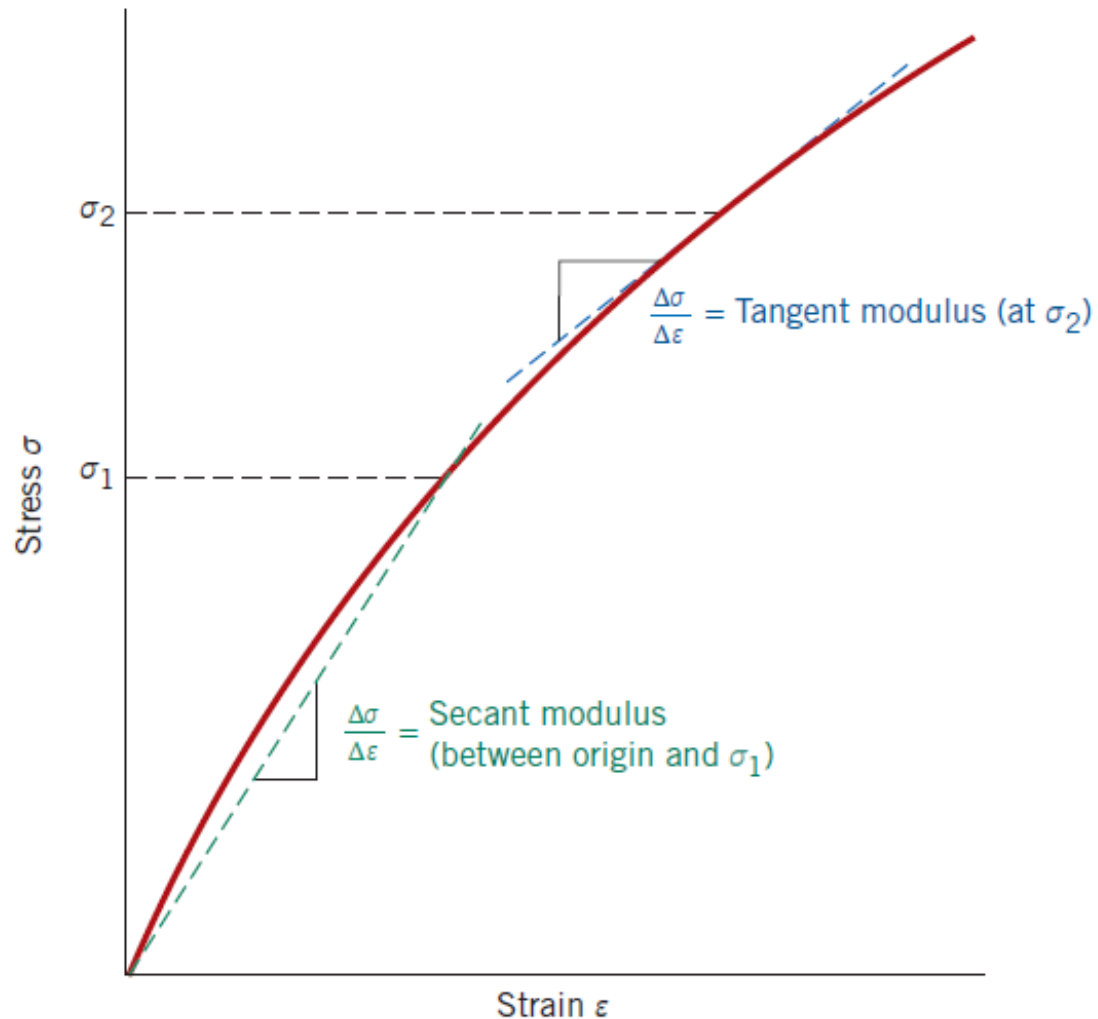
Young's Modulus or Modulus of Elasticity (E in GPa)



Mechanical Properties

Elastic Deformation

gray cast iron, concrete, some plastics etc.
(reversible and non-Linear)

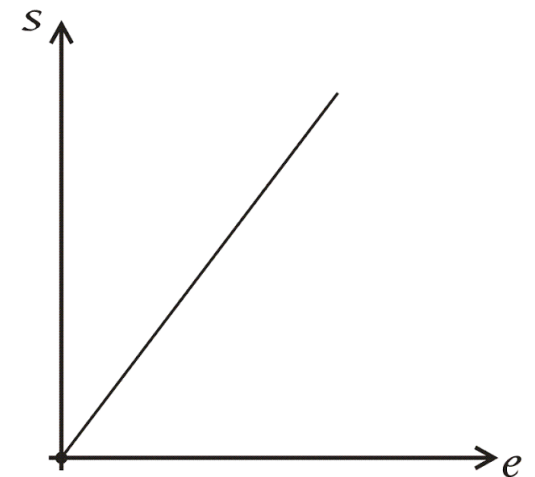
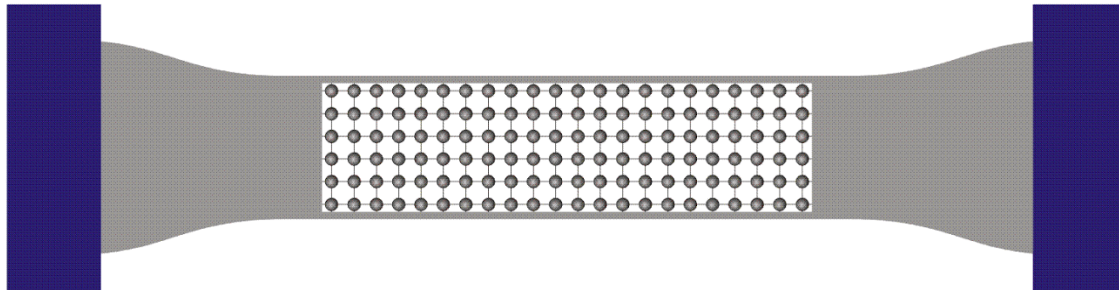
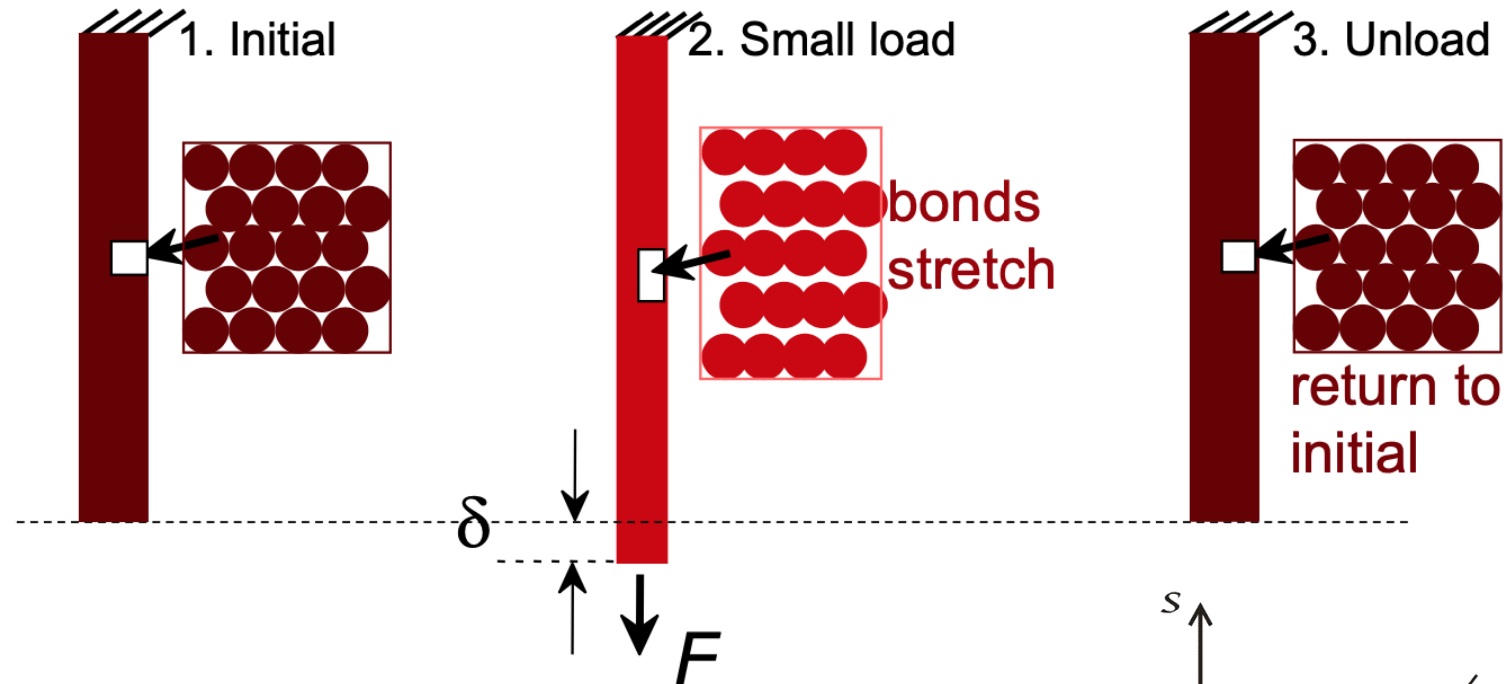




Mechanical Properties

Elastic Deformation

What happens at the atomic level?

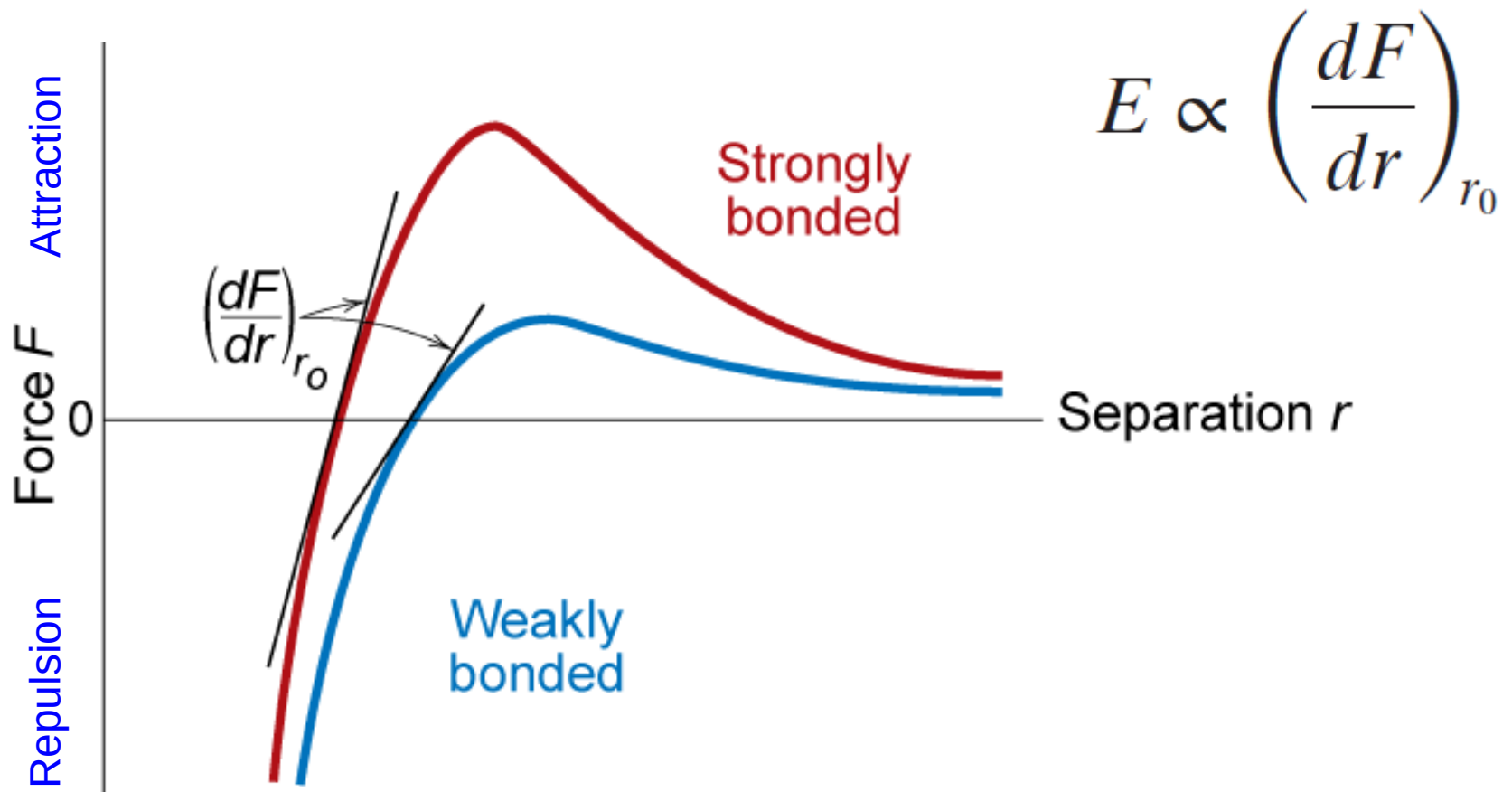




Mechanical Properties

Elastic Deformation

Stiffer bonds will resist deformation more

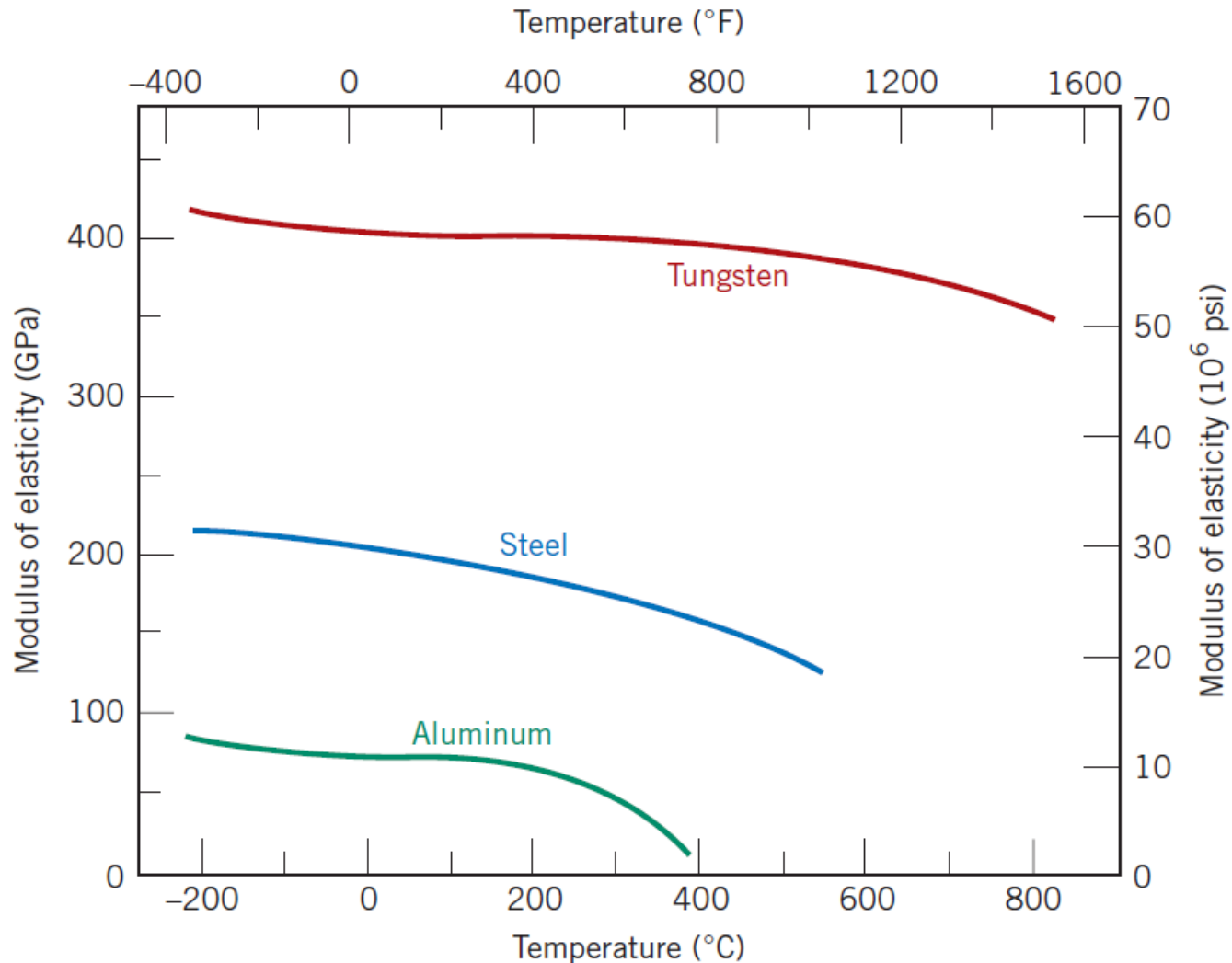




Mechanical Properties

Elastic Deformation

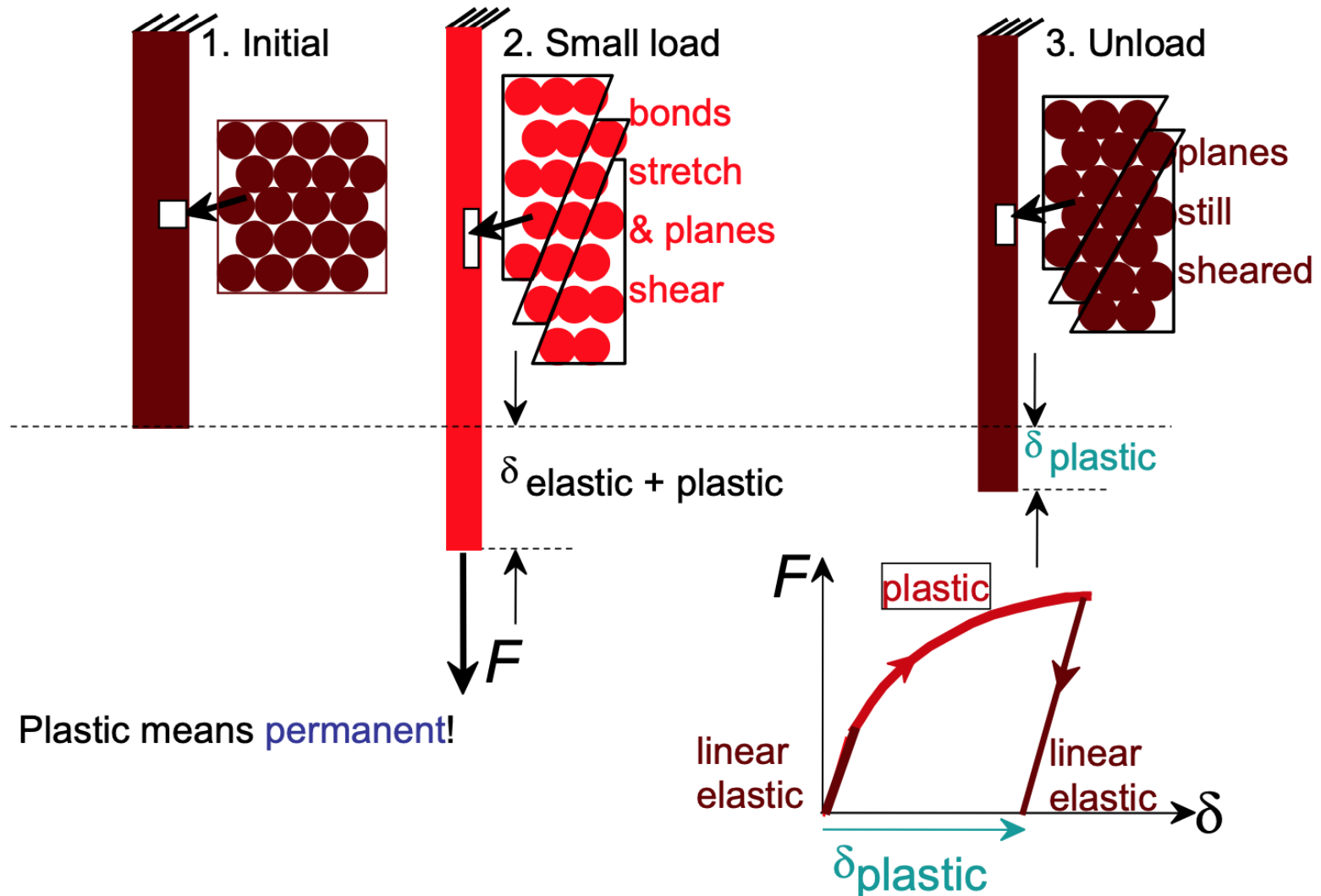
What is the effect of temperature?



Mechanical Properties

Plastic Deformation

What happens at the atomic level?



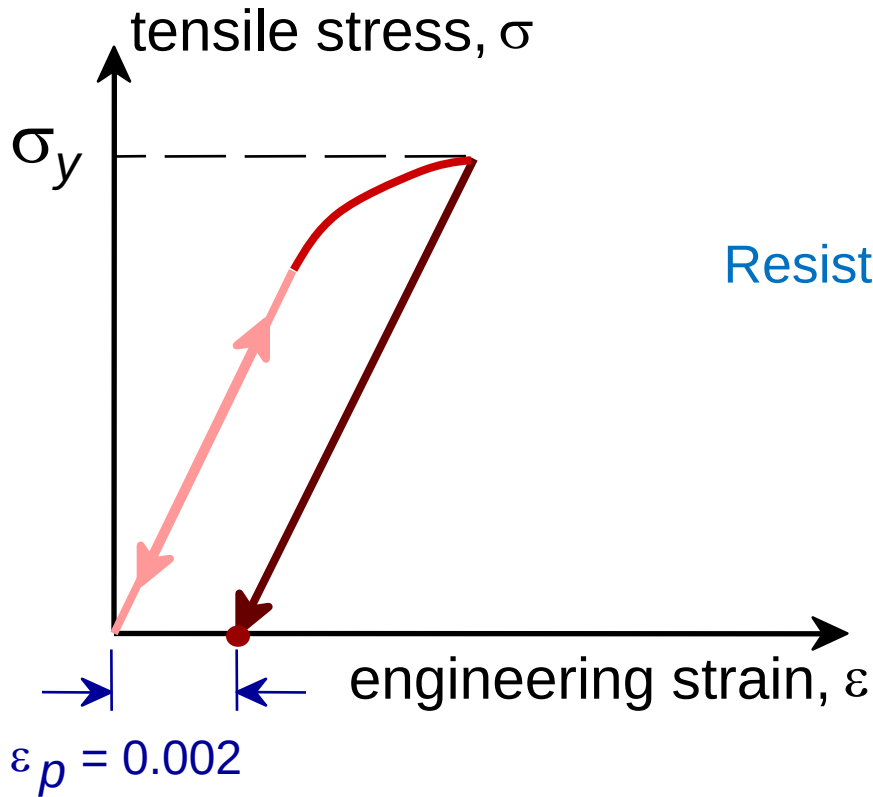


Mechanical Properties

Plastic Deformation

Yield Strength/Point

Stress at which *noticeable* plastic deformation has occurred.



σ_y

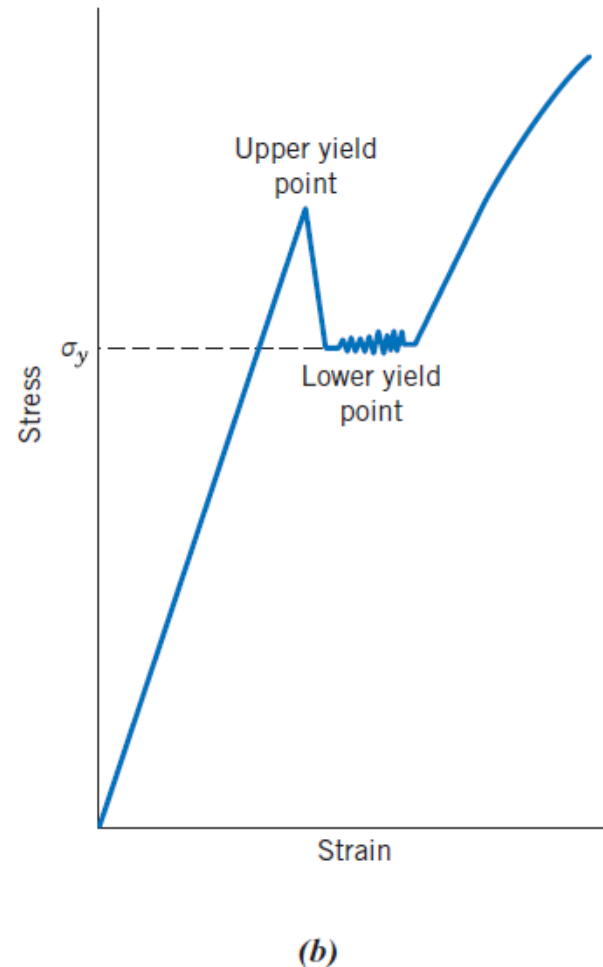
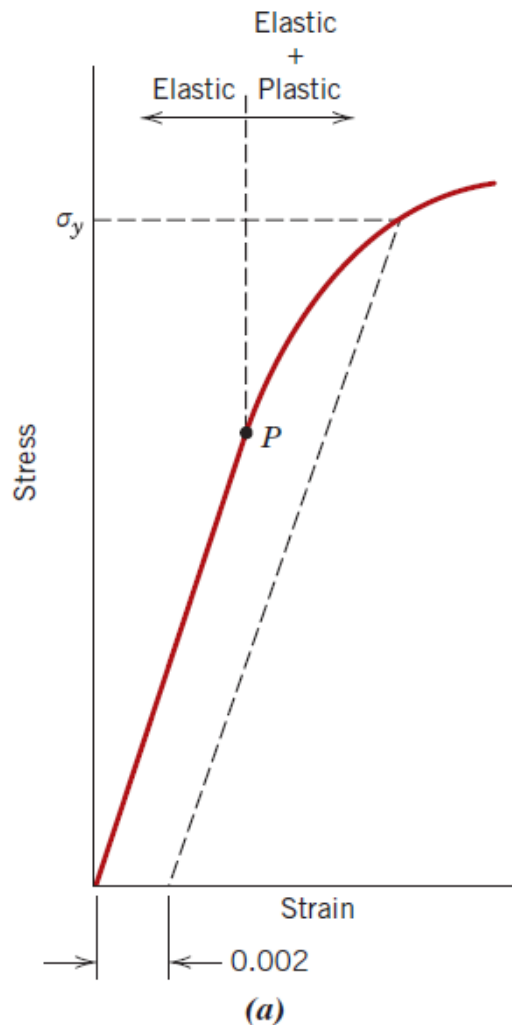
Resistance of material to plastic deformation
is due to **yield strength**

Mechanical Properties

Plastic Deformation

Yield Strength/Point

Yielding





Mechanical Properties

Tensile Stress – Strain

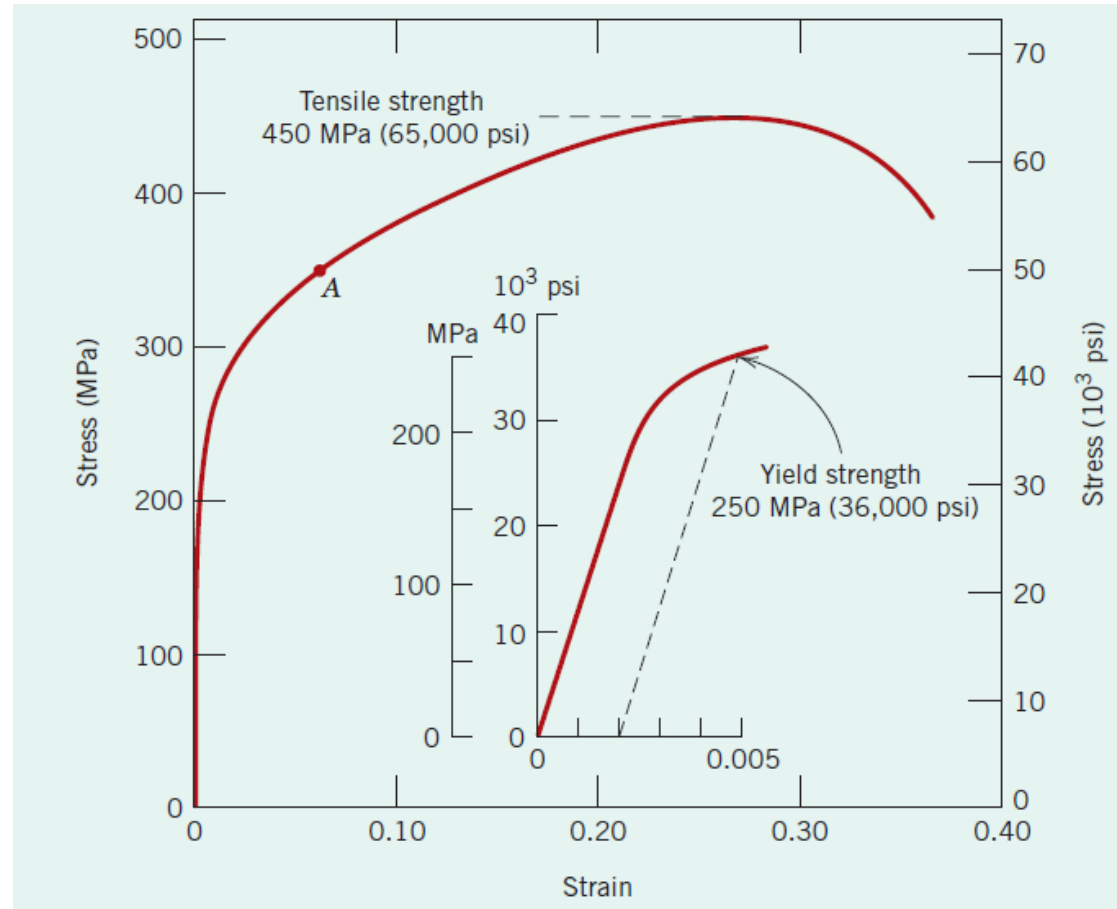
From the tensile stress–strain behavior for the brass specimen shown below, determine:

(a) The modulus of elasticity

$$E = \text{slope} = \frac{\Delta\sigma}{\Delta\varepsilon} = \frac{\sigma_2 - \sigma_1}{\varepsilon_2 - \varepsilon_1}$$

$$E = \frac{(150 - 0) \text{ MPa}}{0.0016 - 0}$$

$$= 93.8 \text{ GPa } (13.6 \times 10^6 \text{ psi})$$





Mechanical Properties

Tensile Stress – Strain

From the tensile stress–strain behavior for the brass specimen shown below, determine:

(b) The yield strength at a strain offset of 0.002

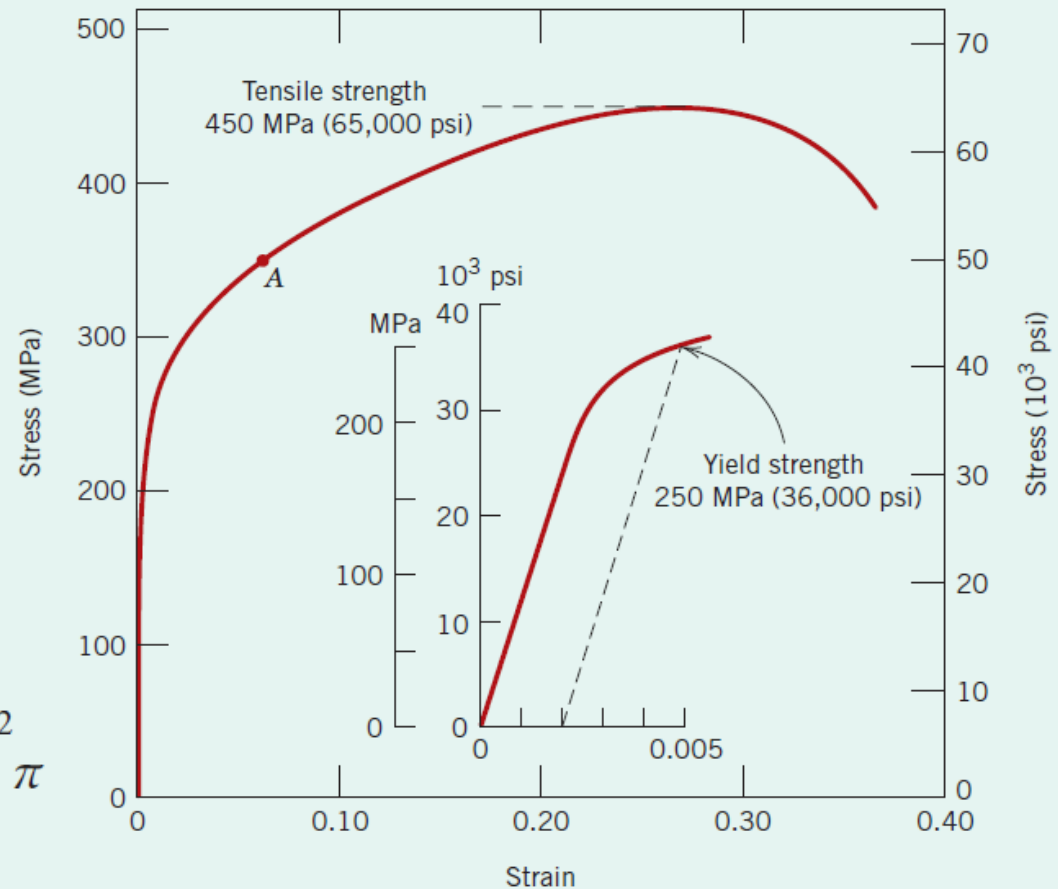
approximately 250 MPa (36,000 psi)

(c) The maximum load that can be sustained by a cylindrical specimen having an original diameter of 12.8 mm (0.505 in.)

$$F = \sigma A_0 = \sigma \left(\frac{d_0}{2} \right)^2 \pi$$

$$= (450 \times 10^6 \text{ N/m}^2) \left(\frac{12.8 \times 10^{-3} \text{ m}}{2} \right)^2 \pi$$

$$= 57,900 \text{ N (13,000 lb}_f\text{)}$$





Mechanical Properties

Tensile Stress – Strain

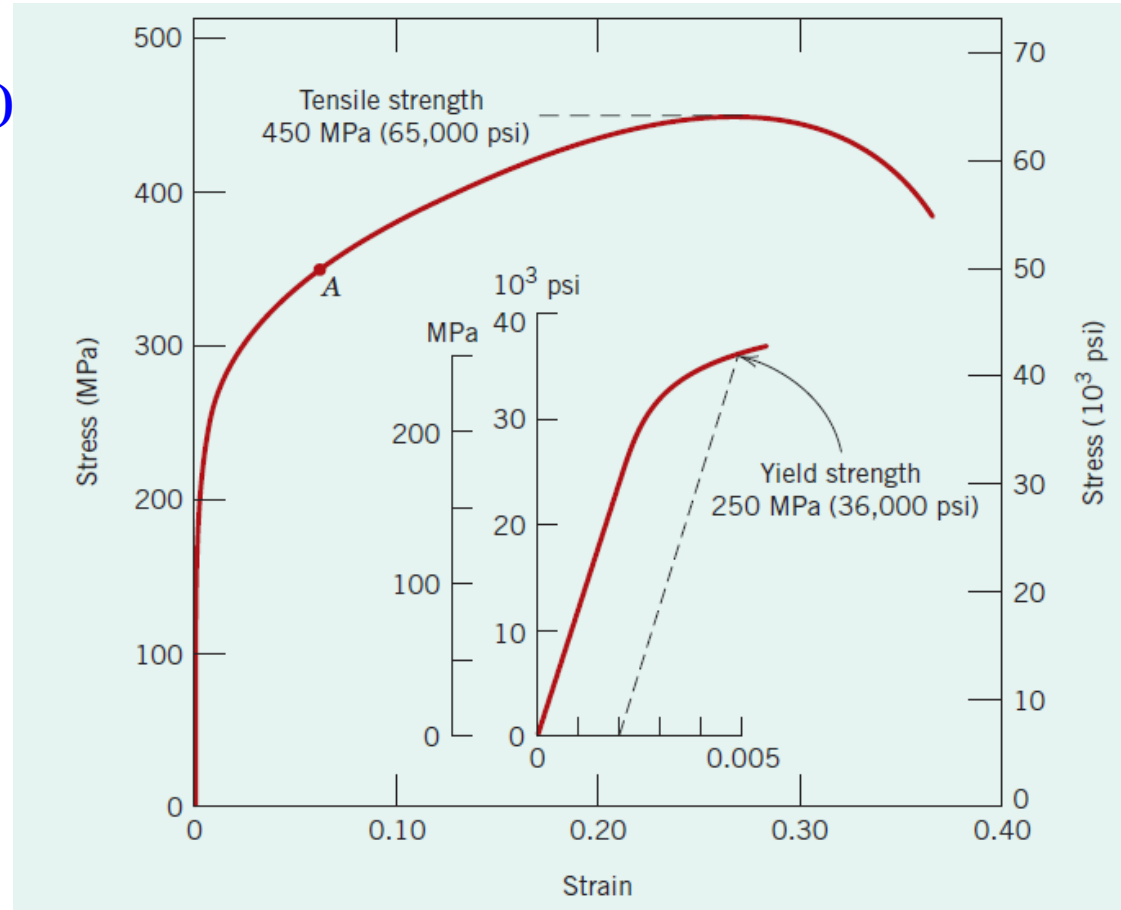
From the tensile stress–strain behavior for the brass specimen shown below, determine:

(d) The change in length of a specimen originally 250 mm (10 in.) long that is subjected to a tensile stress of 345 MPa (50,000 psi)

$$\Delta l = \epsilon l_0$$

$$= (0.06)(250 \text{ mm})$$

$$= 15 \text{ mm (0.6 in.)}$$





Mechanical Properties

Tensile Stress – Strain

A piece of copper originally 305 mm (12 in.) long is pulled in tension with a stress of 276 MPa (40,000 psi). If the deformation is entirely elastic, what will be the resultant elongation? Elastic modulus of Cu is 110 GPa.

$$\sigma = \varepsilon E = \left(\frac{\Delta l}{l_0} \right) E$$

$$\Delta l = \frac{\sigma l_0}{E}$$

$$\Delta l = \frac{(276 \text{ MPa})(305 \text{ mm})}{110 \times 10^3 \text{ MPa}} = 0.77 \text{ mm (0.03 in.)}$$

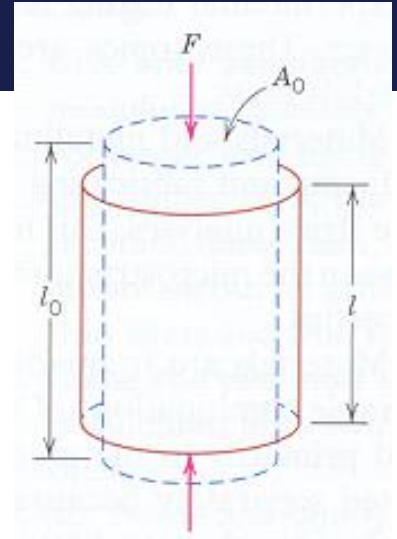


Mechanical Properties

Stress – Strain Tests

1. Compressive stress:

not very common, and mostly used for brittle materials, stress is taken to be negative, strain is already negative
→ compressive is much larger than tensile



2. Shear stress (couple)

$$\tau = \frac{F}{A_0}$$

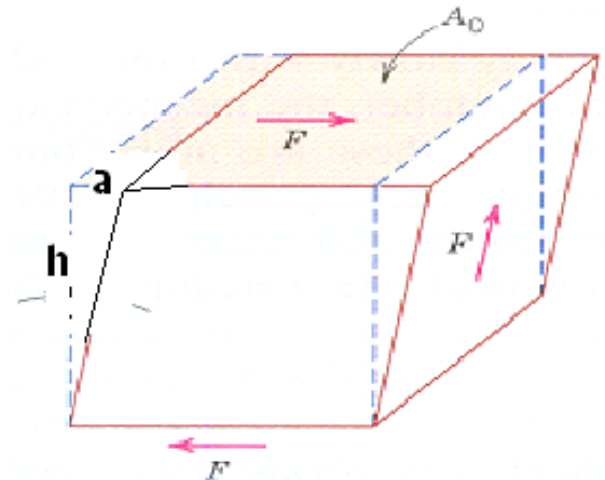
where F is force or load applied to upper and lower faces distance h apart, each of which has an area A_0

The **shear strain** γ is defined as the tangent of the strain angle θ

$$\text{Shear strain} = \gamma = a / h = \tan \theta$$

$$\tau = \gamma G$$

Units?



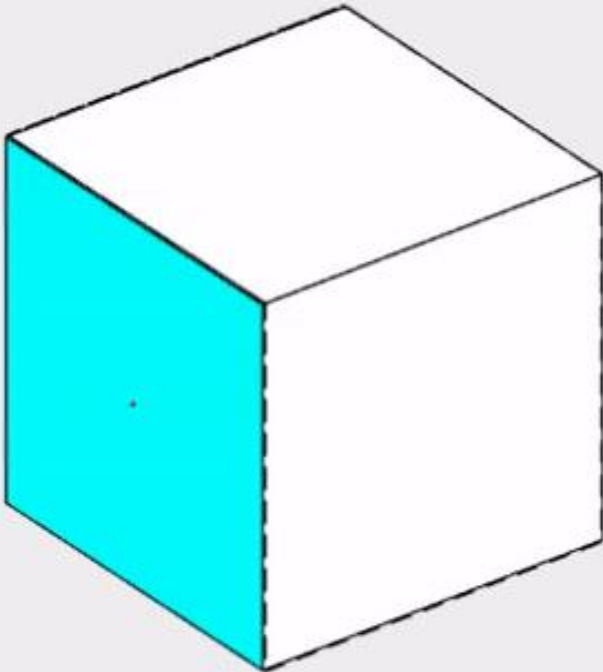
3. Torsion is special form of shear (torque) applied to shaft or axles in rotation



Mechanical Properties

Elastic Deformation

Poisson's Ratio ν is defined as the ratio of the lateral and axial strains



Poisson's Ratio is the parameter that tells us how the material will deform in the lateral directions.

ν = Lateral / Axial strain

$$\nu = -\frac{\epsilon_x}{\epsilon_z} = -\frac{\epsilon_y}{\epsilon_z}$$

Mechanical Properties

Elastic Deformation

Poisson's Ratio ν is defined as the ratio of the lateral and axial strains

Isotropic materials and in elastic region only

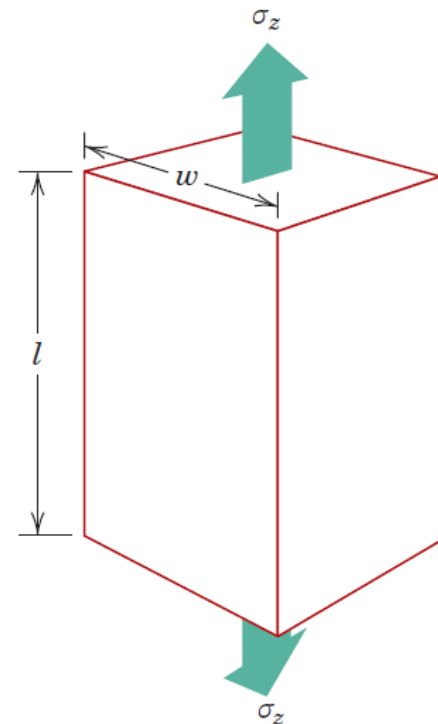
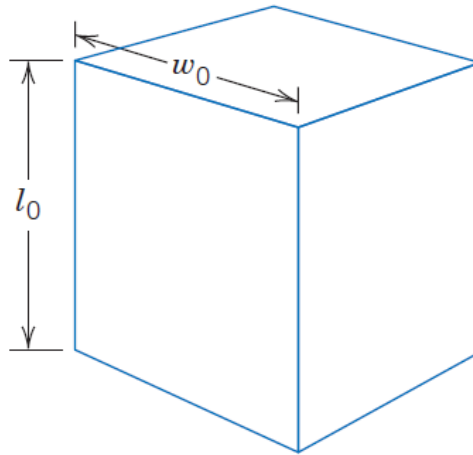
$$\epsilon_z = \frac{l - l_0}{l_0} = \frac{\Delta l}{l_0} > 0$$

$$\epsilon_x = \frac{w - w_0}{w_0} = \frac{\Delta w}{w_0} < 0$$

Poisson's ratio:

$$\nu = -\frac{\epsilon_x}{\epsilon_z}$$

$$E = 2G(1 + \nu)$$



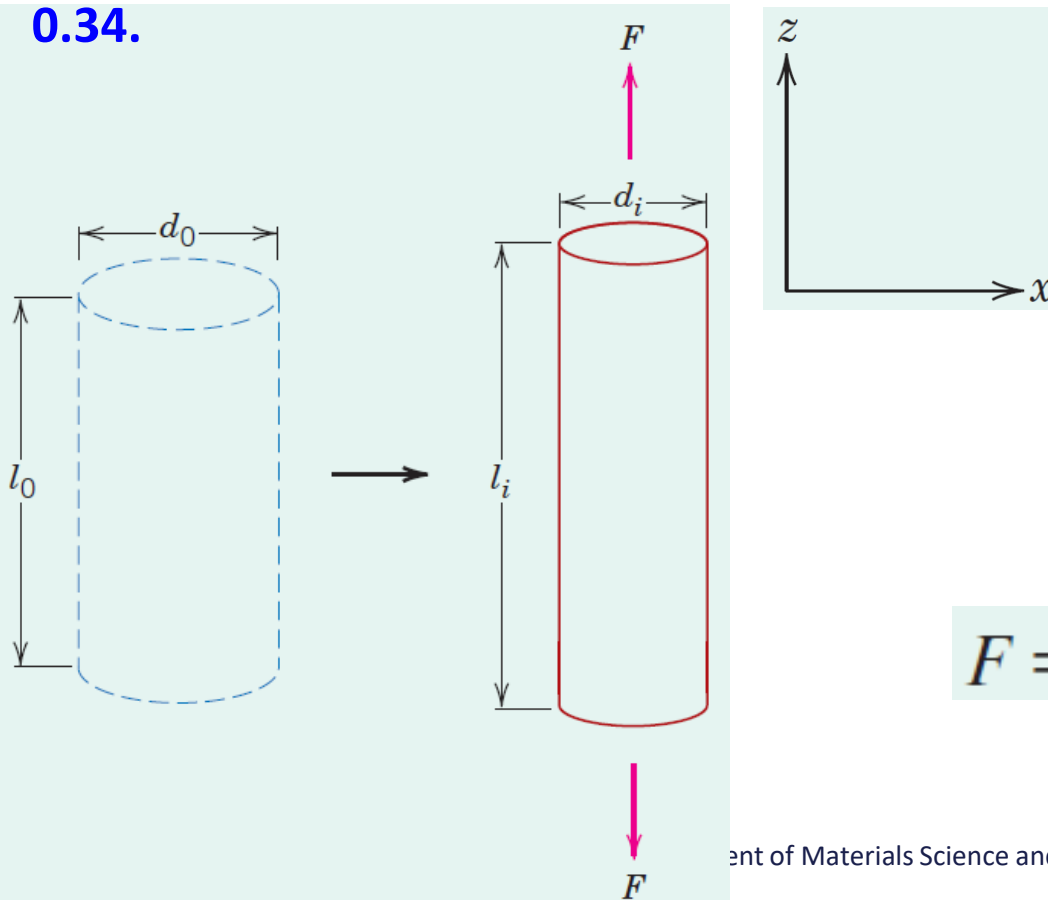
Shear Modulus

Depart

Elastic modulus

Elastic Deformation

A tensile stress is to be applied along the long axis of a cylindrical brass rod that has a diameter of 10 mm (0.4 in.). Determine the magnitude of the load required to produce a 2.5×10^{-3} mm (10^{-4} in.) change in diameter if the deformation is entirely elastic. Elastic modulus is 97 GPa (14×10^6 psi). Poisson's ratio for brass is 0.34.



$$\varepsilon_z = \frac{\Delta l}{l_0} = \frac{l_i - l_0}{l_0}$$
$$\varepsilon_x = \frac{\Delta d}{d_0} = \frac{d_i - d_0}{d_0}$$

$$F = 5600 \text{ N (1293 lb}_f\text{)}$$

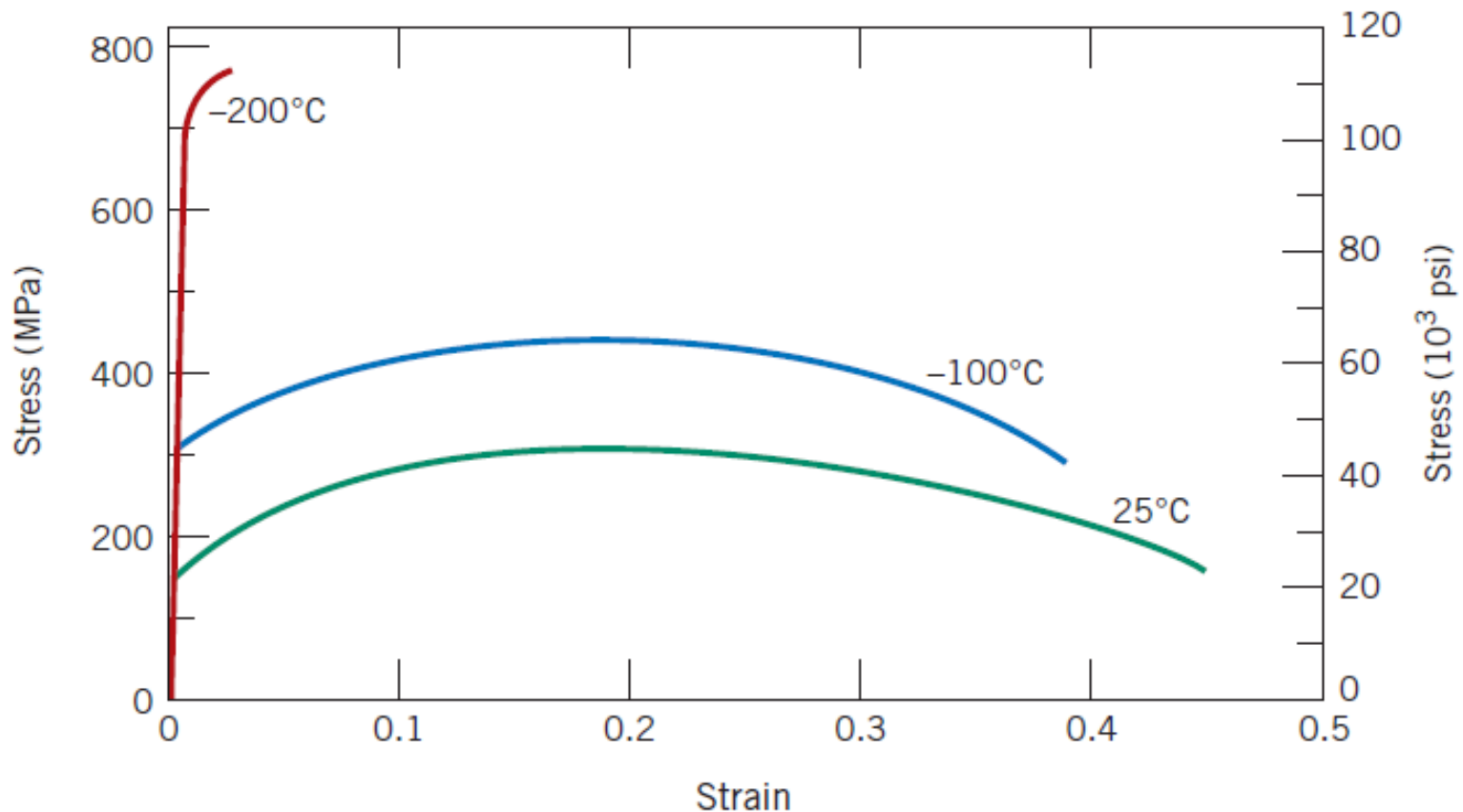


Mechanical Properties

Tensile Stress – Strain

Influence of Temperature:

Example–Iron (ductility increases as T increases. Why?)





Mechanical Properties

Tensile Stress – Strain

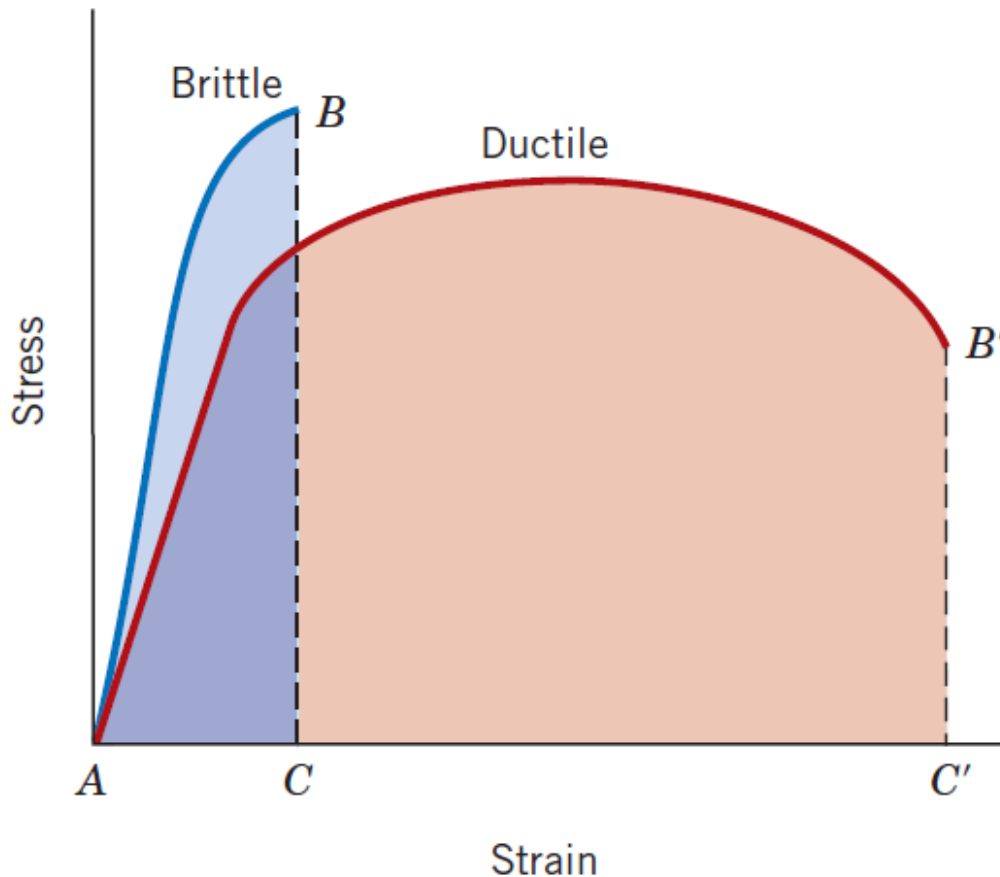
Ductility: measure of the degree of plastic deformation that has been sustained at fracture

% Elongation

$$\%EL = \left(\frac{l_f - l_0}{l_0} \right) \times 100$$

% Reduction in Area

$$\%RA = \left(\frac{A_0 - A_f}{A_0} \right) \times 100$$





Mechanical Properties

Tensile Stress – Strain

Properties of different materials

<i>Metal Alloy</i>	<i>Yield Strength, MPa (ksi)</i>	<i>Tensile Strength, MPa (ksi)</i>	<i>Ductility, %EL [in 50 mm (2 in.)]</i>
Aluminum	35 (5)	90 (13)	40
Copper	69 (10)	200 (29)	45
Brass (70Cu–30Zn)	75 (11)	300 (44)	68
Iron	130 (19)	262 (38)	45
Nickel	138 (20)	480 (70)	40
Steel (1020)	180 (26)	380 (55)	25
Titanium	450 (65)	520 (75)	25
Molybdenum	565 (82)	655 (95)	35

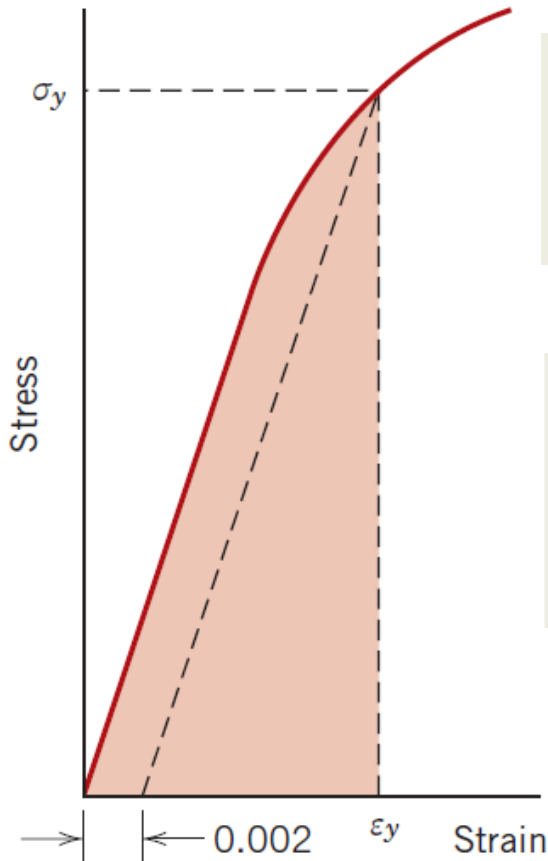


Mechanical Properties

Tensile Stress – Strain

Resilience: the capacity of a material to absorb energy when it is deformed elastically and then, upon unloading, to have this energy recovered

Modulus of Resilience



$$U_r = \int_0^{\epsilon_y} \sigma d\epsilon$$

$$U_r = \frac{1}{2} \sigma_y \epsilon_y$$

Units:
J/m³ or Pa

$$U_r = \frac{1}{2} \sigma_y \epsilon_y = \frac{1}{2} \sigma_y \left(\frac{\sigma_y}{E} \right) = \frac{\sigma_y^2}{2E}$$

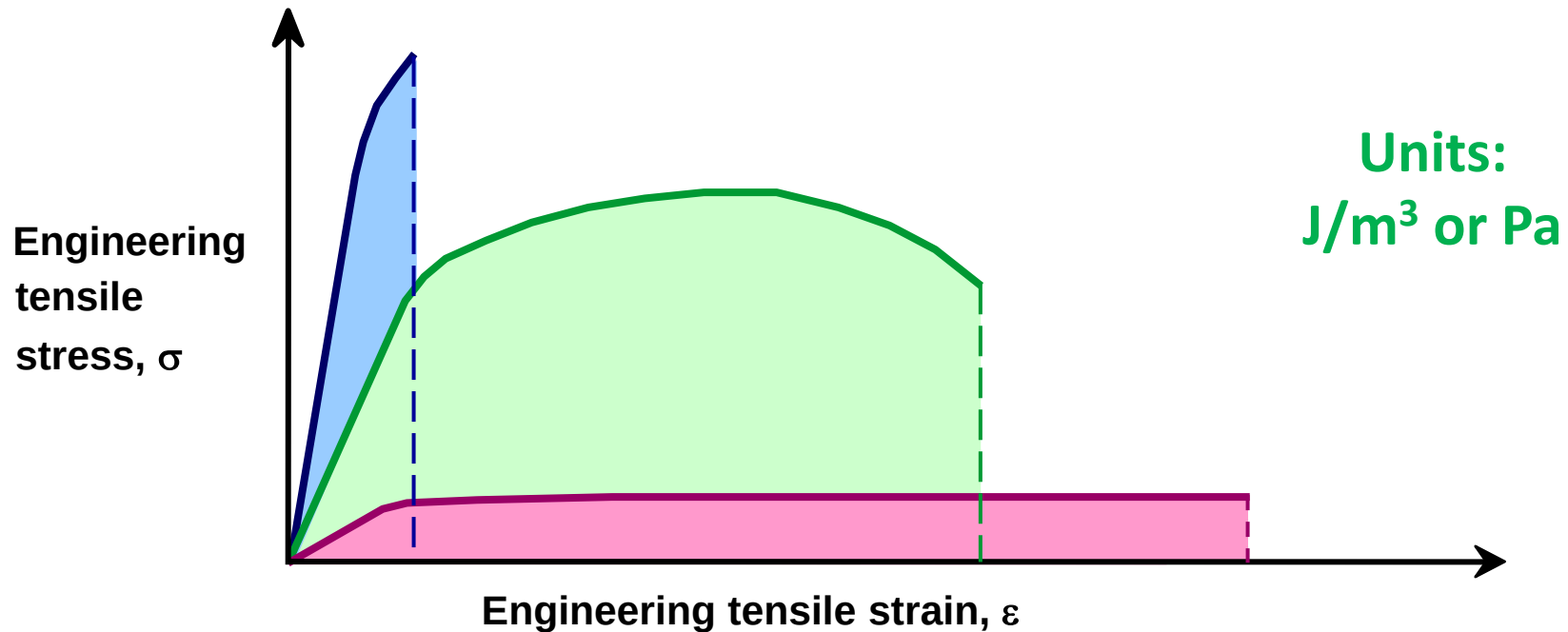
resilient materials are those having high yield strengths and low moduli of elasticity



Mechanical Properties

Tensile Stress – Strain

Toughness: the ability of a material to absorb energy and plastically deform before fracturing



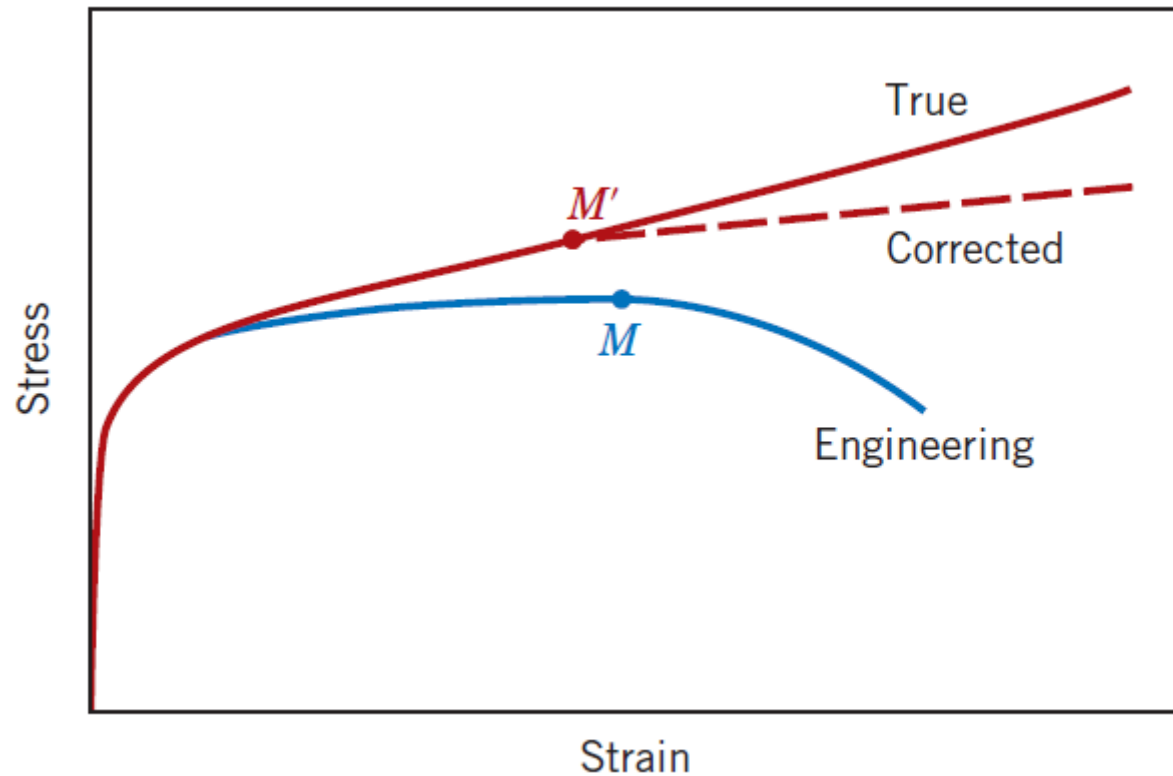
$$\text{Toughness} = \sigma \times \epsilon = \frac{F}{A} \times \frac{\Delta l}{l} = \frac{\text{Energy}}{\text{volume}} = \frac{\text{Joule}}{\text{m}^3}$$



Mechanical Properties

Tensile Stress – Strain

True stress and true strain: instantaneous stress and strain



$$\sigma_T = \frac{F}{A_i}$$

$$\varepsilon_T = \ln \frac{l_i}{l_0}$$

$$\sigma_T = \sigma(1 + \varepsilon)$$

$$\varepsilon_T = \ln(1 + \varepsilon)$$



Mechanical Properties

Tensile Stress – Strain

A cylindrical specimen of steel having an original diameter of 12.8 mm (0.505 in.) is tensile tested to fracture and found to have an engineering fracture strength σ_f of 460 MPa (67,000 psi). If its cross-sectional diameter at fracture is 10.7 mm (0.422 in.), determine:

(a) The ductility in terms of percentage reduction in area

$$\% \text{ RA} = \frac{\left(\frac{12.8 \text{ mm}}{2}\right)^2 \pi - \left(\frac{10.7 \text{ mm}}{2}\right)^2 \pi}{\left(\frac{12.8 \text{ mm}}{2}\right)^2 \pi} \times 100$$

$$= \frac{128.7 \text{ mm}^2 - 89.9 \text{ mm}^2}{128.7 \text{ mm}^2} \times 100 = 30\%$$

(b) The true stress at fracture

$$F = \sigma_f A_0$$

$$= (460 \times 10^6 \text{ N/m}^2)(128.7 \text{ mm}^2) \left(\frac{1 \text{ m}^2}{10^6 \text{ mm}^2}\right)$$

$$= 59,200 \text{ N}$$

$$\sigma_T = \frac{F}{A_f} = \frac{59,200 \text{ N}}{(89.9 \text{ mm}^2) \left(\frac{1 \text{ m}^2}{10^6 \text{ mm}^2}\right)}$$

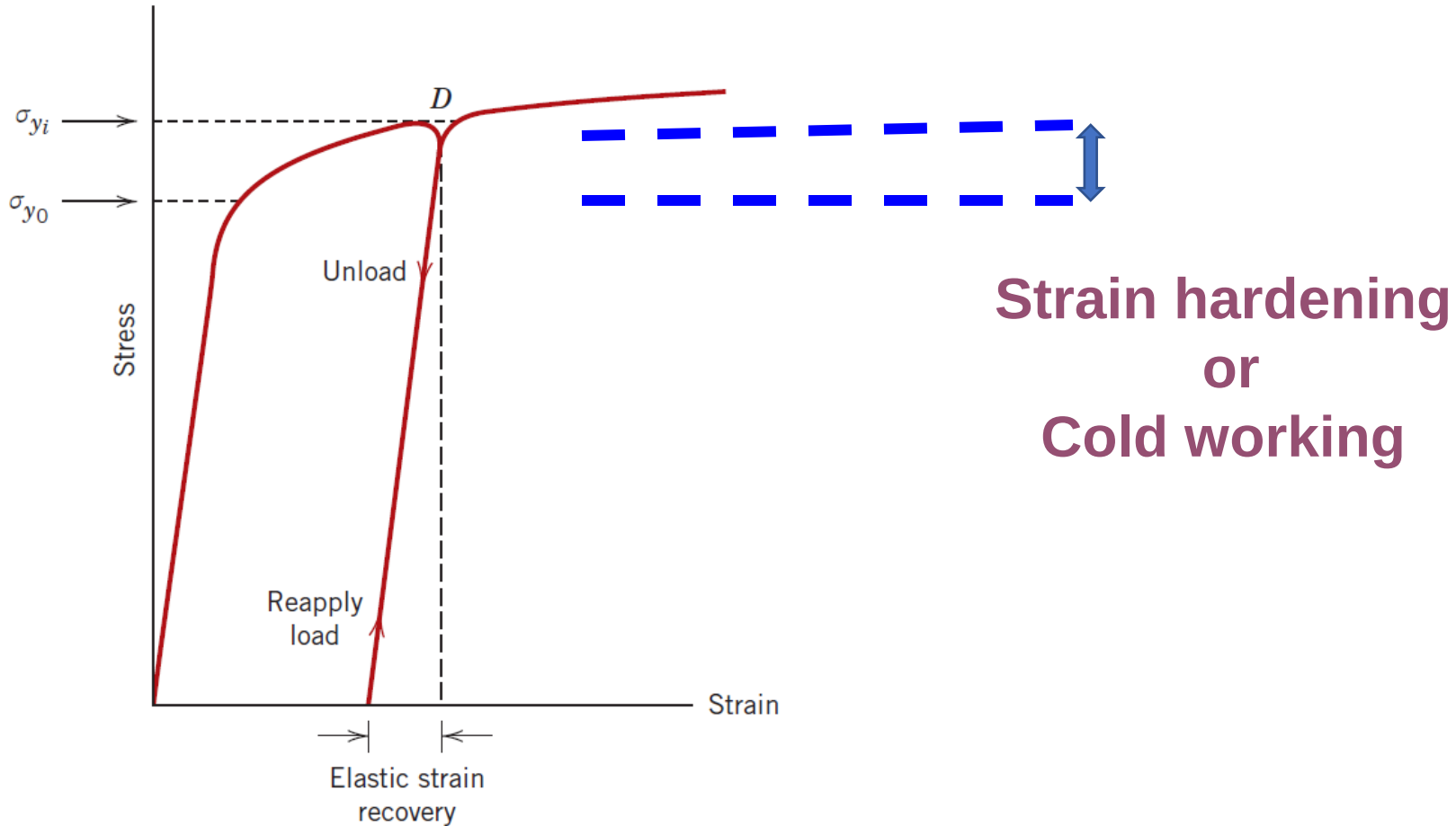
$$= 6.6 \times 10^8 \text{ N/m}^2 = 660 \text{ MPa (95,700 psi)}$$



Mechanical Properties

Tensile Stress – Strain

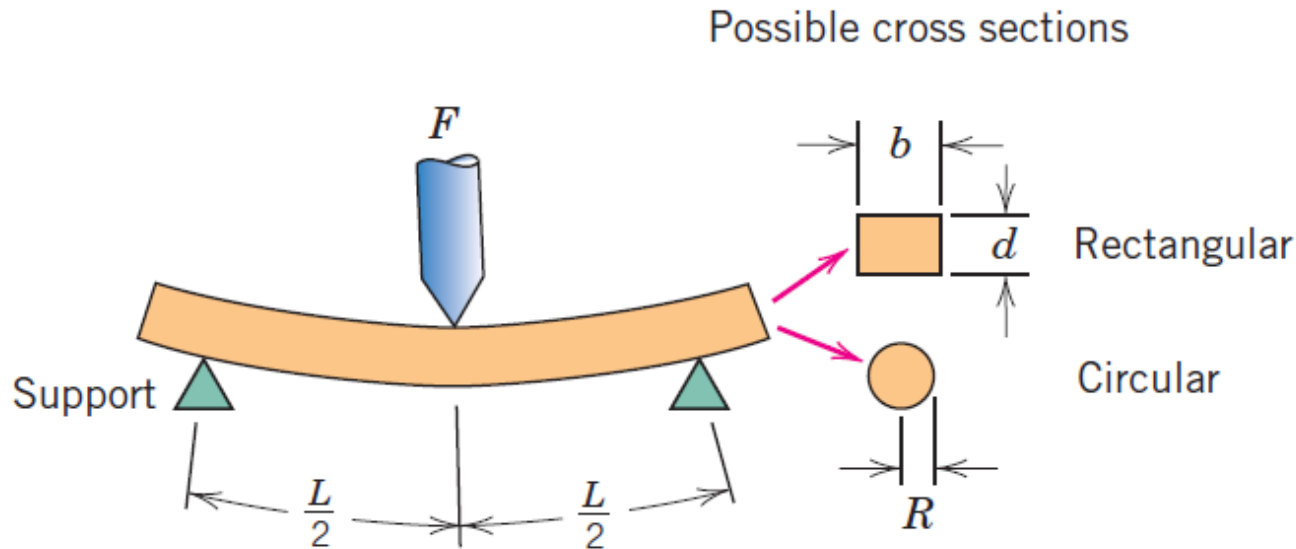
Elastic recovery after plastic deformation





Mechanical Properties

Flexural strength in ceramics



$$\sigma = \text{stress} = \frac{Mc}{I}$$

where M = maximum bending moment

c = distance from center of specimen to outer fibers

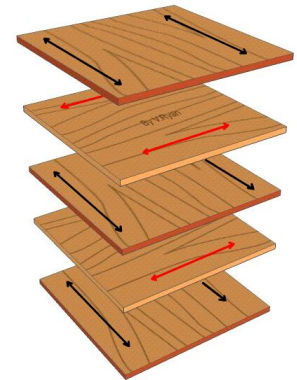
I = moment of inertia of cross section

F = applied load

	$\frac{M}{FL}$	$\frac{c}{d}$	$\frac{I}{bd^3}$	$\frac{\sigma}{\frac{3FL}{2bd^2}}$
Rectangular	$\frac{FL}{4}$	$\frac{d}{2}$	$\frac{bd^3}{12}$	$\frac{3FL}{2bd^2}$
Circular	$\frac{FL}{4}$	R	$\frac{\pi R^4}{4}$	$\frac{FL}{\pi R^3}$

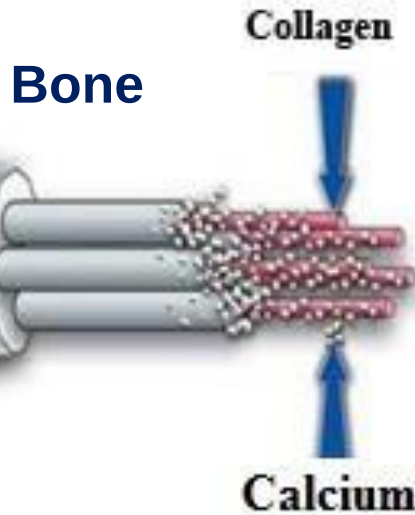
Composites

Reinforced Cement Concrete

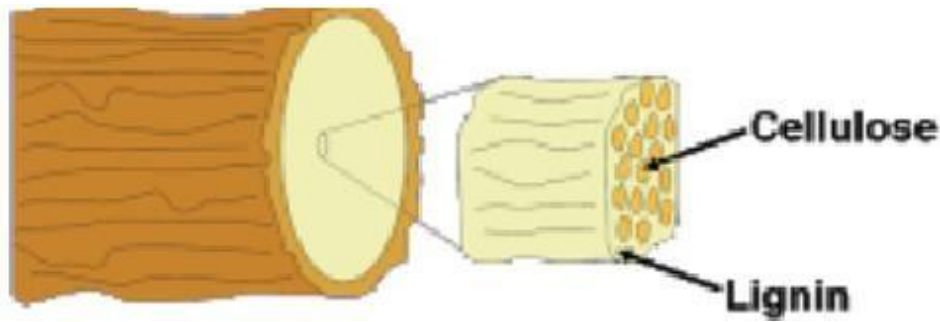


Plywood

Fiberglass water tank (glass fibres reinforce plastic)



Wood (Cellulose fibres in



Composites

FIBER

MATRIX

COMPOSITE



+



=



Mud and Straw Plaster



Mud and Straw Bricks



Old houses



Science and Engineering



Conventional Materials:

Polymers: *Light weight, low cost **but low strength and low moduli**. Also large strains.*

Ceramics: *Strong, stiff. **but brittle..!***

Metals: *Intermediate strength and moduli, along with good ductility, **but heavy..!***

Engineering Applications often require unusual combination of properties

For Example:

Aerospace requires strong, stiff, light weight and abrasion resistant materials.

Available strong materials are dense and heavy (metals/ceramics);

While light weight materials (polymers) are not good abrasion resistant and lack enough strength.

Solution..??

Composites (PMCs and CMCs and MMCs)

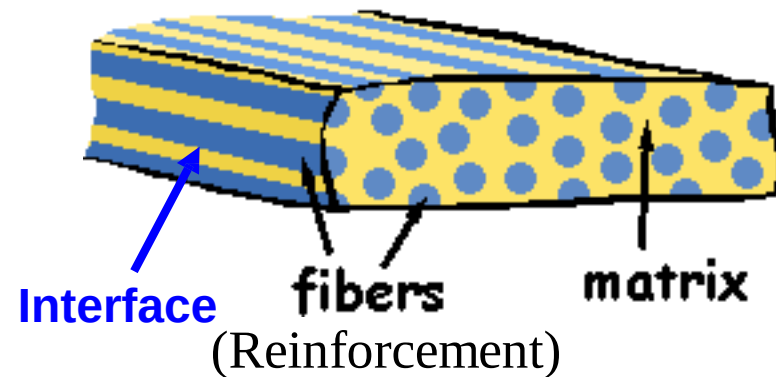
Other Applications: *Automobiles/transportation, sports etc.*



Three important components of a composite material:

1. Matrix (continuous)
2. Reinforcement (dis-continuous)
3. Interface ... chemically dissimilar

a close-up of what
a composite might
look like



FIBER

MATRIX

COMPOSITE



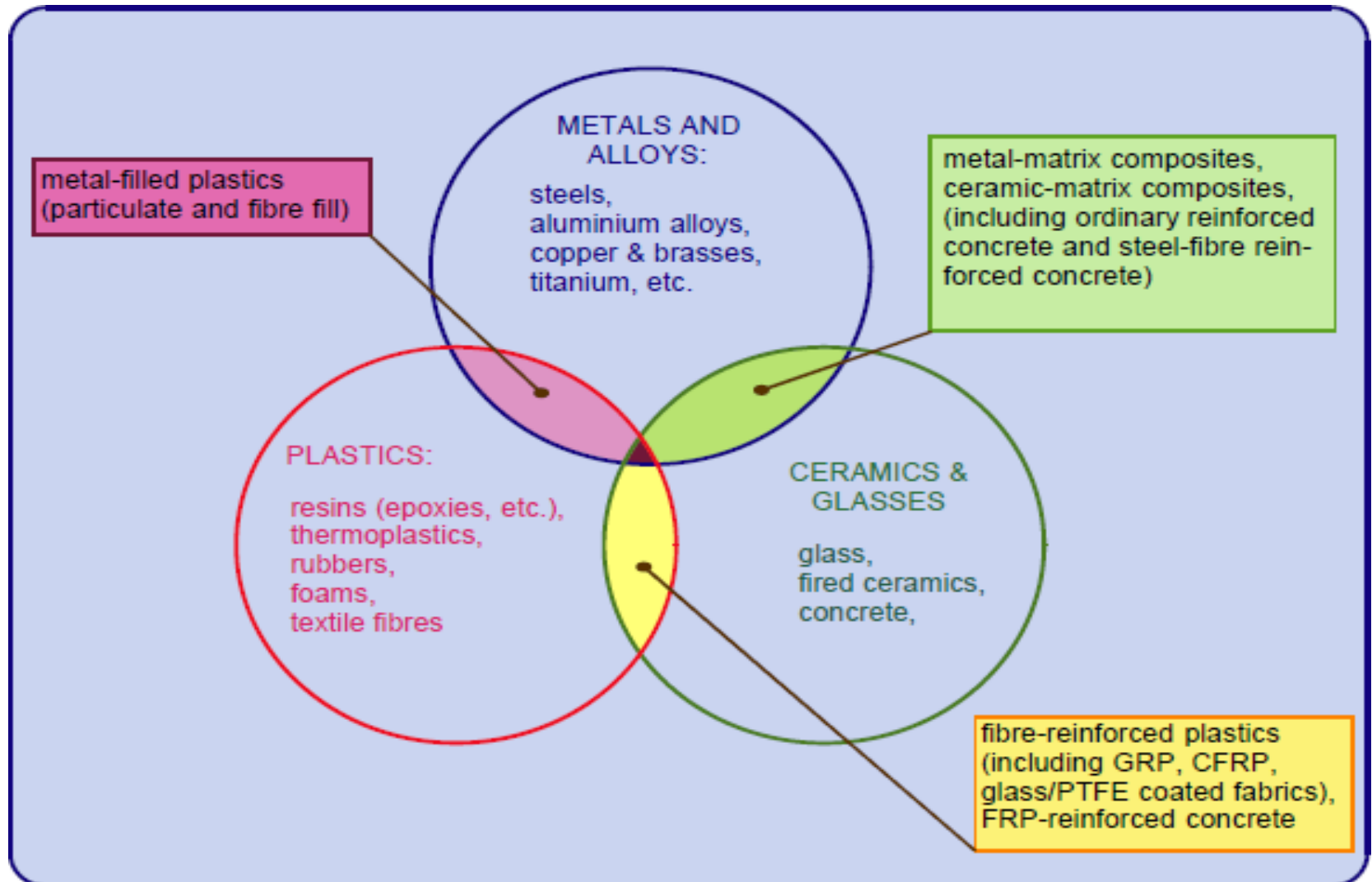
Roles of these components...??

Composites



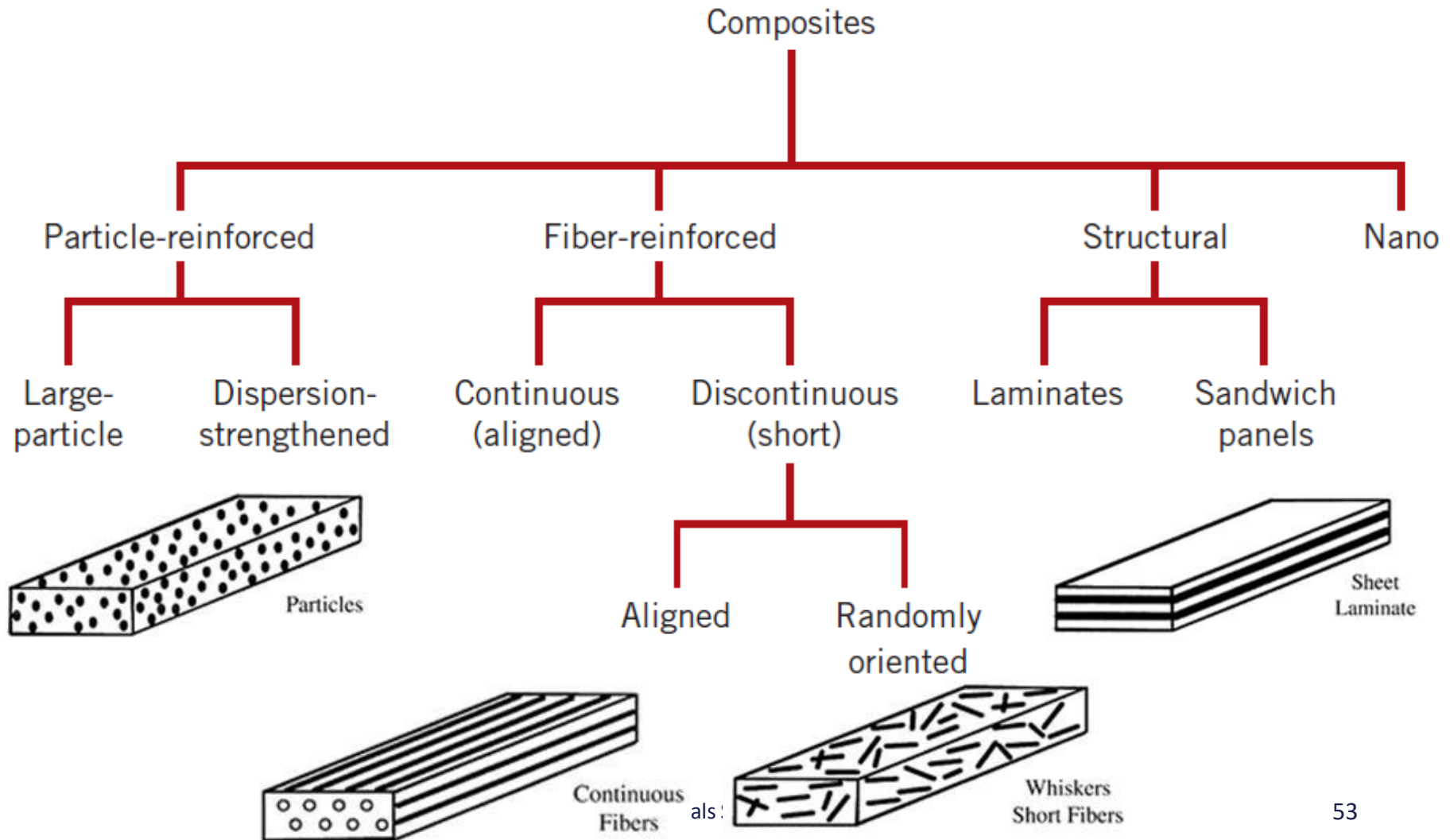
Classification

Polymer Matrix (PMC), Metal Matrix (MMC), Ceramic Matrix (CMC)



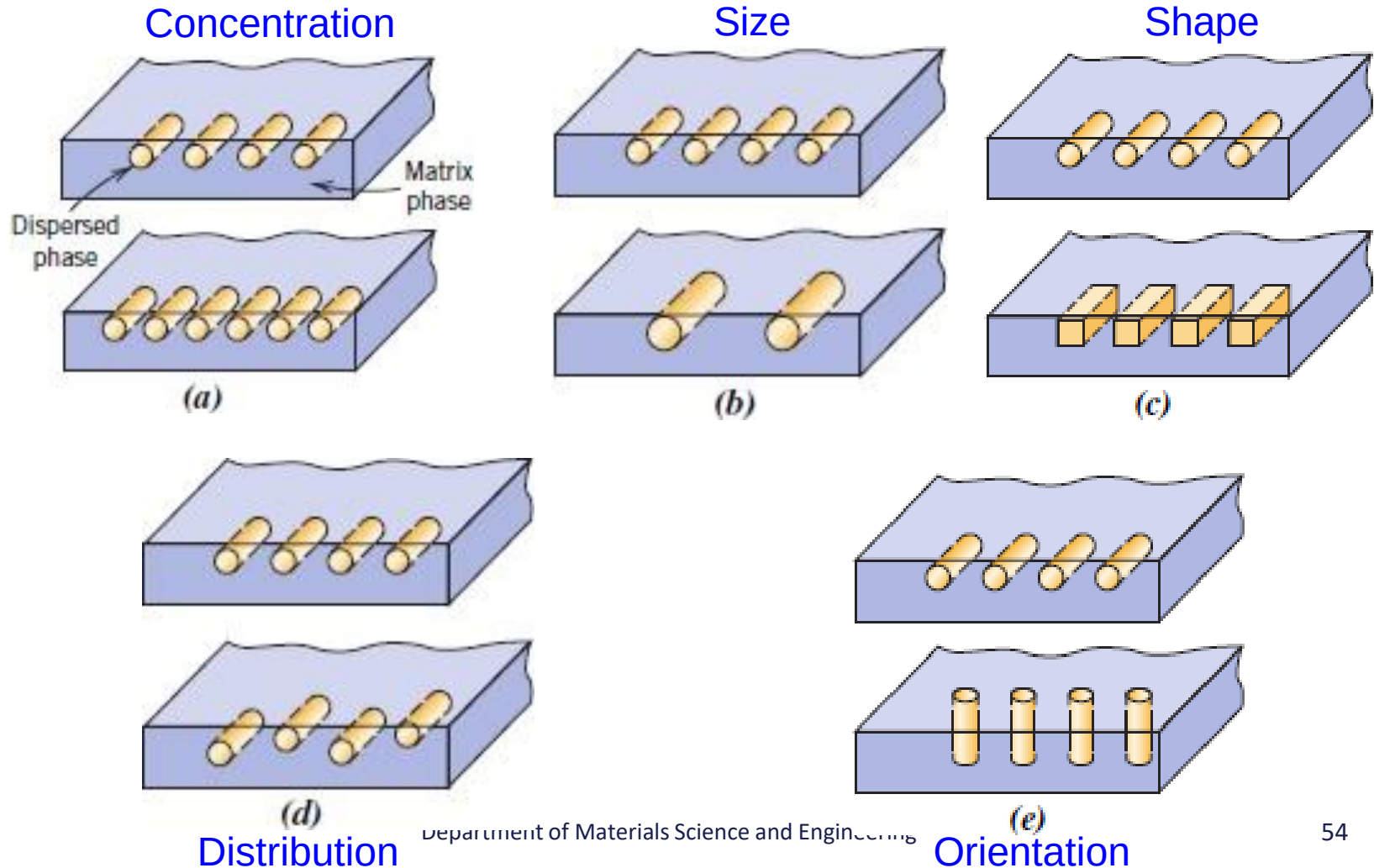
Classification

Based on Reinforcement Phase (Morphology: Size, Shape, Orientation)

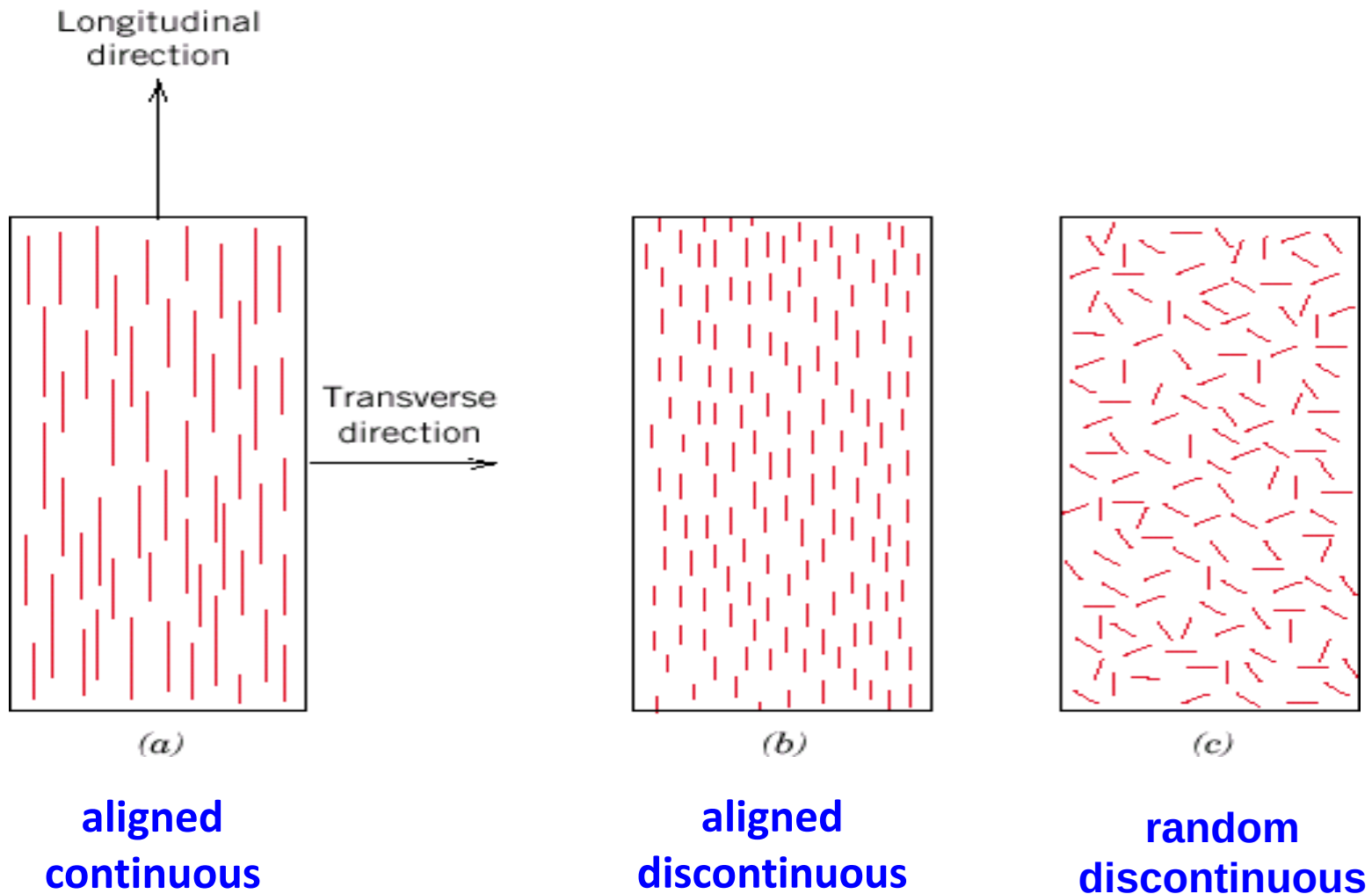


Classification

Based on Reinforcement Phase (Geometry : Concentration, Size, Shape, Distribution, Orientation)



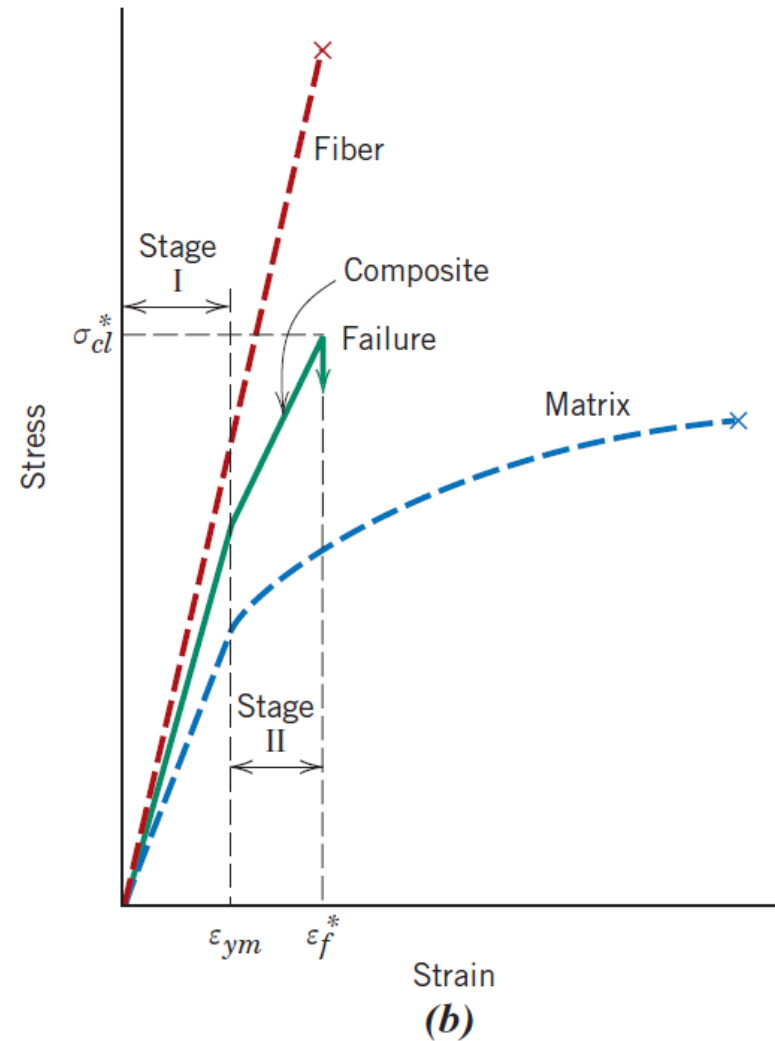
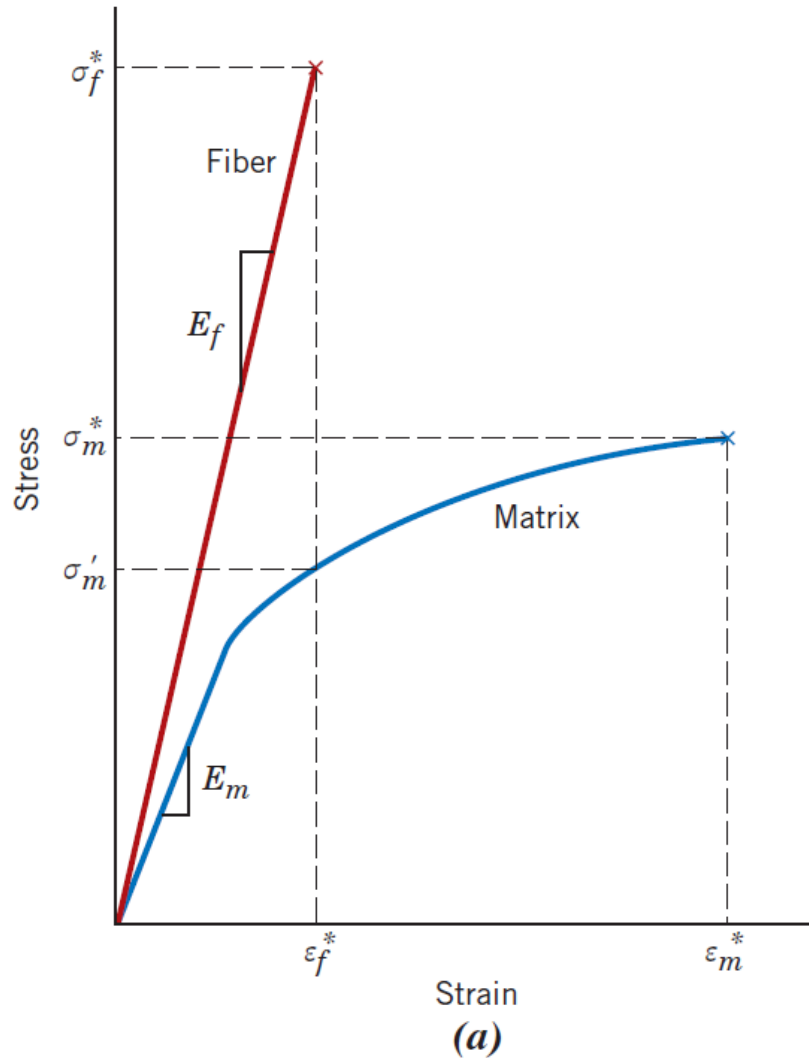
Stress-strain - Influence of Fiber orientation





Composites

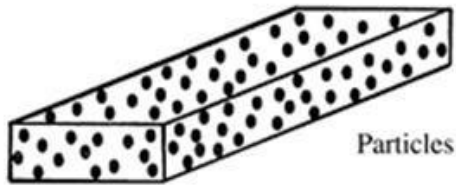
Stress-strain - Influence of Fiber orientation



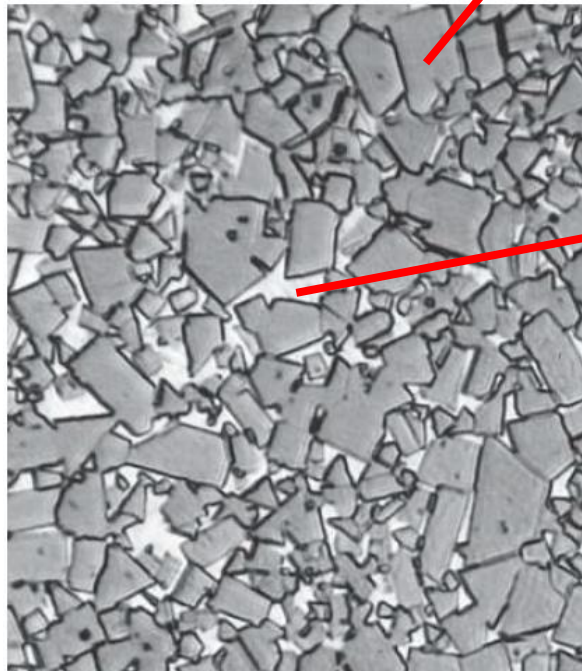
Composites

Particulate Reinforced

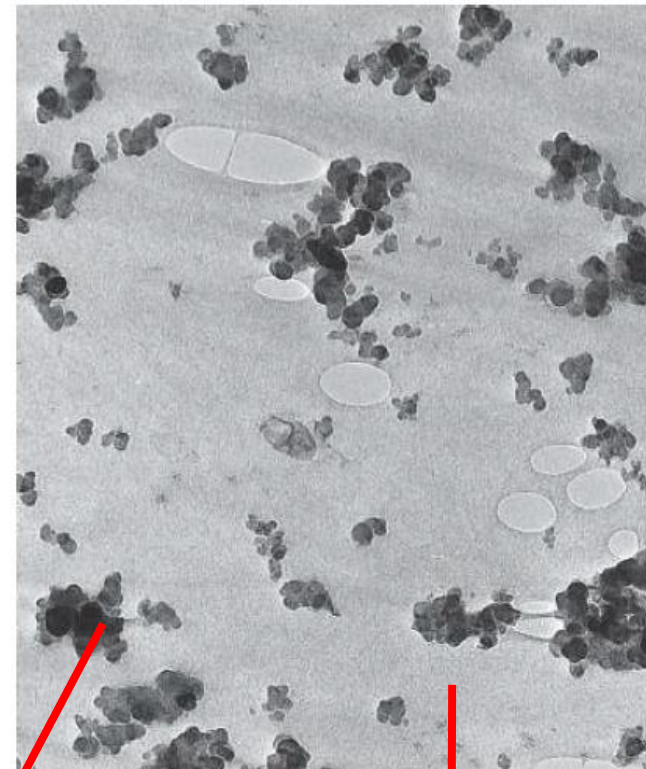
Large particle ($> 0.1 \mu\text{m}$) and dispersion strengthened ($0.01 - 0.1 \mu\text{m}$)



**WC
(90 vol%)**



Co



Carbon black particles

Synthetic rubber

100 nm

Concrete ???

100 μm



Composites

Particulate Reinforced

Rule of Mixtures: to estimate the properties of composite by volume fraction of matrix and reinforcements

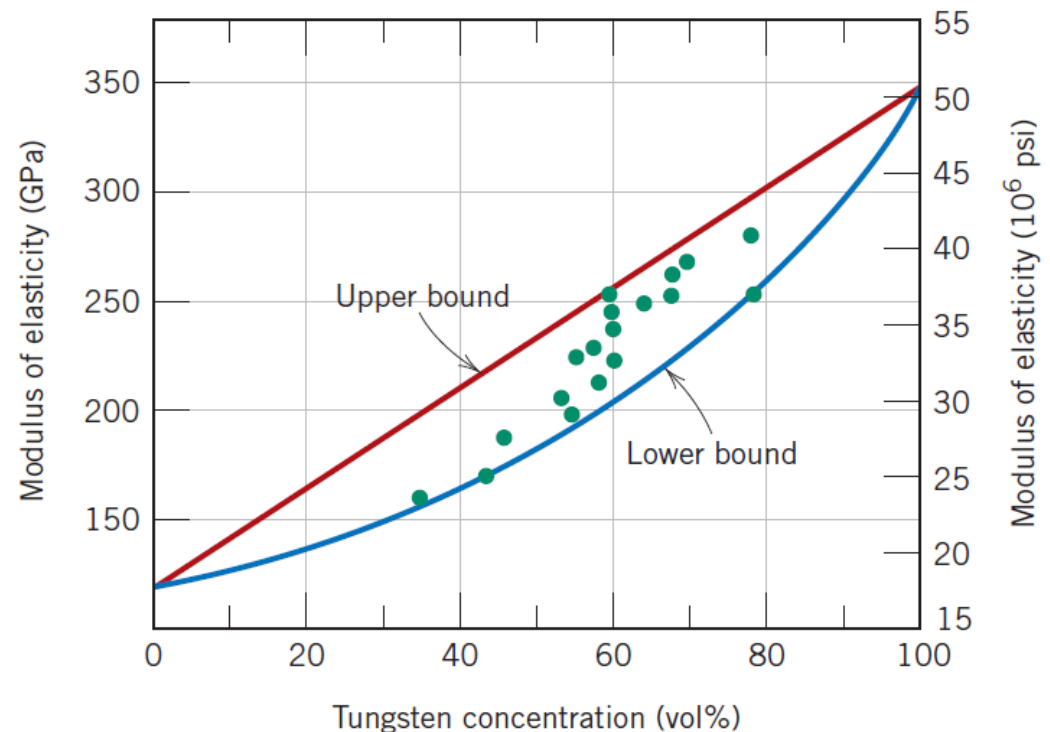
Elastic modulus falls between upper and lower limits

Upper Bound

$$E_c(u) = E_m V_m + E_p V_p$$

Lower Bound

$$E_c(l) = \frac{E_m E_p}{V_m E_p + V_p E_m}$$





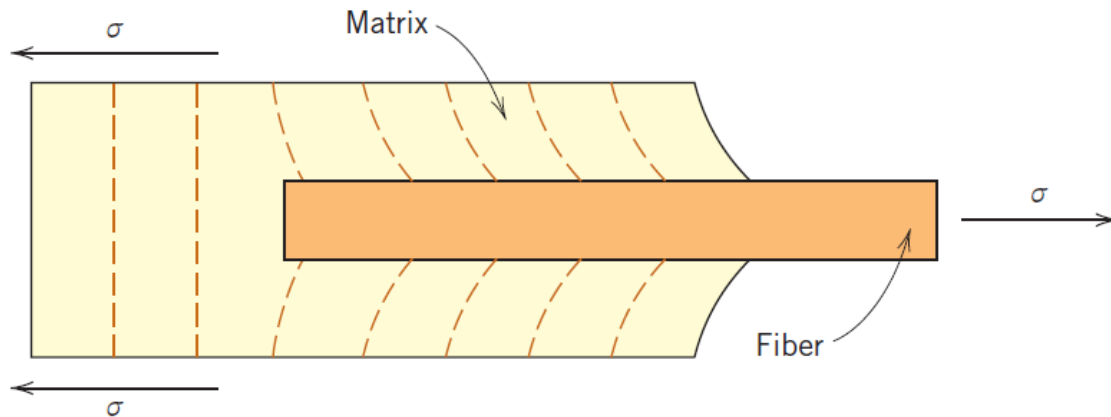
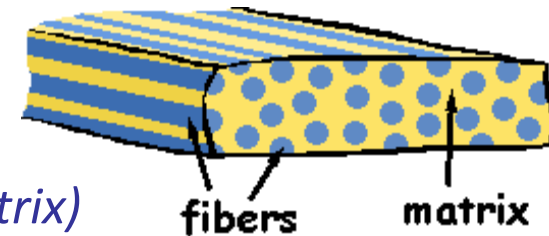
Composites

Fiber Reinforced

(High strength and stiffness to weight)

Mechanical properties of the composite depends on:

- Mechanical properties of the fiber
- How much load the matrix can transmit to the fibers
(depends on interfacial bond between the fiber and the matrix)



$$l_c = \frac{\sigma_f^* d}{2\tau_c}$$

Critical fiber length:

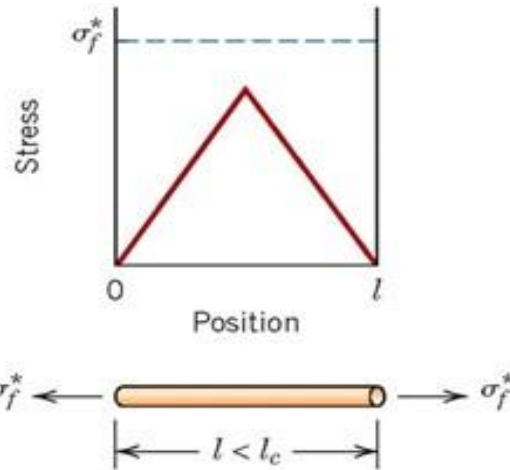
- minimum fiber length at which there is maximum possible load transfer from the matrix to the fiber
- necessary for effective strengthening and stiffening of the composite material

Composites

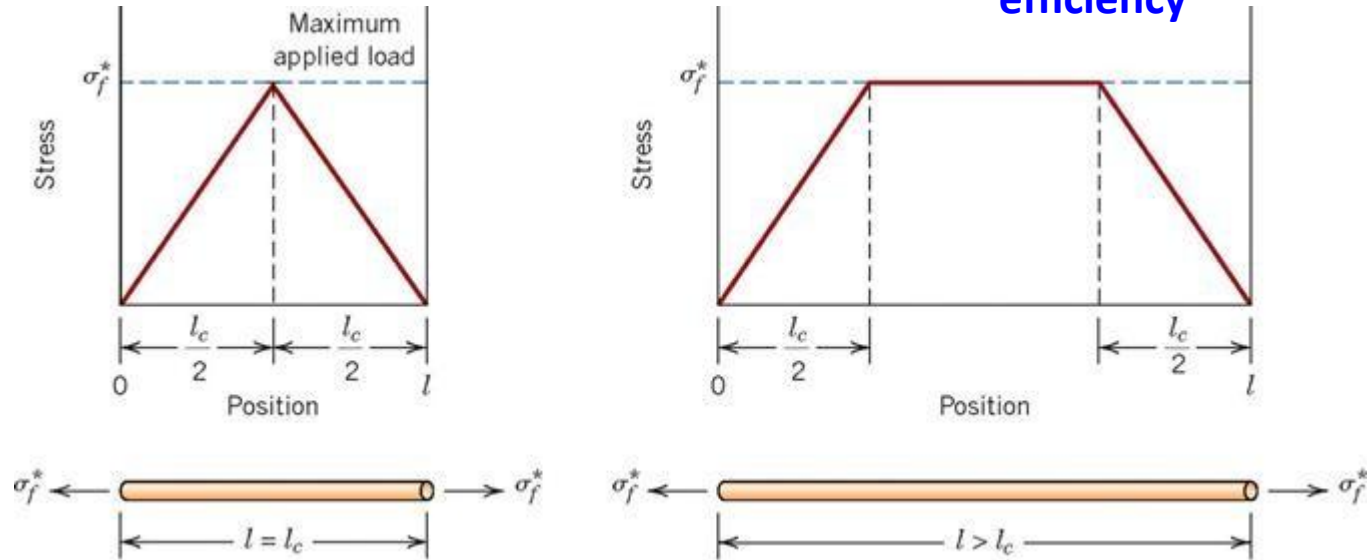


Fiber Reinforced

Poorer fiber efficiency



Better fiber efficiency



$$l_c = \frac{\sigma_f^* d}{2\tau_c}$$

- Continuous fibers: $l \gg 15l_c$
- Discontinuous/short fibers: $l < 15l_c$

d = diameter of the fiber, τ_c = fiber-matrix bond strength

σ_f = fiber fracture/tensile strength

Fiber Reinforced – continuous and aligned

Mechanical properties in this case depend on:

- Stress-strain behavior of fiber and matrix
- Phase volume fractions
- Direction of applied stress/load

Under tensile load in longitudinal direction

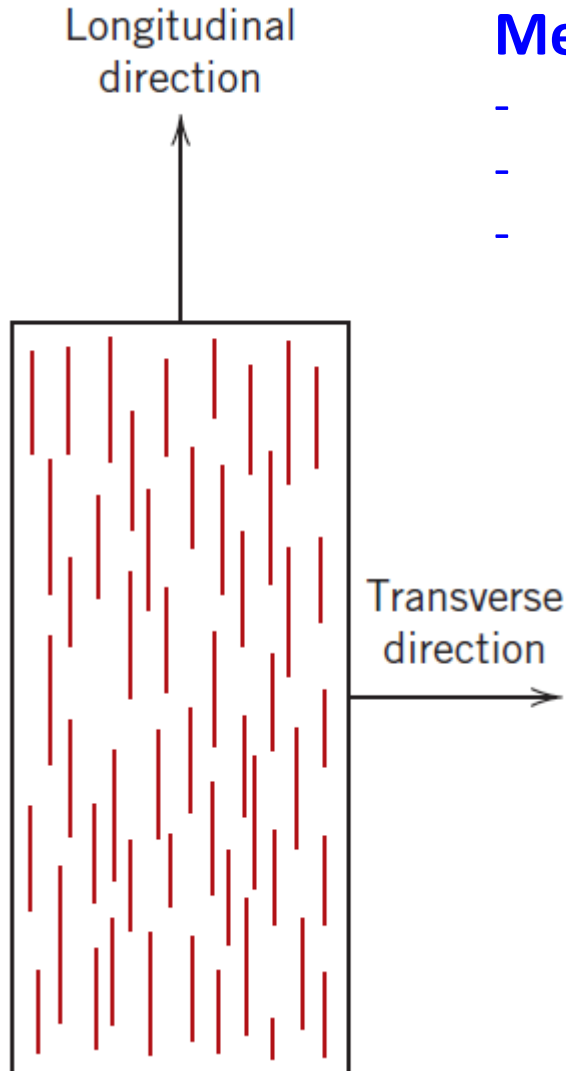
Rule of mixtures

$$E_{cl} = E_m V_m + E_f V_f$$

$$E_{cl} = E_m(1 - V_f) + E_f V_f$$

Similarly:

$$\frac{F_f}{F_m} = \frac{E_f V_f}{E_m V_m}$$





Composites

Fiber Reinforced

A continuous and aligned glass fiber–reinforced composite consists of 40 vol% glass fibers having a modulus of elasticity of 69 GPa (10×10^6 psi) and 60 vol% polyester resin that, when hardened, displays a modulus of 3.4 GPa (0.5×10^6 psi).

(a) Compute the modulus of elasticity of this composite in the longitudinal direction.

$$\begin{aligned} E_{cl} &= (3.4 \text{ GPa})(0.6) + (69 \text{ GPa})(0.4) \\ &= 30 \text{ GPa} (4.3 \times 10^6 \text{ psi}) \end{aligned}$$

(b) If the cross-sectional area is 250 mm^2 (0.4 in.^2) and a stress of 50 MPa (7250 psi) is applied in this longitudinal direction, compute the magnitude of the load carried by each of the fiber and matrix phases.

$$\frac{F_f}{F_m} = \frac{(69 \text{ GPa})(0.4)}{(3.4 \text{ GPa})(0.6)} = 13.5 \quad F_f = 13.5 F_m$$

$$F_c = A_c \sigma = (250 \text{ mm}^2)(50 \text{ MPa}) = 12,500 \text{ N}$$

$$F_c = F_f + F_m = 12,500 \text{ N}$$

$$13.5 F_m + F_m = 12,500 \text{ N}$$

$$F_m = 860 \text{ N} (200 \text{ lb}_f)$$

$$\text{Dep: } F_f = F_c - F_m = 12,500 \text{ N} - 860 \text{ N} = 11,640 \text{ N}$$



Composites

Fiber Reinforced

A continuous and aligned glass fiber–reinforced composite consists of 40 vol% glass fibers having a modulus of elasticity of 69 GPa (10×10^6 psi) and 60 vol% polyester resin that, when hardened, displays a modulus of 3.4 GPa (0.5×10^6 psi).

(c) Determine the strain that is sustained by each phase when the stress in part (b) is applied.

Elastic Behavior in Transverse loading

$$E_{ct} = \frac{E_m E_f}{V_m E_f + V_f E_m} = \frac{E_m E_f}{(1 - V_f) E_f + V_f E_m}$$

DO IT YOURSELF



Composites

Rule of Mixtures: Generalized Form

- E. Modulus $E_c = E_m V_m + E_f V_f$

- Density

$$\rho = \rho_m V_m + \rho_f V_f$$

- Thermal Conductivity

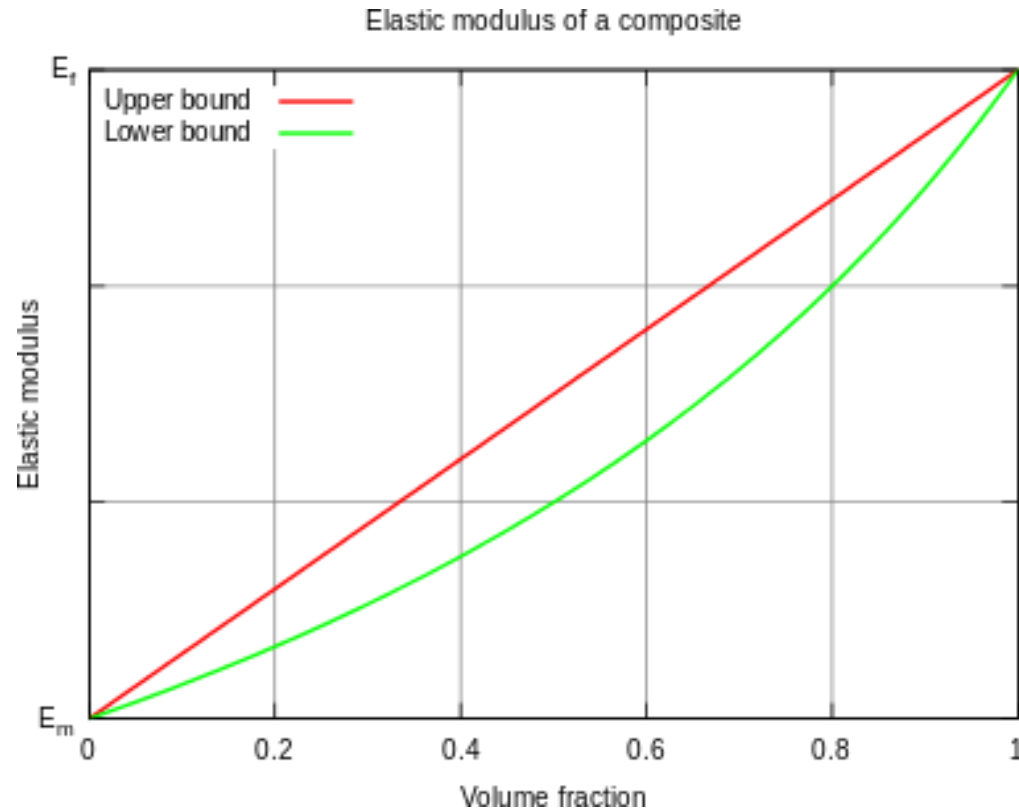
$$k_c = k_m V_m + k_f V_f$$

- Electrical Conductivity

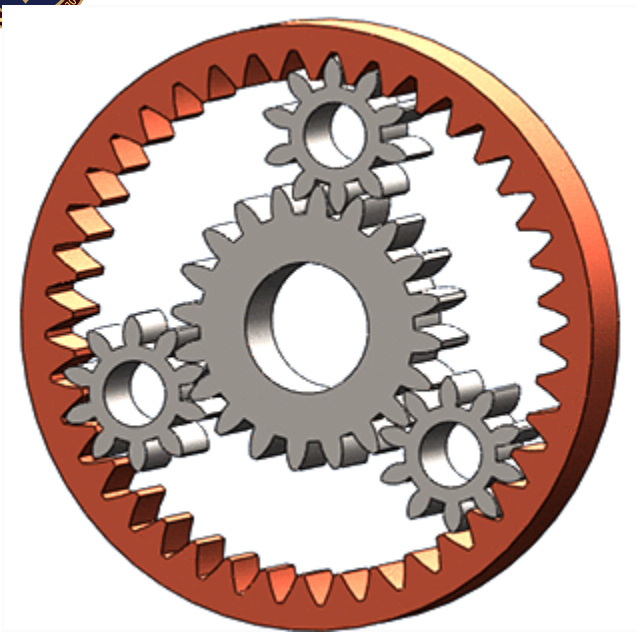
$$\sigma_c = \sigma_m V_m + \sigma_f V_f$$

Generalized form

$$X_c = X_m V_m + X_f V_f$$



Diffusion in Solids



**Surface hardness
enhanced through
diffusion of Carbon (C)
at high temperature**



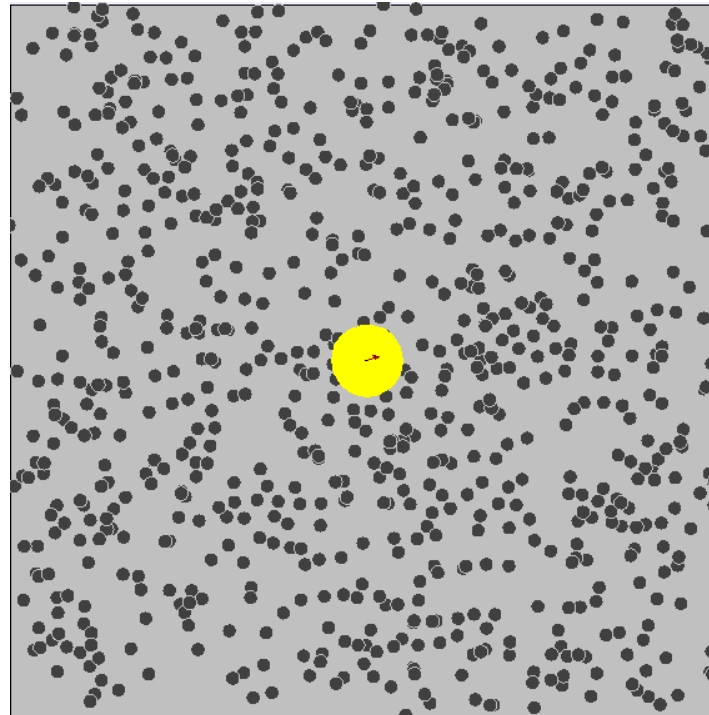
Diffusion in Solids



- **Diffusion:** Mass transport by atomic motion
- **atoms are restless**

Mechanism of Diffusion

- Gases & Liquids – random (**Brownian**) motion





Mechanism of Diffusion

How do atoms move in Solids ?

- **Diffusion**: atoms moving from one lattice site or interstitial site to another in a stepwise manner
- **Two conditions** are to be met:
 - Vacancy at an adjacent site
 - Enough energy to free bonded atom for diffusion
- **What is the energy source?**
 - It is the available **HEAT** = kT
- **What drives diffusion in a particular direction?**

Concentration gradient!

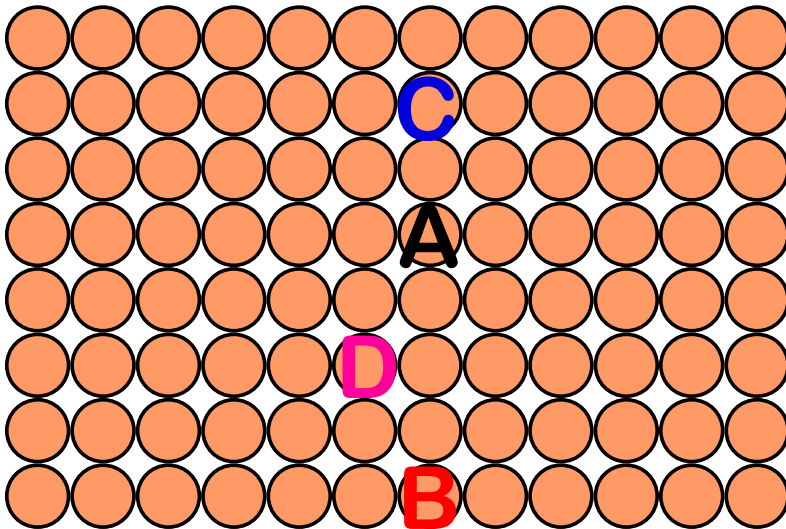


Mechanism of Diffusion

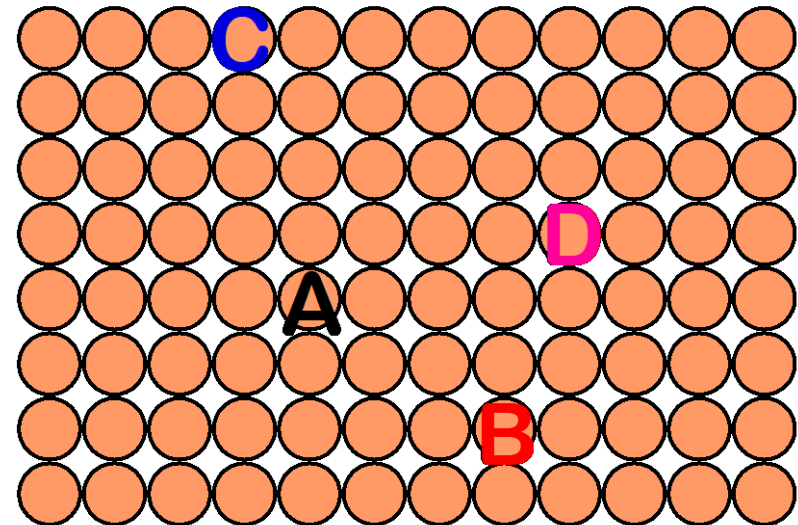
Self Diffusion

- Atoms in pure elemental solid also migrate

Label some atoms



After some time

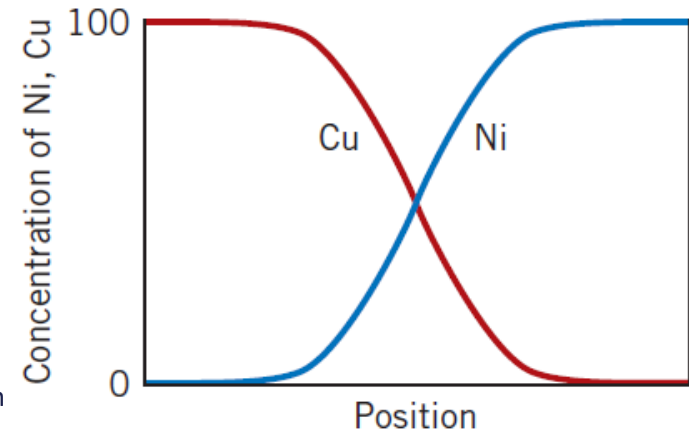
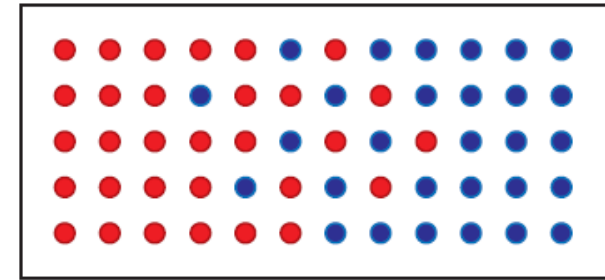
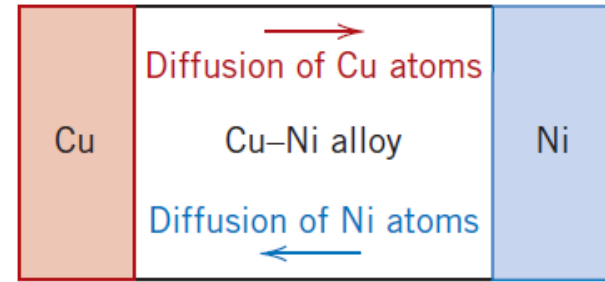
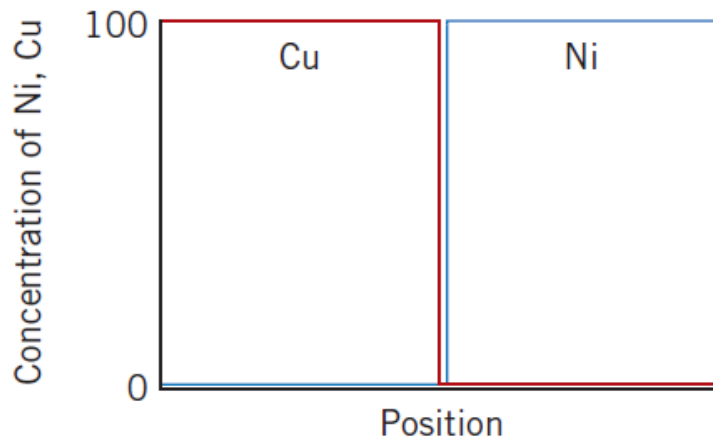
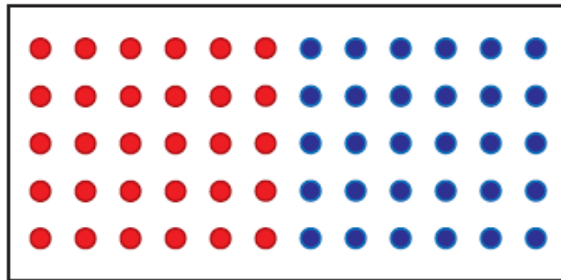


Diffusion in Solids



Mechanism of Diffusion

Inter-Diffusion: for atoms in an alloy (Example: CuNi)

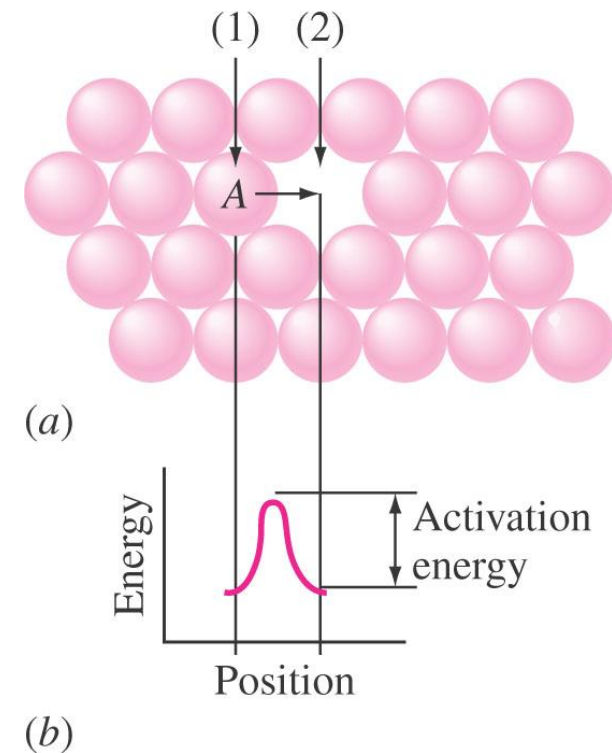
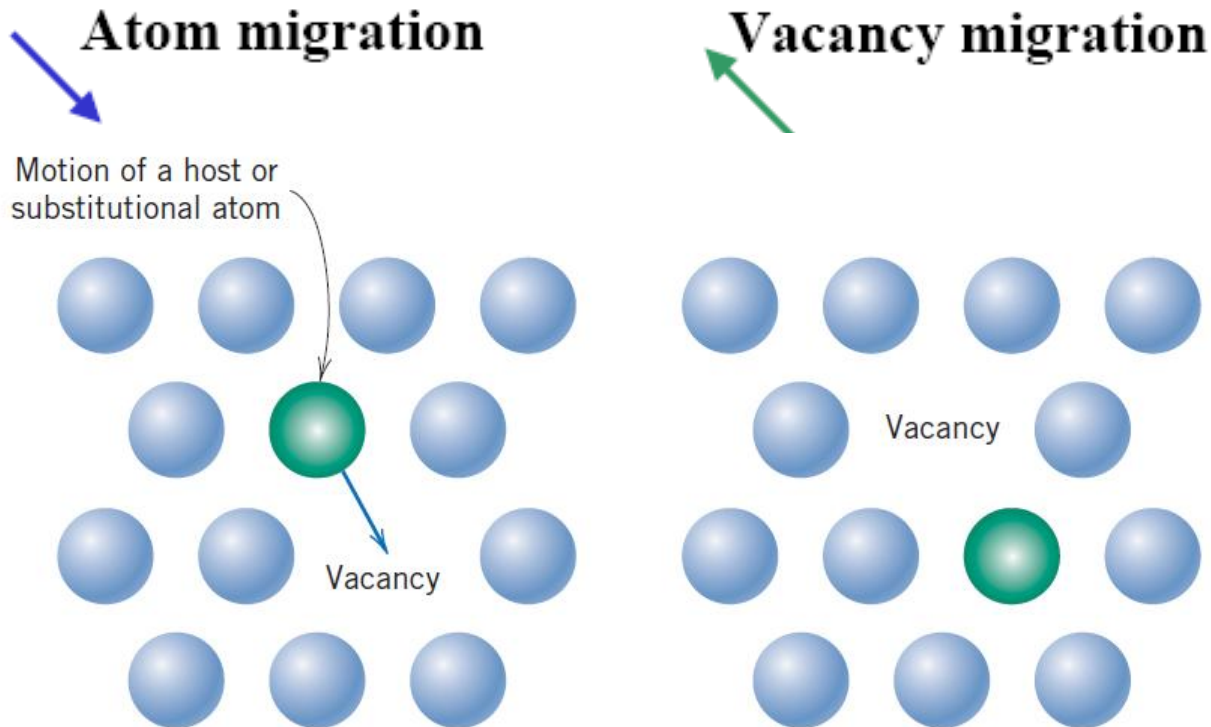


Diffusion in Solids

Mechanism of Diffusion

Vacancy vs Interstitial vs Substitutional

Vacancy Diffusion



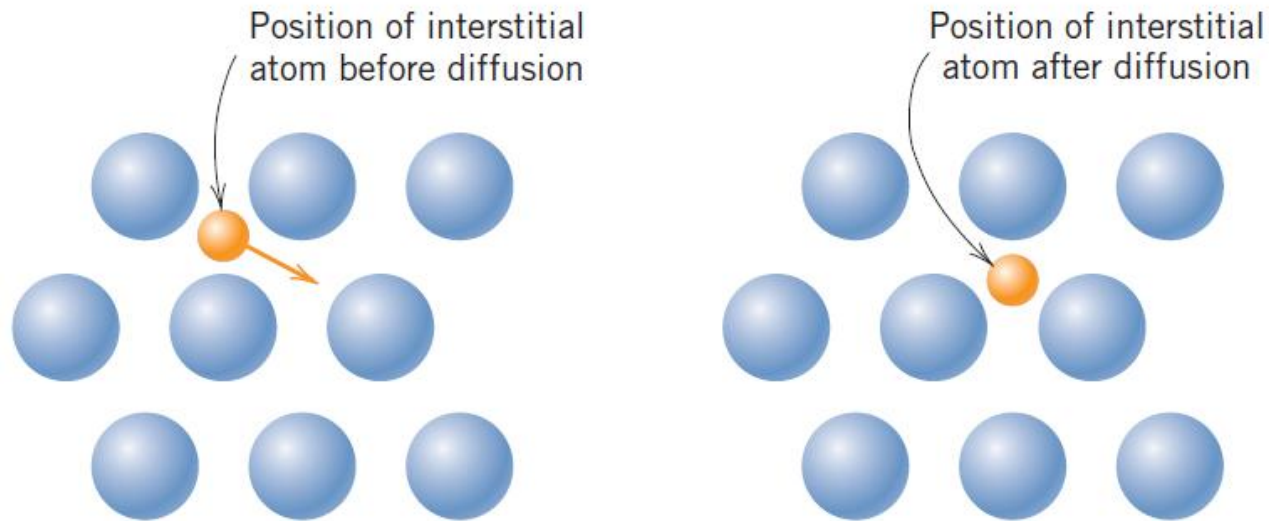
Where does energy come from for atom to leave its place?

What are the relative directions of the motions of vacancy and atom?



Mechanism of Diffusion

Vacancy vs Interstitial vs Substitutional Interstitial Diffusion



Much faster than vacancy diffusion, why ?

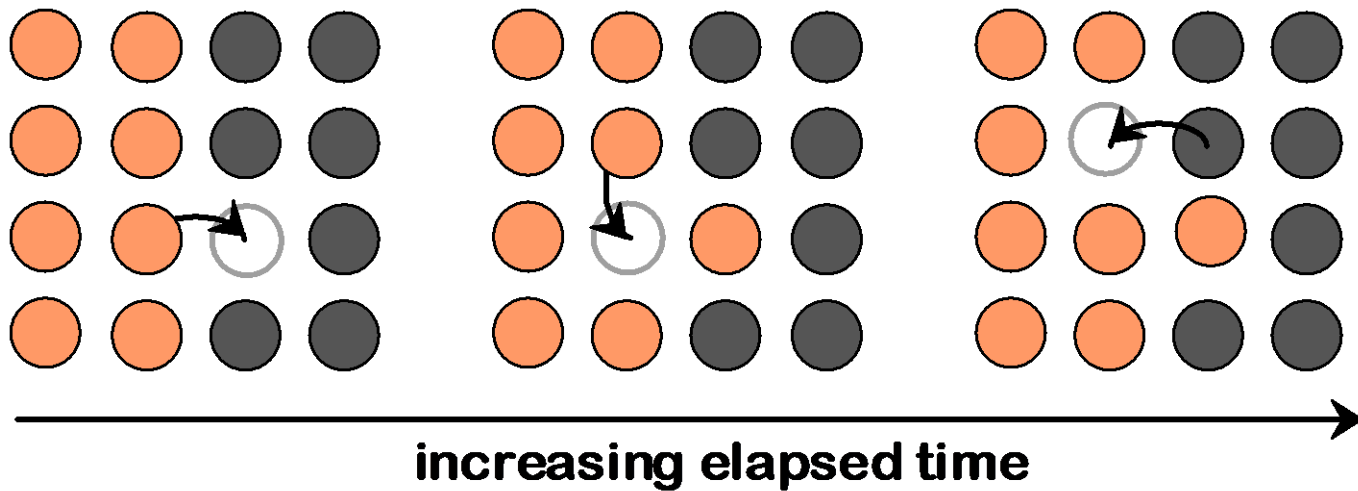
1. Smaller atoms like B, C, H, O, hence more mobile
Weaker interaction with the larger atoms
2. More vacant interstitial sites, no need to create a vacancy!

Diffusion in Solids



Mechanism of Diffusion

Vacancy vs Interstitial vs Substitutional Substitutional Diffusion



- applies to substitution impurities
- atoms exchange with vacancies
- rate depends on:
 - number of vacancies
 - activation energy to exchange

Diffusion in Solids



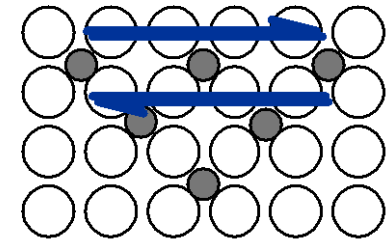
Surface hardness enhanced through diffusion of Carbon (C) into the host iron atoms on the surface



Case Hardening

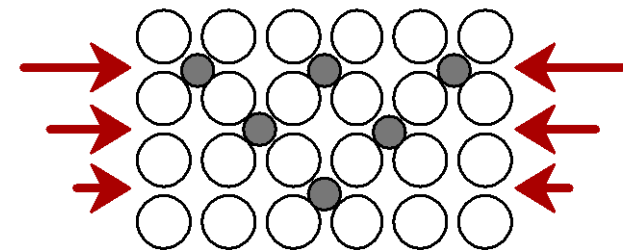
--hard to deform:

C atoms "lock" planes from shearing.



--hard to crack:

C atoms put the surface in compression.





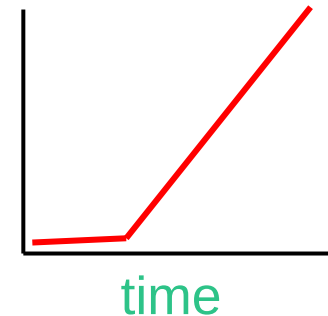
Diffusion in Solids

Diffusion – how to express

How do we quantify the amount or rate of diffusion?

$$J = \frac{M}{At} = \frac{l}{A} \frac{dM}{dt}$$

M = mass
diffused



$$J \equiv \text{Flux} \equiv \frac{\text{moles (or mass) diffusing}}{(\text{surface area})(\text{time})} = \frac{\text{mol}}{\text{cm}^2\text{s}} \text{ or } \frac{\text{kg}}{\text{m}^2\text{s}}$$

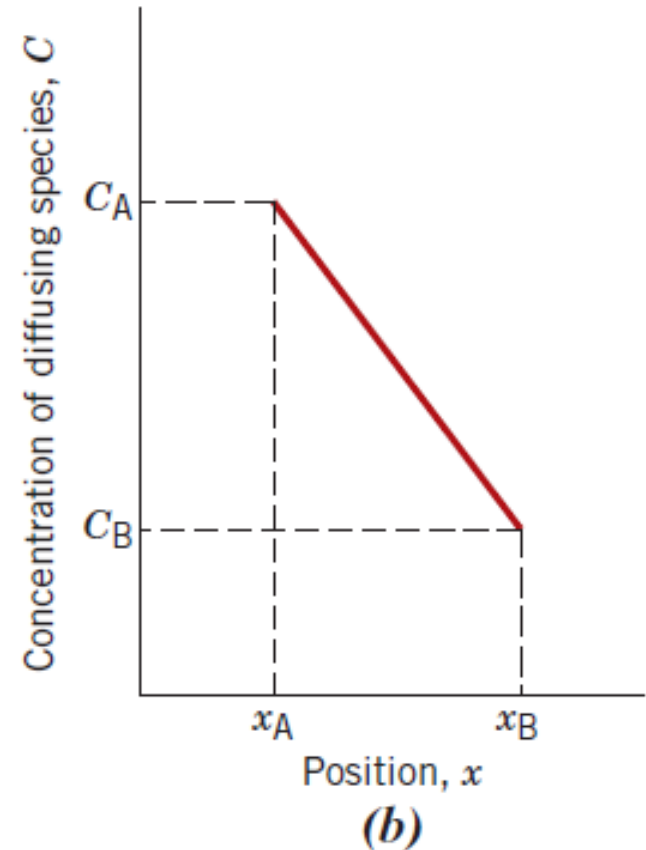
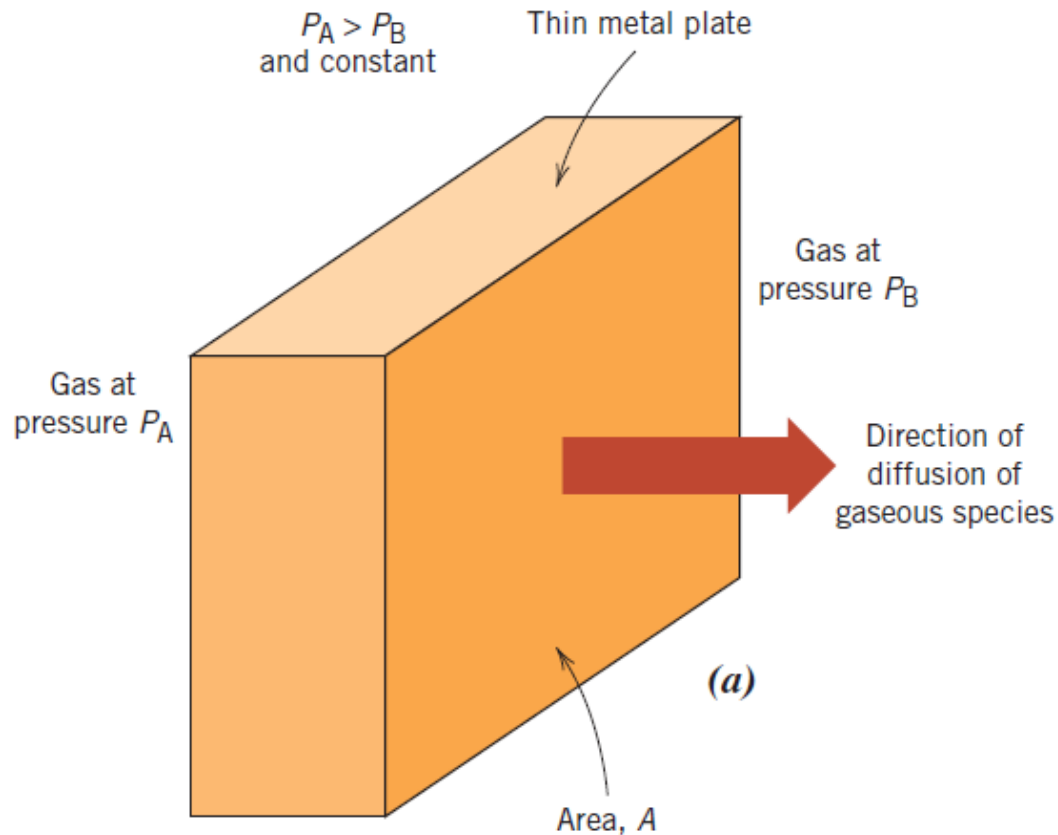
- **Measured empirically**

- Make thin film (membrane) of known surface area
- Impose concentration gradient
- Measure how fast atoms or molecules diffuse through the membrane

Diffusion in Solids

Concentration Gradient

$$\text{concentration gradient} = \frac{dC}{dx} = \frac{\Delta C}{\Delta x} = \frac{C_A - C_B}{x_A - x_B}$$



Class Activity



$$J = \frac{M}{At}$$

$$J = -D \frac{dC}{dx}$$

$$\frac{C_x - C_0}{C_s - C_0} = 1 - \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

$$D = D_0 \exp\left(-\frac{Q_d}{RT}\right)$$

$$\frac{dC}{dx} = \frac{\Delta C}{\Delta x} = \frac{C_A - C_B}{x_A - x_B}$$

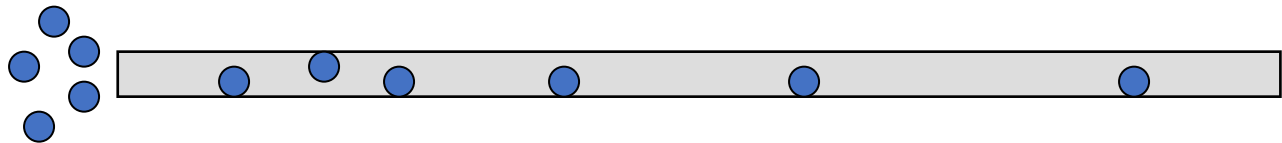
$$\frac{C(x,t) - C_0}{C_s - C_0} = 1 - \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

z	$\operatorname{erf}(z)$
0.85	0.7707
0.90	0.7970
0.95	0.8209
1.0	0.8427
1.1	0.8802

Diffusion in Solids



Generally, concentration of diffusing species is a function of **position x and time t** $\rightarrow C(x, t)$ in which we are interested



Steady state diffusion

Fick's First Law applies when C is function of x : **$C(x)$**

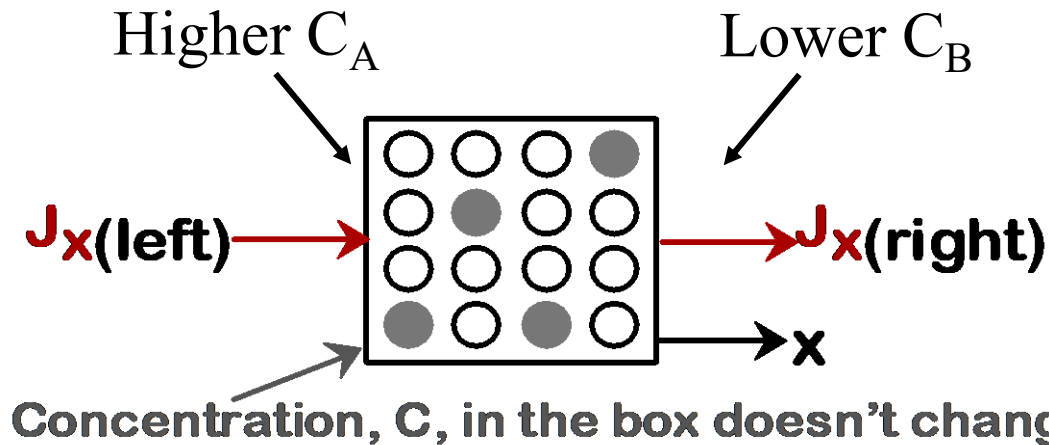
Rate of diffusion independent of time

Non-steady state diffusion

Fick's Second law applies when C is function of x and t : **$C(x, t)$**

Diffusion in Solids

Steady State Diffusion (Fick's first law)



Steady State:

$$J_x(\text{left}) = J_x(\text{right})$$

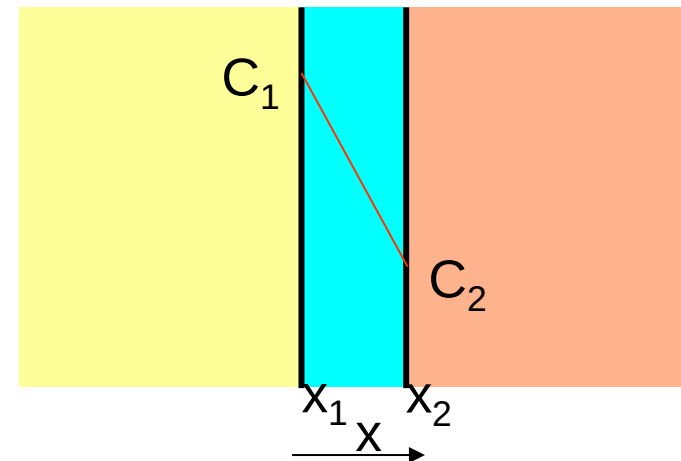
Apply Fick's First Law:

$$\text{Flux} \propto \frac{dC}{dx}$$

$$J = -D \frac{dC}{dx}$$

if linear

$$\frac{dC}{dx} \cong \frac{\Delta C}{\Delta x} = \frac{C_2 - C_1}{x_2 - x_1}$$



(i.e., concentration at any x is independent of time!)



Steady State Diffusion (Fick's first law)

Diffusion constant 'D' in Fick's first Law depends on:

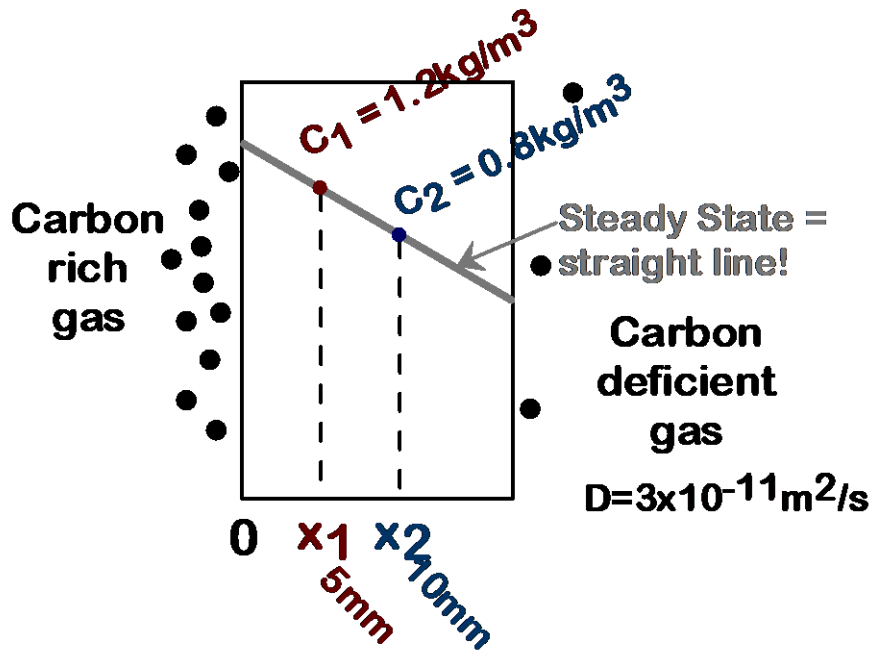
$$J_x = -D \frac{dC}{dx}$$

- Diffusion mechanism (Substitutional vs interstitial)
- Temperature
- Type of crystal structure of the host lattice
 - Interstitial diffusion easier in BCC than in FCC
- Type of crystal imperfections
 - Diffusion takes place faster along grain boundaries than elsewhere in a crystal
 - Diffusion is faster along dislocation lines than through bulk crystal
 - Excess vacancies will enhance diffusion

Diffusion in Solids

Steady State Diffusion (Fick's first law)

A plate of iron is exposed to a carburizing (carbon-rich) atmosphere on one side and a decarburizing (carbon-deficient) atmosphere on the other side at 700°C (1300°F). If a condition of steady state is achieved, calculate the diffusion flux of carbon through the plate if the concentrations of carbon at positions of 5 and 10 mm (5×10^{-3} and 10^{-2} m) beneath the carburizing surface are 1.2 and 0.8 kg/m³, respectively. Assume a diffusion coefficient of 3×10^{-11} m²/s at this temperature.



$$J = -D \frac{C_A - C_B}{x_A - x_B}$$

$$= -(3 \times 10^{-11} \text{ m}^2/\text{s}) \frac{(1.2 - 0.8) \text{ kg/m}^3}{(5 \times 10^{-3} - 10^{-2}) \text{ m}}$$

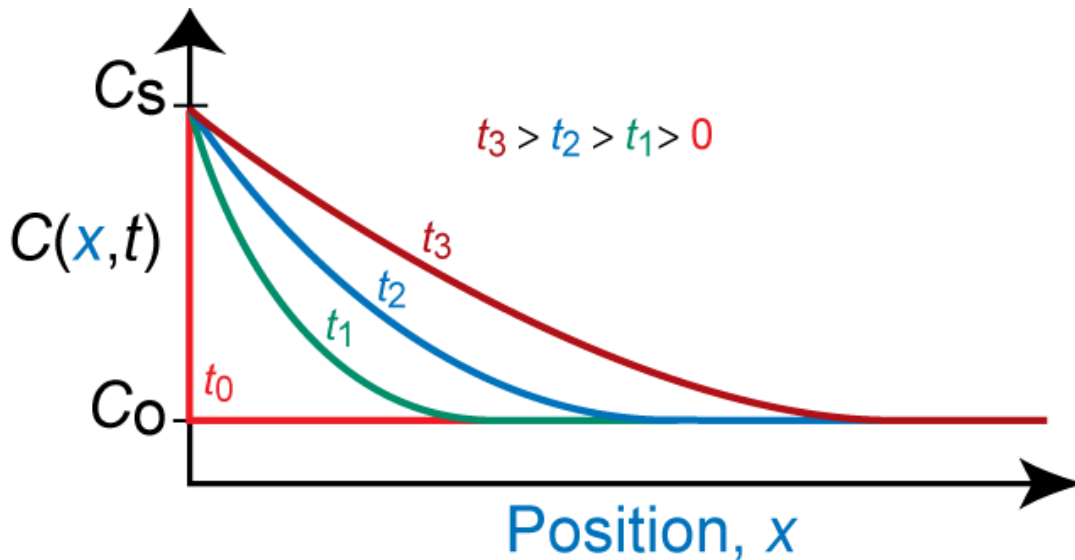
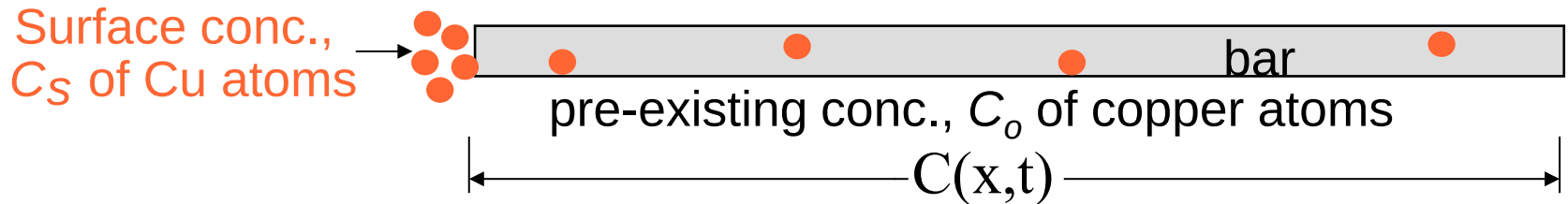
$$= 2.4 \times 10^{-9} \text{ kg/m}^2 \cdot \text{s}$$



Diffusion in Solids

Non-Steady State Diffusion (Fick's second law)

Copper diffuses into a bar of aluminum



$$\frac{\partial C}{\partial t} = \frac{\partial}{\partial x} \left(D \frac{\partial C}{\partial x} \right)$$

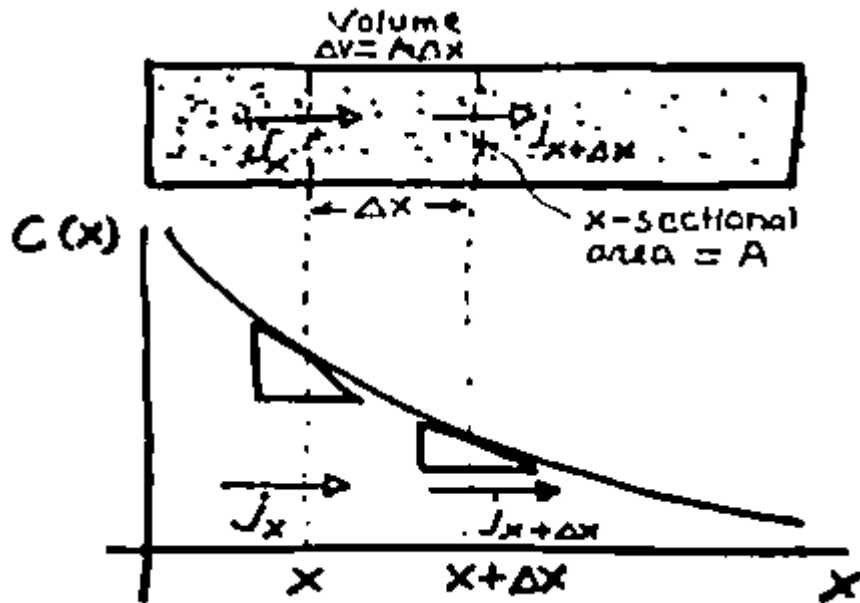
If diffusion co-efficient is independent of composition, then:

$$\frac{\partial C}{\partial t} = D \frac{\partial^2 C}{\partial x^2}$$

Note: After some time, concentration profile may become independent of time

Diffusion in Solids

Non-Steady State Diffusion (Fick's second law)



Mass entering ΔV at x
in time interval Δt

$$m_x = j_x A \Delta t$$

Mass leaving ΔV at $x + \Delta x$
in time interval Δt

$$m_{x+\Delta x} = j_{x+\Delta x} A \Delta t$$

Mass accumulating in ΔV in time Δt

$$\begin{aligned} \Delta m &= m_x - m_{x+\Delta x} \\ &= j_x A \Delta t - j_{x+\Delta x} A \Delta t \\ &= (j_x - j_{x+\Delta x}) A \Delta t \end{aligned}$$

Concentration change in ΔV in time Δt

$$\begin{aligned} \Delta C &= \frac{\Delta m}{\Delta V} \\ &= \frac{-\Delta j A \Delta t}{A \Delta x} \\ &= -\frac{\Delta j}{\Delta x} \Delta t \\ \rightarrow \frac{\Delta C}{\Delta t} &= -\frac{\Delta j}{\Delta x} \end{aligned}$$

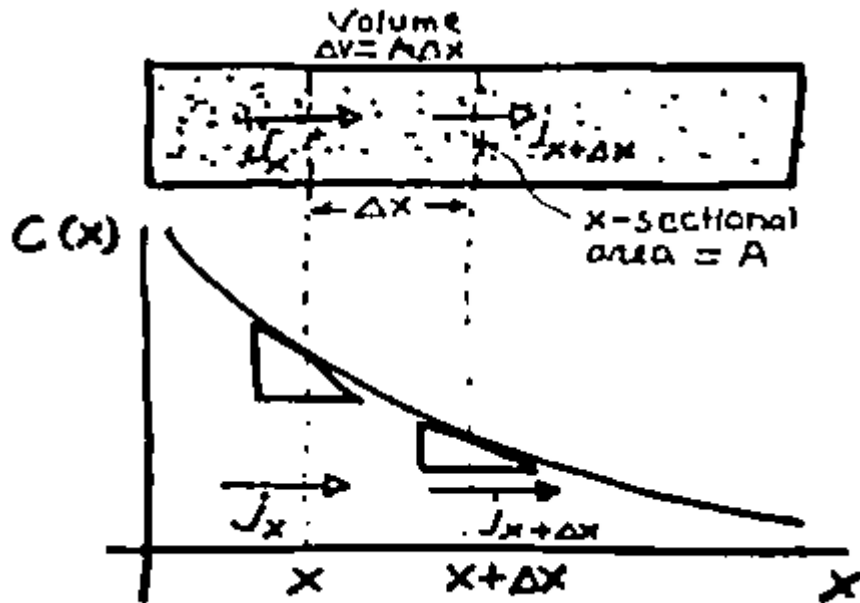
$$\lim_{\Delta t \rightarrow 0} \left(\frac{\Delta C}{\Delta t} \right) = - \lim_{\Delta x \rightarrow 0} \left(\frac{\Delta j}{\Delta x} \right)$$

$$\boxed{\frac{\partial C}{\partial t} = - \frac{\partial j}{\partial x}}$$

Fick's 2nd. Law
1st form.

Diffusion in Solids

Non-Steady State Diffusion (Fick's second law)



$$\boxed{\frac{\partial C}{\partial t} = - \frac{\partial j}{\partial x}} \quad \text{Fick's 2nd. Law 1st form.}$$

$$\frac{\partial C}{\partial t} = - \frac{\partial}{\partial x} \left(- D \frac{\partial C}{\partial x} \right)$$

$$= D \frac{\partial^2 C}{\partial x^2}$$

$$\frac{\partial C}{\partial t} = D \frac{\partial^2 C}{\partial x^2}$$

We can obtain another form of Fick's 2nd. Law by replacing j using Fick's 1st. Law

$$j = - D \frac{\partial C}{\partial x}$$



Diffusion in Solids

Non-Steady State Diffusion (Fick's second law)

$$\frac{\partial C}{\partial t} = \frac{\partial}{\partial x} \left(D \frac{\partial C}{\partial x} \right) = D \frac{\partial^2 C}{\partial x^2}$$
$$\frac{C(x,t) - C_o}{C_s - C_o} = 1 - \operatorname{erf} \left(\frac{x}{2\sqrt{Dt}} \right)$$
$$= 1 - \operatorname{erf}(z)$$

Use proper boundary conditions:

$$t=0, C = C_o, \text{ at } 0 \leq x \leq \infty$$
$$t>0, C = C_s, \text{ at } x = 0$$
$$C = C_o, \text{ at } x = \infty$$

$C(x,t)$ = Concentration at point x and at time t

If everything except D is given in the above equation, D can be calculated

Where:

- C_o = concentration of diffusing species before diffusion starts
- C_s = Concentration of diffusing species at the surface
- x is 0 at the surface and increases with distance into the solid

$\operatorname{erf}(z)$ = error function

$$= \frac{2}{\sqrt{\pi}} \int_0^z e^{-y^2} dy$$

How do you get $\operatorname{erf}(z)$?



Diffusion in Solids

Non-Steady State Diffusion (Fick's second law)

z	$erf(z)$	z	$erf(z)$	z	$erf(z)$
0	0	0.55	0.5633	1.3	0.9340
0.025	0.0282	0.60	0.6039	1.4	0.9523
0.05	0.0564	0.65	0.6420	1.5	0.9661
0.10	0.1125	0.70	0.6778	1.6	0.9763
0.15	0.1680	0.75	0.7112	1.7	0.9838
0.20	0.2227	0.80	0.7421	1.8	0.9891
0.25	0.2763	0.85	0.7707	1.9	0.9928
0.30	0.3286	0.90	0.7970	2.0	0.9953
0.35	0.3794	0.95	0.8209	2.2	0.9981
0.40	0.4284	1.0	0.8427	2.4	0.9993
0.45	0.4755	1.1	0.8802	2.6	0.9998
0.50	0.5205	1.2	0.9103	2.8	0.9999

' z ' can be calculated using
interpolation method

If $erf(z)=0.8125$, then

$$\frac{z - 0.90}{0.95 - 0.90} = \frac{0.8125 - 0.7970}{0.8209 - 0.7970}$$

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Diffusion in Solids

Non-Steady State Diffusion (Fick's second law)

An FCC iron-carbon alloy initially containing 0.20 wt% C is carburized at an elevated temperature and in an atmosphere that gives a surface carbon concentration constant at 1.0 wt%. If after 49.5 h the concentration of carbon is 0.35 wt% at a position 4.0 mm below the surface, determine the temperature (in °C) at which the treatment was carried out. Note that $D_0 = 2.3 \times 10^{-5}$, $Q_d = 148$ kJ/mol and $R = 8.314$ J/mol-K.

$$\frac{C(x,t) - C_o}{C_s - C_o} = 1 - \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right) \\ = 1 - \operatorname{erf}(z)$$

- $t = 49.5$ h
- $x = 4 \times 10^{-3}$ m

- $C_x = 0.35$ wt%
- $C_s = 1.0$ wt%
- $C_o = 0.20$ wt%

$$\frac{C(x,t) - C_o}{C_s - C_o} = \frac{0.35 - 0.20}{1.0 - 0.20} = 1 - \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right) = 1 - \operatorname{erf}(z)$$

$$\therefore \operatorname{erf}(z) = 1 - 0.15 / 0.80 = 0.8125$$

Note: Find Z from table, then D from Z and lastly T from

$$D = D_0 \operatorname{Exp}(-Q_d / RT)$$



Diffusion in Solids

Non-Steady State Diffusion (Fick's second law)

Numerical continued

z	$\text{erf}(z)$	
0.90	0.7970	$\frac{z - 0.90}{0.95 - 0.90} = \frac{0.8125 - 0.7970}{0.8209 - 0.7970}$
z	0.8125	
0.95	0.8209	$z = 0.93$

Now solve for D

$$z = \frac{x}{2\sqrt{Dt}} \Rightarrow D = \frac{x^2}{4z^2t}$$

$$\therefore D = \left(\frac{x^2}{4z^2t} \right) = \frac{(4 \times 10^{-3} \text{ m})^2}{(4)(0.93)^2(49.5 \text{ h})} \frac{1 \text{ h}}{3600 \text{ s}} = 2.6 \times 10^{-11} \text{ m}^2/\text{s}$$



Diffusion in Solids

Temperature dependence of Diffusion

Diffusion coefficient increases with increasing T

$$D = D_o \exp \left(- \frac{Q_d}{R T} \right)$$

D = diffusion coefficient [m^2/s]

D_o = pre-exponential [m^2/s]

Q_d = activation energy [J/mol or eV/atom]

R = gas constant [8.314 J/mol-K]

T = absolute temperature [K]

D depends on T because:

- Vacancies increase with T
- Available Energy to break bonds increases with T

The activation energy may be thought of as that energy required to produce the diffusive motion of one mole of atoms. A large activation energy results in a relatively small diffusion coefficient



Diffusion in Solids

Temperature dependence of Diffusion

Numerical continued

$$T = \frac{Q_d}{R(\ln D_o - \ln D)}$$

for diffusion of C in FCC Fe

$$D_o = 2.3 \times 10^{-5} \text{ m}^2/\text{s} \quad \text{and} \quad Q_d = 148,000 \text{ J/mol}$$

$$\therefore T = \frac{148,000 \text{ J/mol}}{(8.314 \text{ J/mol} \cdot \text{K})(\ln 2.3 \times 10^{-5} \text{ m}^2/\text{s} - \ln 2.6 \times 10^{-11} \text{ m}^2/\text{s})}$$

$$T = 1300 \text{ K} = 1027^\circ\text{C}$$

Diffusion in Solids

Temperature dependence of Diffusion

$$D = D_0 \exp\left(-\frac{Q_d}{RT}\right)$$

Taking ln or log on both sides

$$\ln D = \ln D_0 - \frac{Q_d}{R} \left(\frac{1}{T}\right)$$

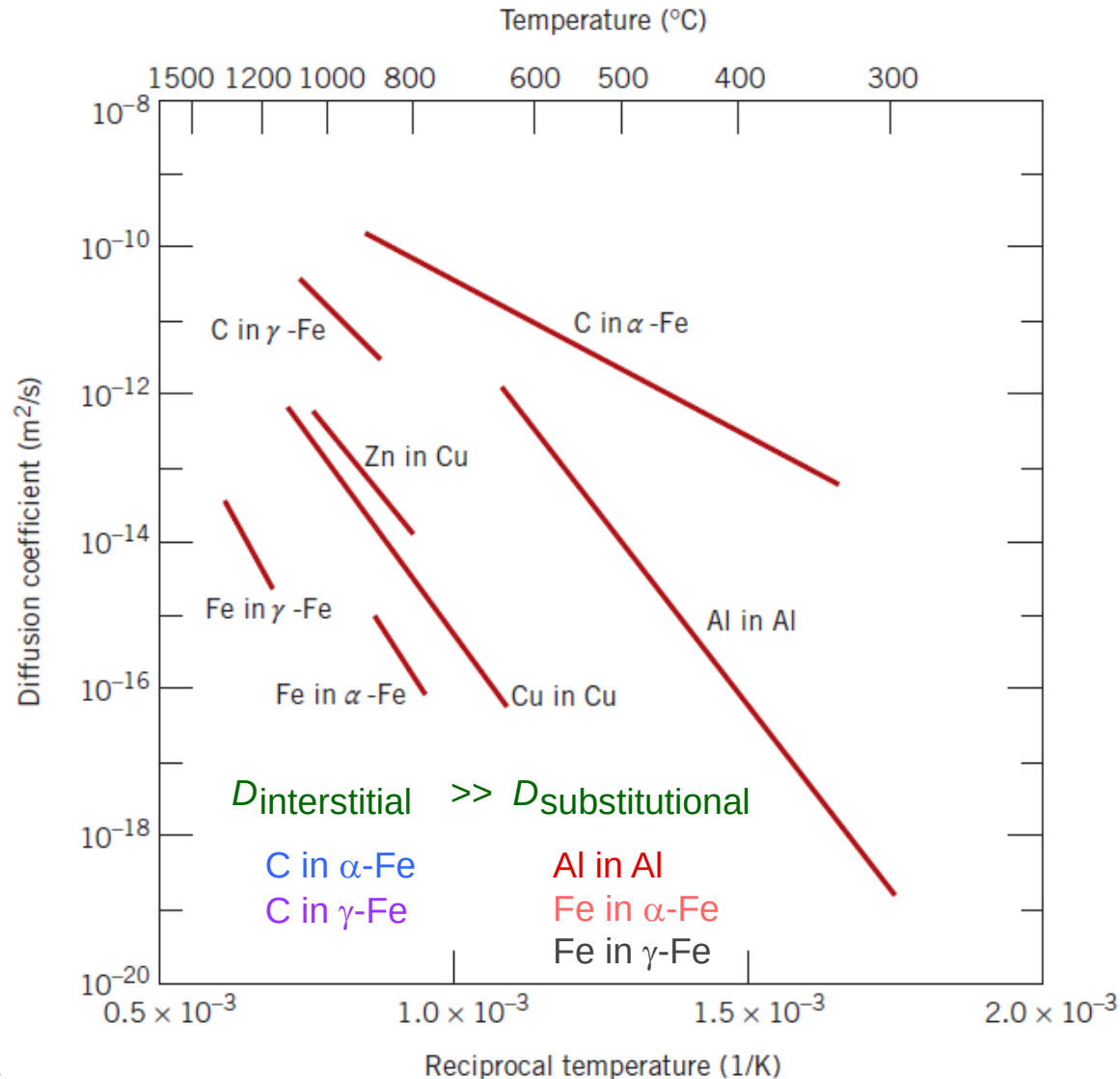
$$\log D = \log D_0 - \frac{Q_d}{2.3R} \left(\frac{1}{T}\right)$$

This is similar to equation of straight line

$$y = b + mx$$

Where $y = \log D$ and $x = 1/T$

Depart





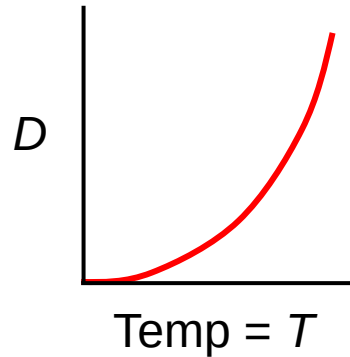
Diffusion in Solids

Temperature dependence of Diffusion

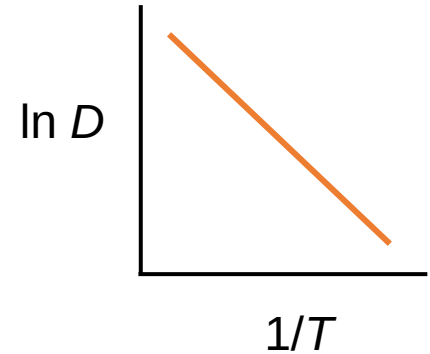
At 300°C the diffusion coefficient and activation energy for Cu in Si are $D(300^\circ\text{C}) = 7.8 \times 10^{-11} \text{ m}^2/\text{s}$; $Q_d = 41.5 \text{ kJ/mol}$. What is the diffusion coefficient at 350°C?

$$D = D_0 \exp\left(-\frac{Q_d}{RT}\right)$$

$$\ln D = \ln D_0 - \frac{Q_d}{R} \left(\frac{1}{T}\right)$$



transform
data →



$$\ln D_2 = \ln D_0 - \frac{Q_d}{R} \left(\frac{1}{T_2}\right) \quad \text{and} \quad \ln D_1 = \ln D_0 - \frac{Q_d}{R} \left(\frac{1}{T_1}\right)$$

$$\therefore \ln D_2 - \ln D_1 = \ln \frac{D_2}{D_1} = -\frac{Q_d}{R} \left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

$$D_2 = D_1 \exp \left[-\frac{Q_d}{R} \left(\frac{1}{T_2} - \frac{1}{T_1}\right) \right]$$



Diffusion in Solids

Temperature dependence of Diffusion

At 300°C the diffusion coefficient and activation energy for Cu in Si are $D(300^\circ\text{C}) = 7.8 \times 10^{-11} \text{ m}^2/\text{s}$; $Q_d = 41.5 \text{ kJ/mol}$. What is the diffusion coefficient at 350°C?

$$D_2 = D_1 \exp \left[-\frac{Q_d}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right) \right]$$

$$T_1 = 273 + 300 = 573 \text{ K}$$

$$T_2 = 273 + 350 = 623 \text{ K}$$

$$D_2 = (7.8 \times 10^{-11} \text{ m}^2/\text{s}) \exp \left[\frac{-41,500 \text{ J/mol}}{8.314 \text{ J/mol} \cdot \text{K}} \left(\frac{1}{623 \text{ K}} - \frac{1}{573 \text{ K}} \right) \right]$$

$$D_2 = 15.7 \times 10^{-11} \text{ m}^2/\text{s}$$

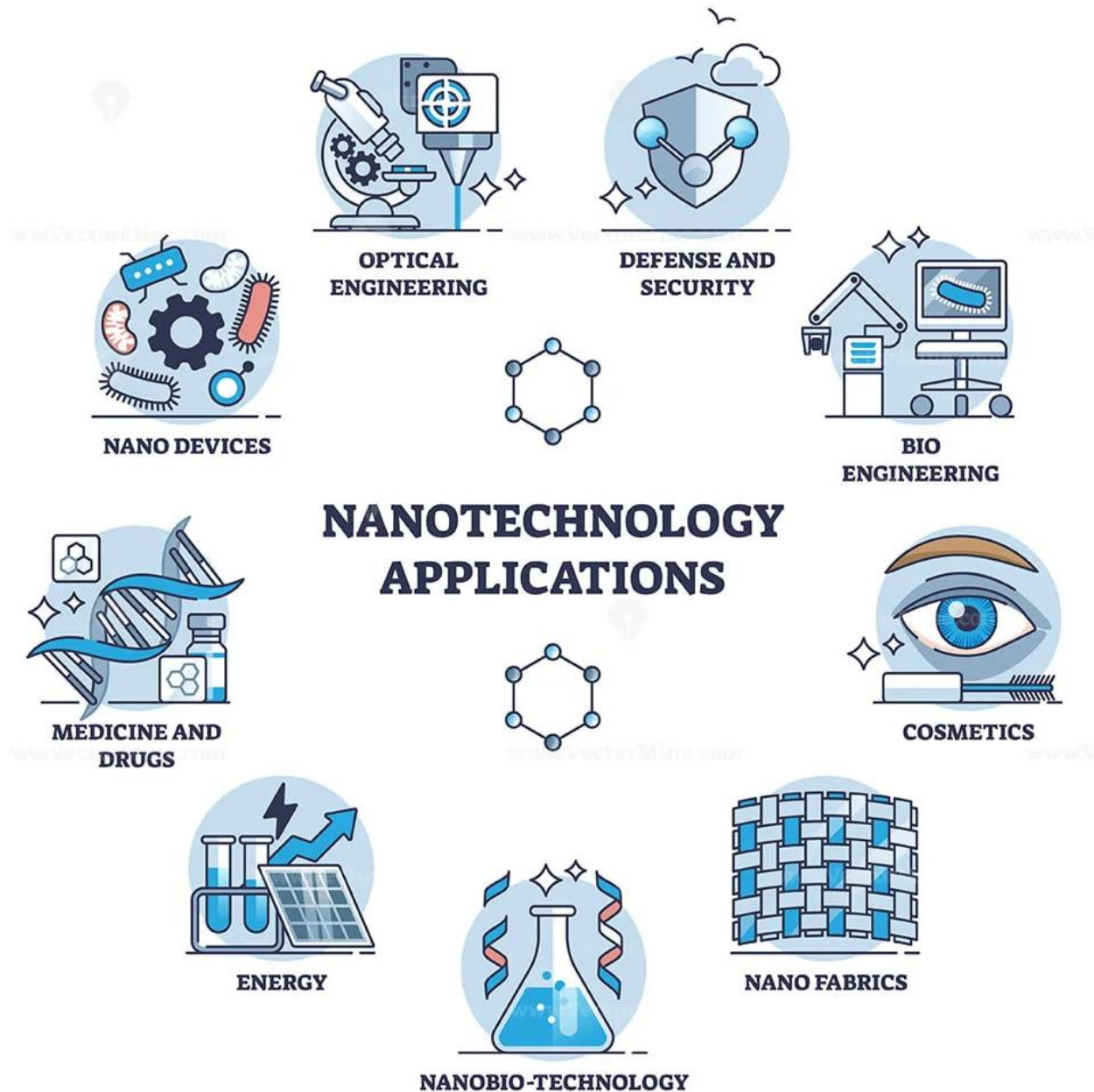
Nanotechnology



Nanotechnology refers to the branch of science and engineering devoted to designing, producing, and using structures, devices, and systems by manipulating atoms and molecules at nanoscale, i.e., having one or more dimensions of the order of 100 nanometres (100 millionth of a millimeter) or less



Nanotechnology



Nanotechnology – how small is nano?



Macro level

Human

1 m



Mouse

10 cm



Fly

1 cm



Ant

5 mm



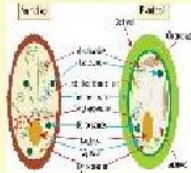
Head of a pin

1-2 mm



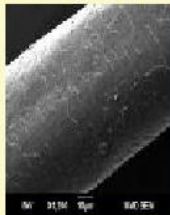
Plant and animal cell

100 μ m



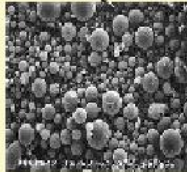
Human hair

60 -120 μ m



Fly ash

10- 20 μ m



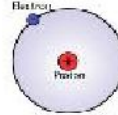
Red blood cell with white cell

2-5 μ m

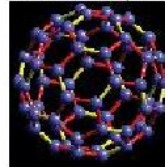


Hydrogen atom

0.1 nm



Buckminster fullerene(C60)



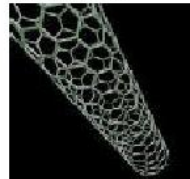
DNA (width)

2.0 nm



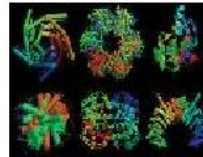
Nanotube

3-30 nm



Proteins

5-50 nm



Nano level

Quantum Dots (of CdSe)

8.0 nm



Dip pen nanolithography

10-15 nm



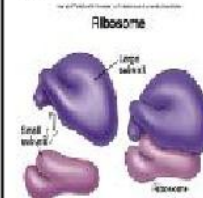
Dendrimers

10 nm



Ribosome

25 nm



Quantum Dots

8.0 nm

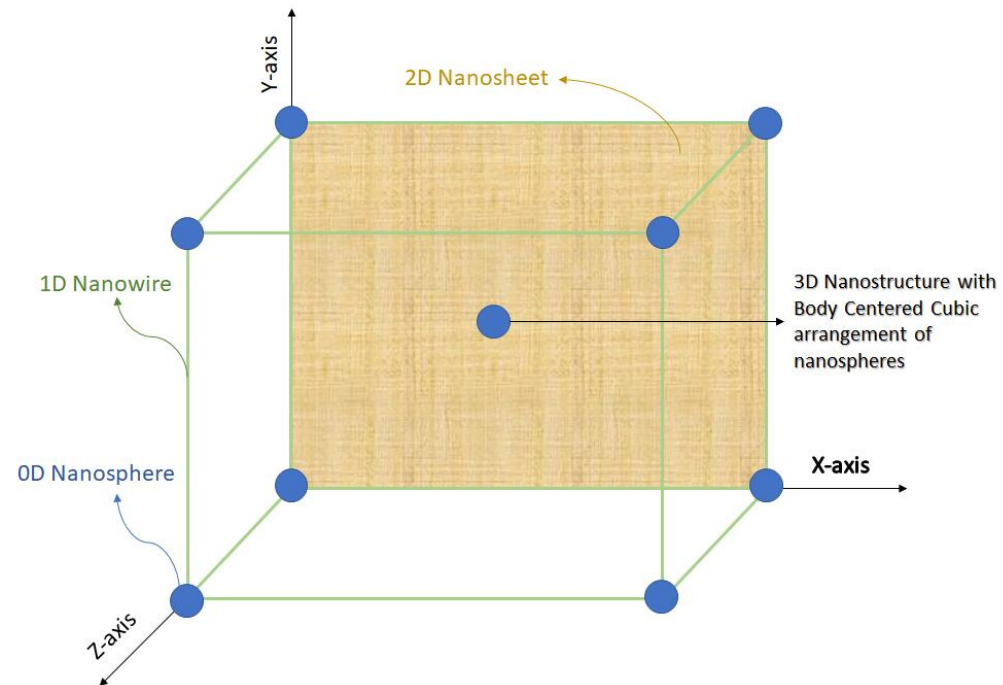
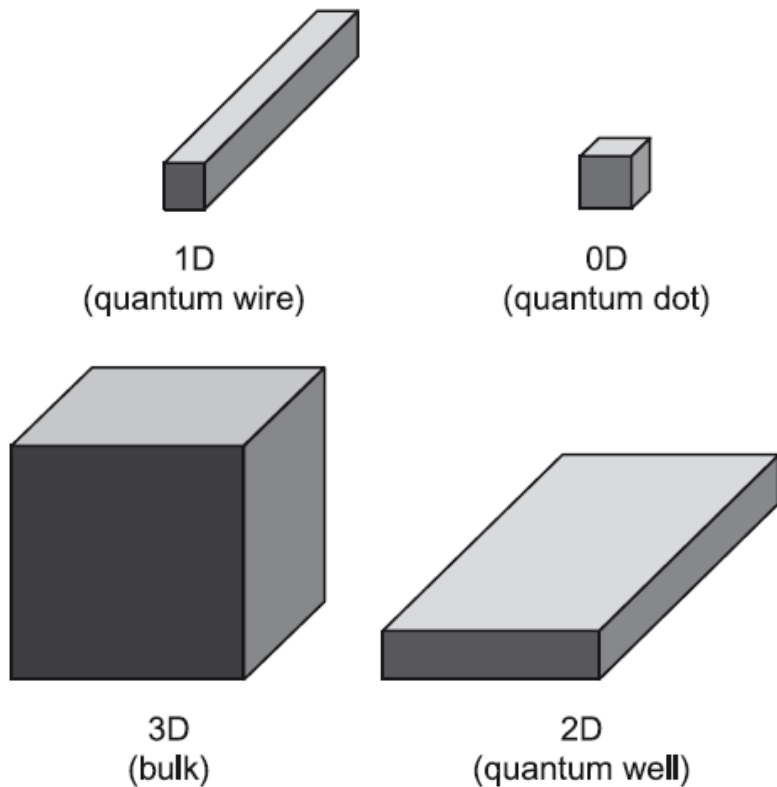




Nanomaterials

Nanomaterials may be classified as those materials which have at least one of their dimensions in the nanometric range.

- Below which there is significant variation in the property of interest compared to microcrystalline materials.
- Nanocrystalline materials have a grain size of the order of 1–100 nm.

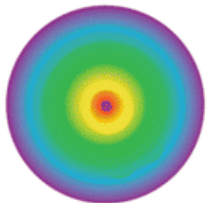


Nanotechnology

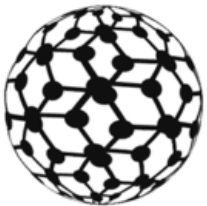
Classification of Nanostructured materials

0D

Nanospheres,
clusters



Quantum dots



Fullerenes



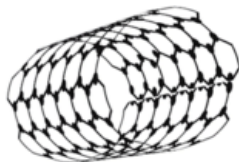
Gold nanoparticles

1D

Nanotubes,
wires, rods



Metal nanorods,
Ceramic crystals



Carbon nanotubes,
Metallic nanotubes



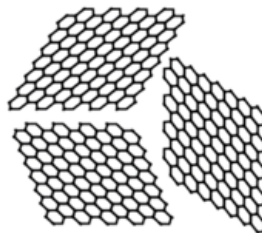
Gold nanowires,
Polymeric nanofibers,
Self assembled structures

2D

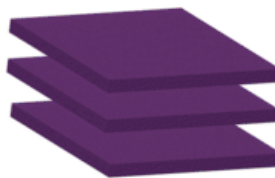
Thin films, plates,
layered structures



Carbon coated
nanoplates



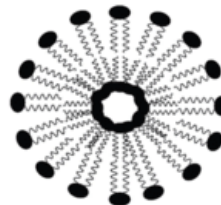
Graphene sheets



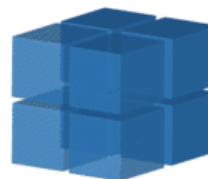
Layered nanomaterials

3D

Bulk NMs,
polycrystals



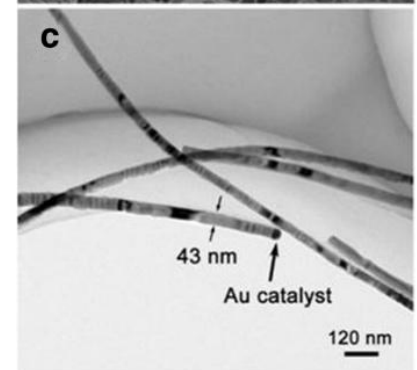
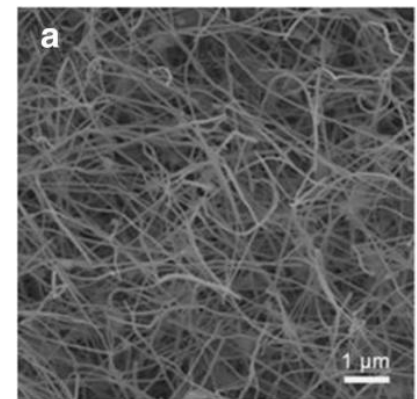
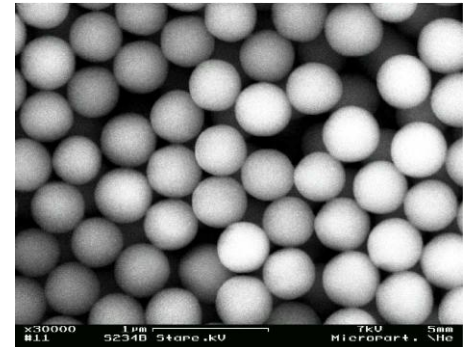
Liposome



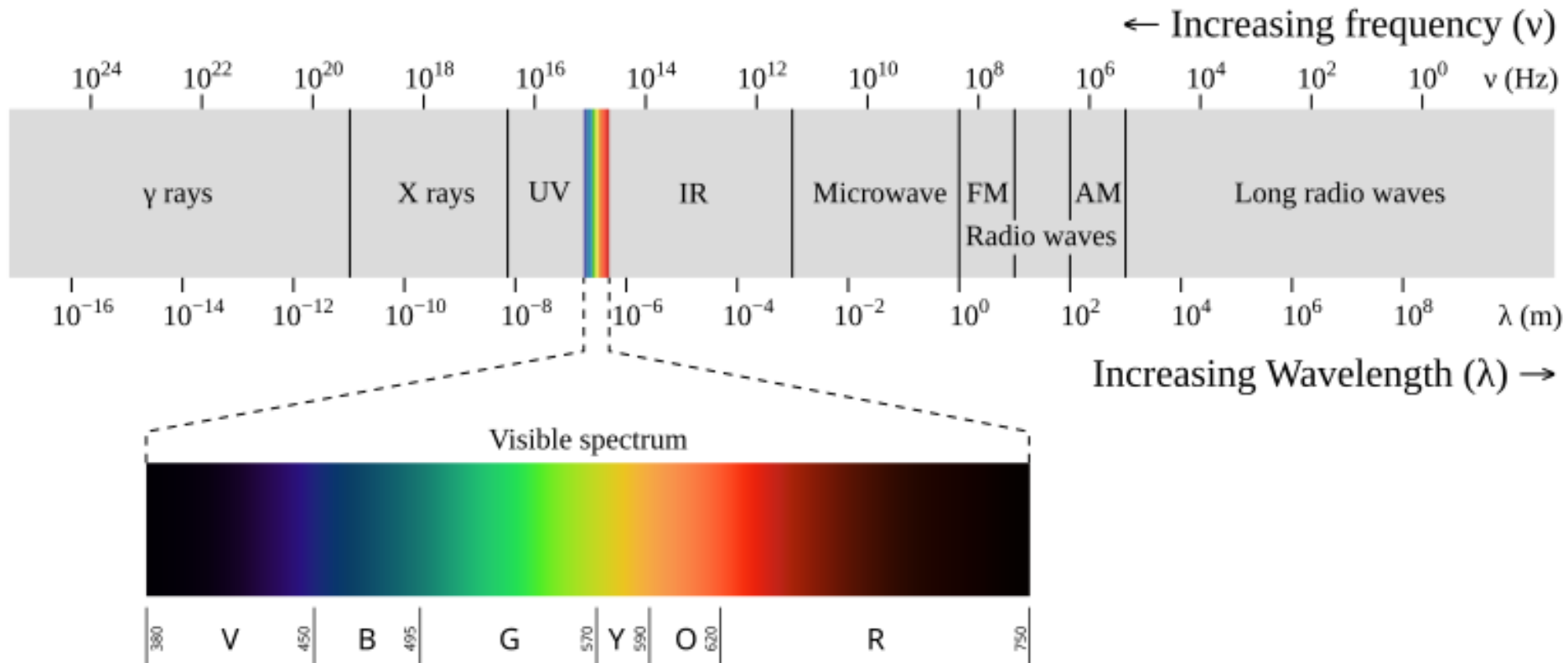
Polycrystalline



Dendrimer



Spectrum of light



Quantum Dots



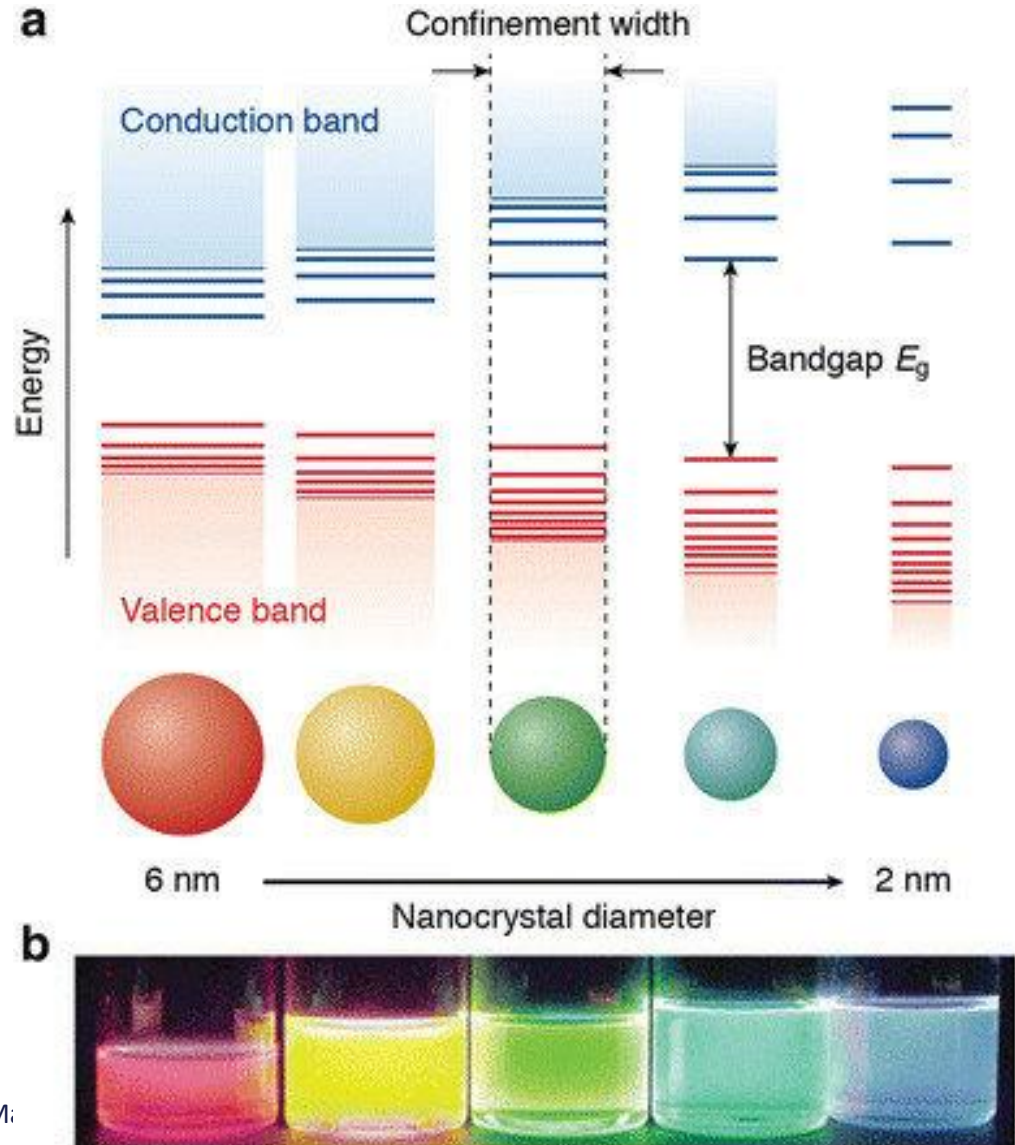
Quantum dots (QDs)

semiconductor nanocrystal that exhibits unique optical and electronic properties, including the ability to transport electrons and emit light of various colors when exposed to UV light

At very small sizes physical properties (magnetic, electric and optical) of materials can change dramatically.

Department of M:

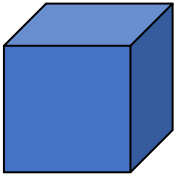
semiconductor nanocrystal (2 – 10 nm)



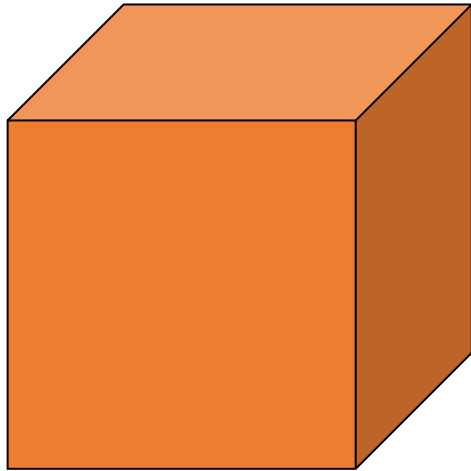


What makes technology at the nanoscale different from technology at the macroscale?

surface area to volume ratio



2 cm (edge length) has a surface area of 24 cm² and a volume of 8 cm³ (ratio = 3:1)



10 cm (edge length) has a surface area of 600 cm² and a volume of 1000 cm³ (ratio = 0.6:1)

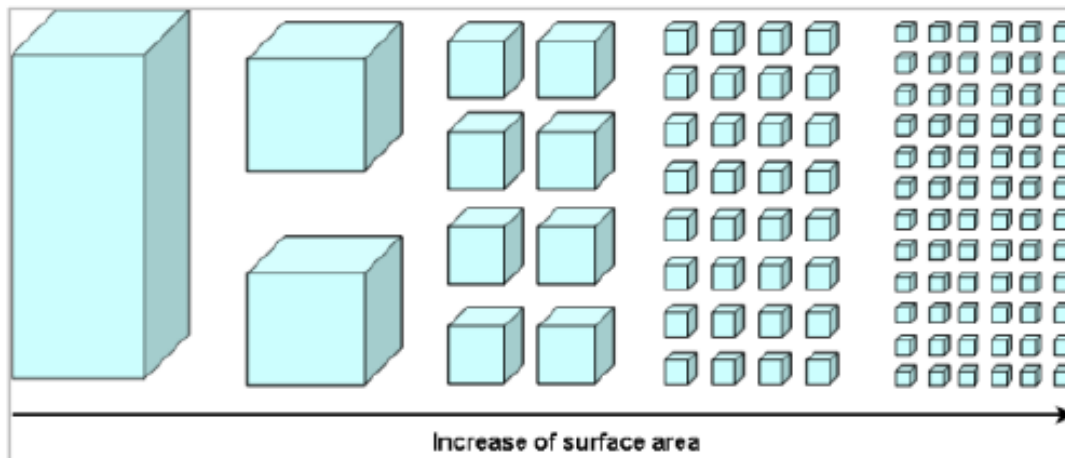


Nanotechnology

What makes technology at the nanoscale different from technology at the macroscale?

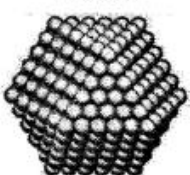
As objects get smaller, they have a much greater surface area to volume ratio

- Materials at the nanoscale have a greater surface area to volume ratio than their bulk counterparts.
- The relative concentration of under-coordinated surface atoms is thus greater for nanomaterials compared to their bulk form

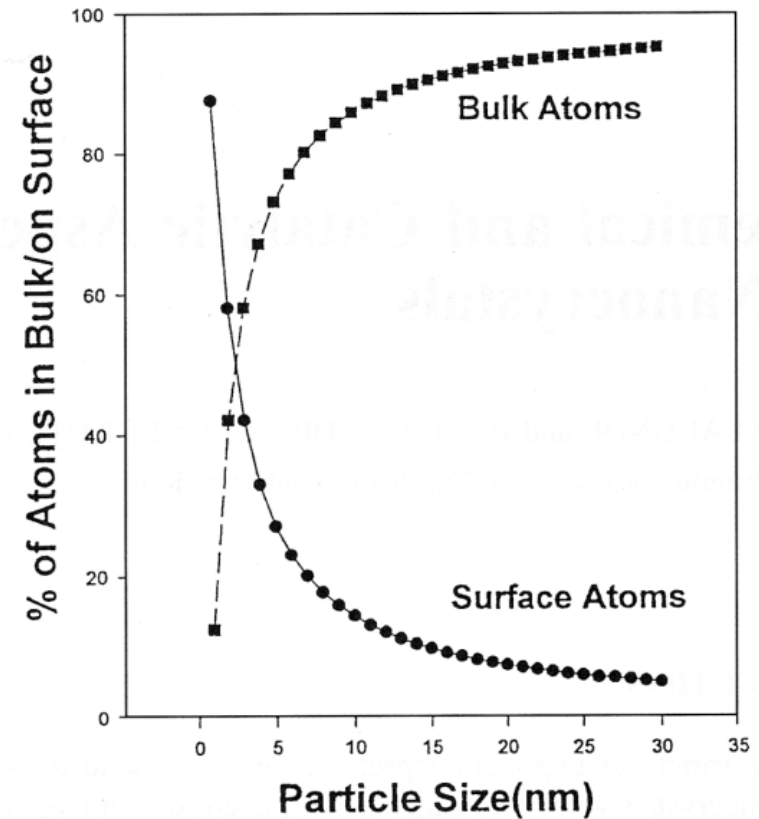


The total volume remains the same, the collective surface area is greatly increased.



Full-shell Clusters		Total Number of Atoms	Surface Atoms (%)
1. Shell		13	92
2 Shells		55	76
3 Shells		147	63
4 Shells		309	52
5 Shells		561	45

Surface to bulk atoms ratio



$$\text{Ratio} = (\text{No. of atoms on surface of a particle}) / (\text{Total No. of atoms in the particle})$$



- **Cluster**
 - A collection of units (atoms or reactive molecules) of up to about 50 units
- **Nanocrystal**
 - A solid particle that is a single crystal in the nanometer range
- **Nanoparticle**
 - A solid particle in the 1-100 nm range that could be non-crystalline, an aggregate of crystallites or a single crystallite
- **Colloids**
 - A stable liquid phase containing particles in the 1-1000 nm range. A colloid particle is one such 1-1000 nm particle



Calculate the ratio of surface atoms to volume atoms

Consider 1 cm³ material (Cu) with Density= 8.4 g/cm³, At. Mass = 63.5 g/mol and Radius = 128 pm

Ratio = (No. of atoms on surface of a particle) / (Total No. of atoms in the particle)

$$N \text{ (atoms in 1cm}^3 \text{)} = [(8.4 \text{ g/cm}^3) / (63.5 \text{ g/mol})] [6.02 \times 10^{23} / \text{mol}] = 8 \times 10^{22} \sim 10^{23} / \text{cm}^3$$

$$\text{Surface area of 1 cm}^3 = 1 \text{ cm} \times 1 \text{ cm} \times 6 = 6 \text{ cm}^2$$

$$\text{Area occupied by one atom} = 0.05 \text{ nm}^2$$

$$\text{Atoms per face} = [\text{atoms} / 0.05 \text{ nm}^2] [1 \text{ cm}^2] = (\text{Atoms} / 0.05 \text{ nm}^2) \times 10^{14} \text{ nm}^2 = 2.0 \times 10^{15} \text{ atoms per face}$$

$$\text{No. of atoms on total surface} = 12.0 \times 10^{15}$$

$$\text{Ratio} = 12 \times 10^{15} \text{ atoms on surface} / 10^{23} \text{ atoms total in 1 cm}^3 = 12 \times 10^{15} / 10^8 \times 10^{15} = 12/10^8$$

Calculate the ratio for 1 nm³

$$\text{No. of atoms in 1 nm}^3 = \text{Atoms in 1 cm}^3 / 10^{23} / 10^{21}$$

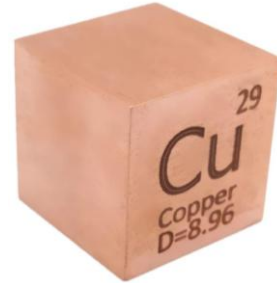
$$\text{Surface area of 1 nm}^3 = 1 \text{ nm} \times 1 \text{ nm} \times 6 = 6 \text{ nm}^2$$

$$\text{Area occupied by one atom} = 0.05 \text{ nm}^2$$

$$\text{Atoms per face} = [\text{atoms} / 0.05 \text{ nm}^2] [1 \text{ nm}^2] = (\text{Atoms} / 0.05 \text{ nm}^2) \times 1 \text{ nm}^2 = 20 \text{ atoms per face}$$

$$\text{No. of atoms on total surface} = 120$$

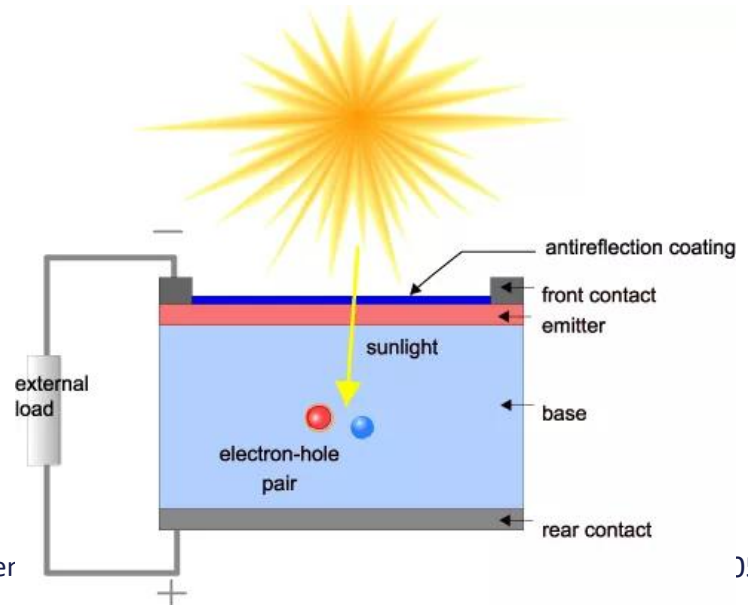
$$\text{Ratio} = 120 \text{ atoms on surface} / 100 \text{ atoms total in 1 nm}^3 = 1.2$$



Electrical Properties of Materials



Photovoltaics





Electrical Properties of Materials

Electrical Conduction

- Ohm's Law:

voltage drop (Volts V , = J/C)
C = Coulomb

$$V = IR$$

resistance (Ohms)

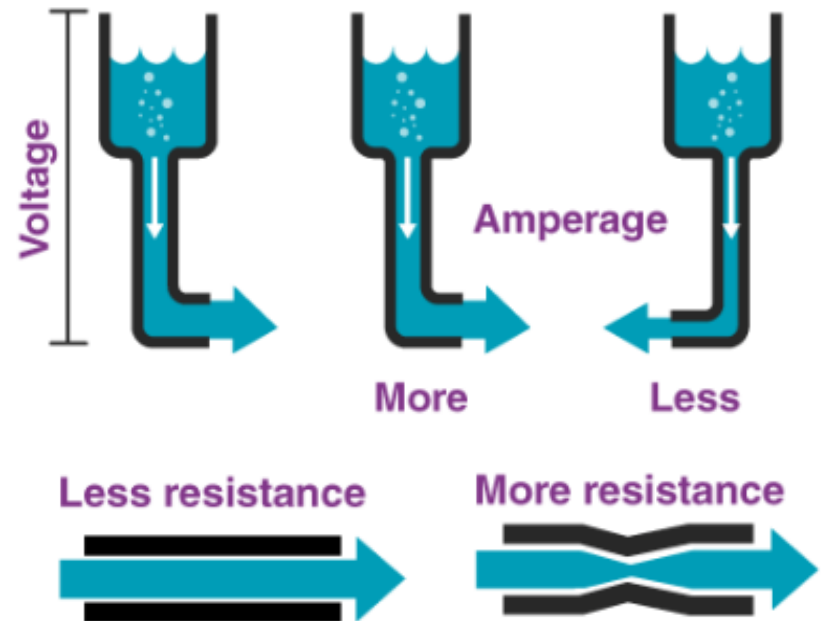
current (Ampere, $A = C/s$)

Ohm's law states that the voltage across a conductor is directly proportional to the current flowing through it, provided all **physical conditions and temperatures remain constant**

Resistance: Measure of opposition to the flow of electrical current

Conductor: A component that allows flow of electrical current (conventional/electronic)

Resistor: A component in the electrical circuit for offering predetermined resistance to electrical current





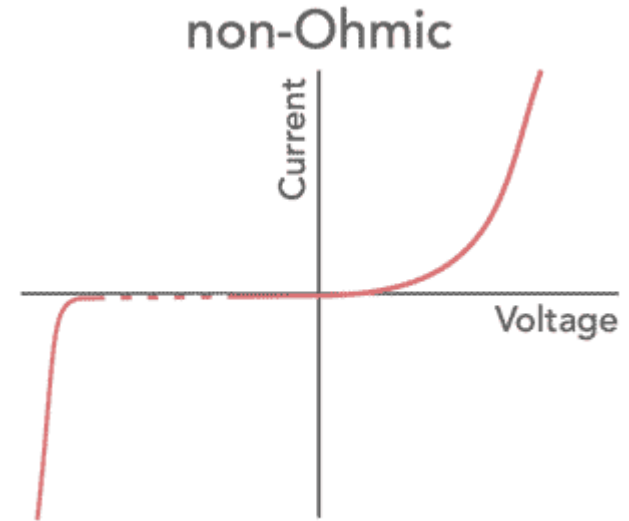
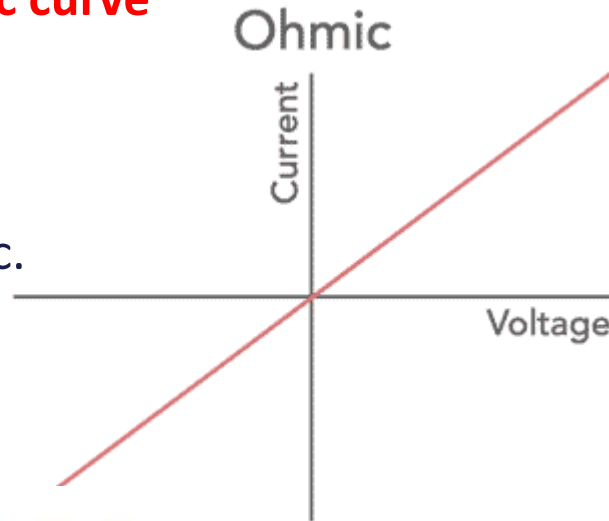
Electrical Properties of Materials

https://www.electronics-notes.com/articles/basic_concepts/resistance/ohmic-nonohmic-linear-conductors.php

I-V characteristic curve

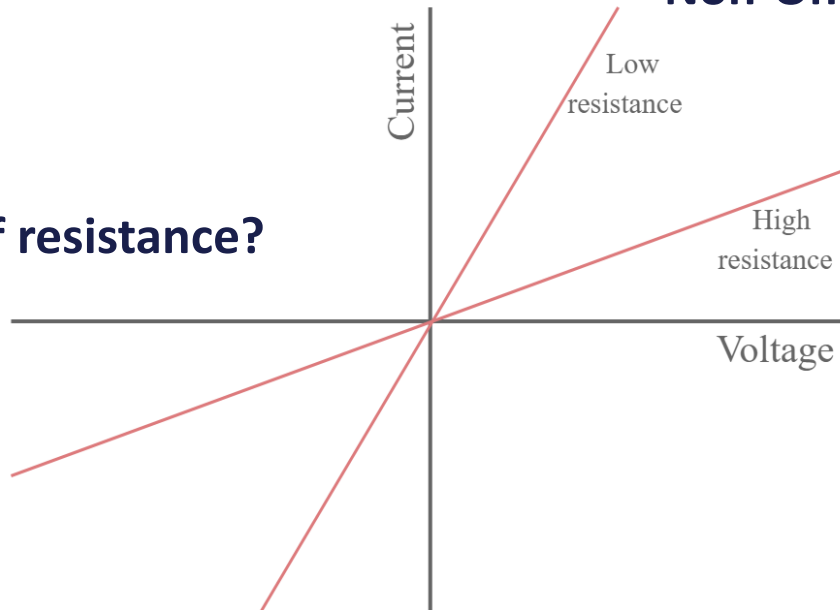
Ohmic: $V \propto I$

Conducting metals etc.



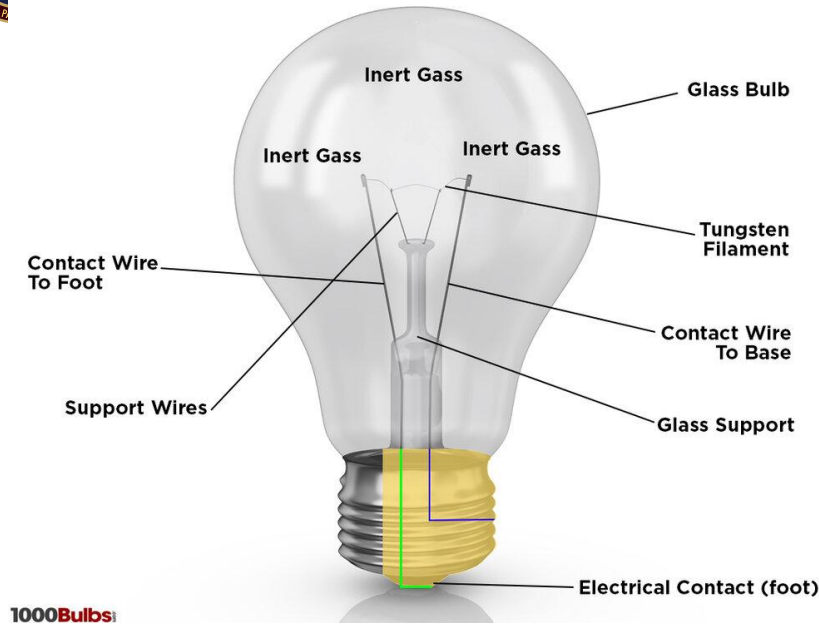
Non-Ohmic: semiconductors, insulators ..

Impact of resistance?



Electrical Properties of Materials

https://www.electronics-notes.com/articles/basic_concepts/resistance/ohmic-nonohmic-linear-conductors.php

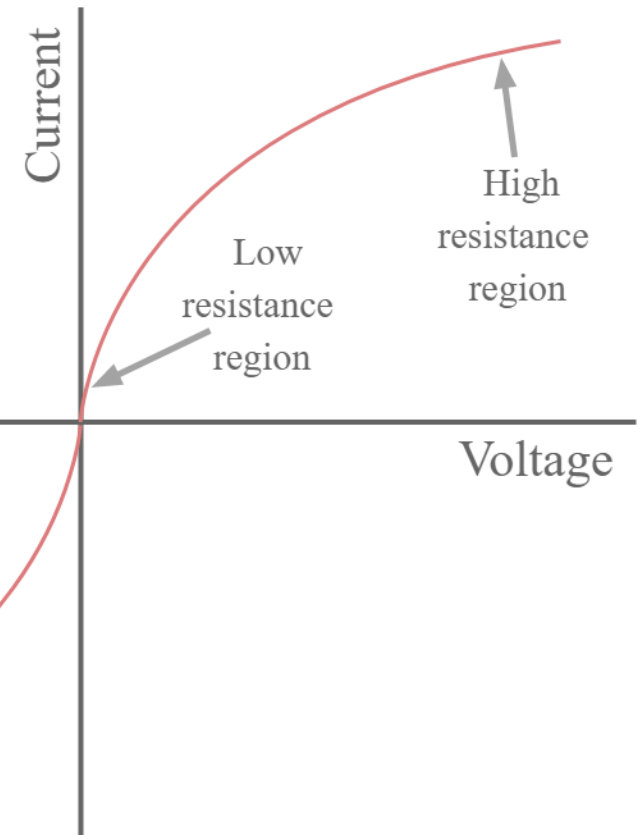


Incandescent Light Bulb

Ohmic or non-Ohmic?

Heat-related increase in resistance

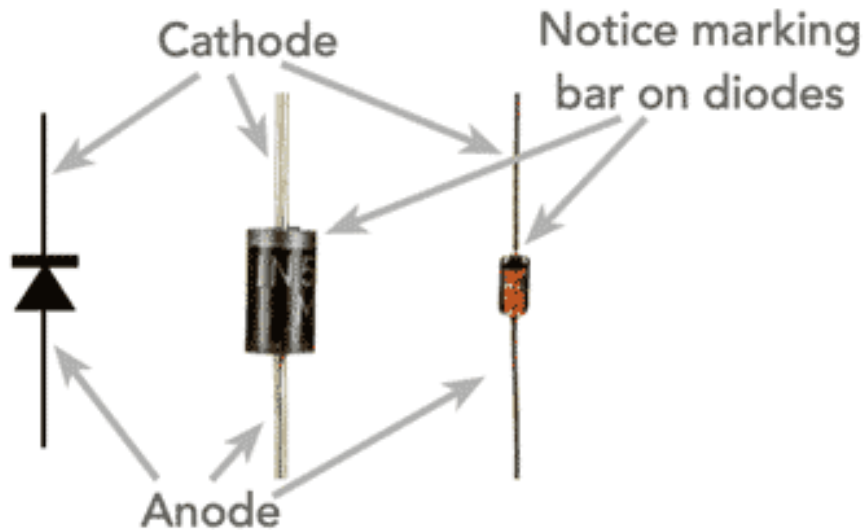
Non-Ohmic





Electrical Properties of Materials

https://www.electronics-notes.com/articles/basic_concepts/resistance/ohmic-nonohmic-linear-conductors.php

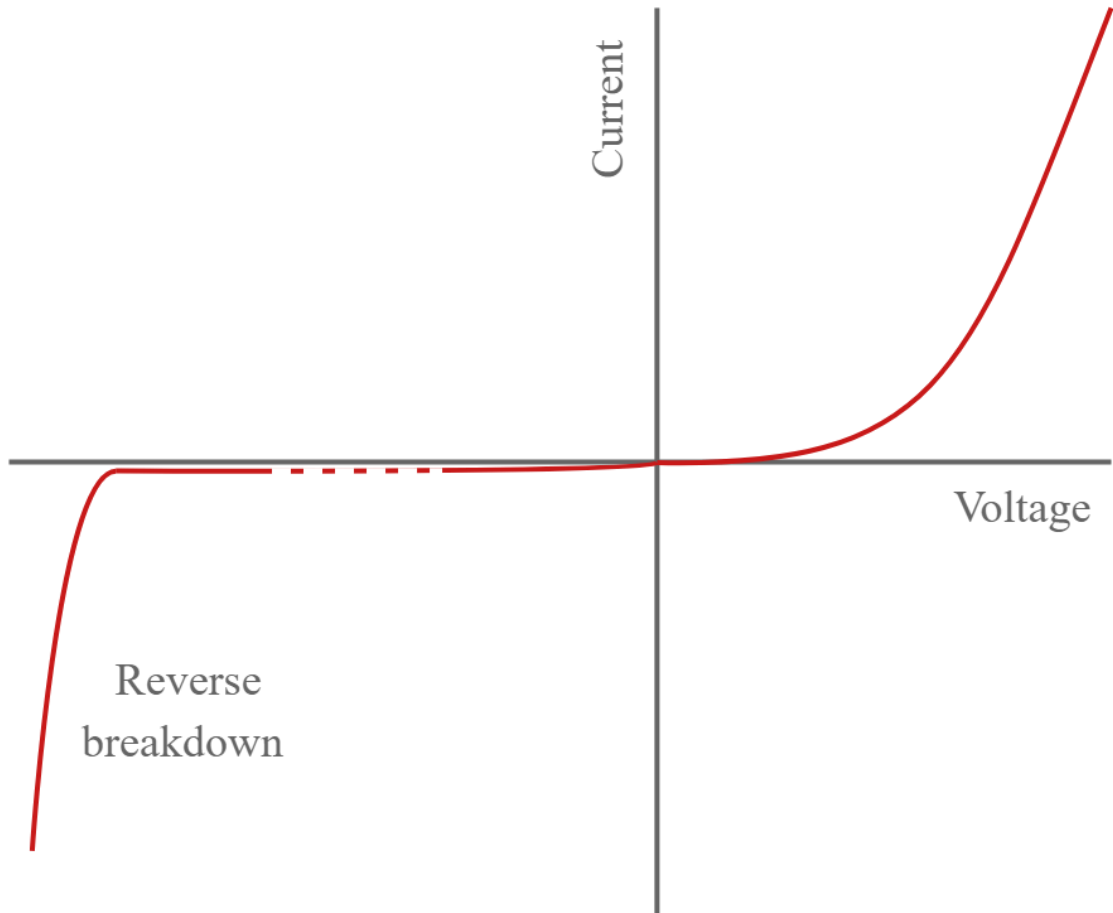


Diode (P-N junction)

Directional: current flows in one direction only

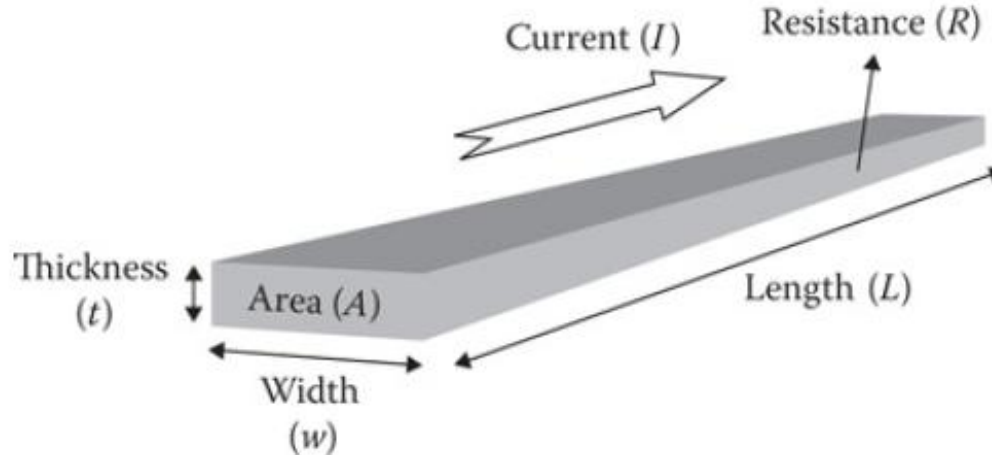
Semiconductors have similar behavior

Non-Ohmic



Electrical Properties of Materials

Electrical Conduction



- Resistivity ρ :

- a material property that is independent of sample size and geometry

$$\rho = \frac{RA}{L}$$

surface area of current flow

current flow path length

- Conductivity σ

$$\sigma = \frac{1}{\rho}$$

Electrical Properties of Materials



SI DERIVED BASIC ELECTRICAL UNITS

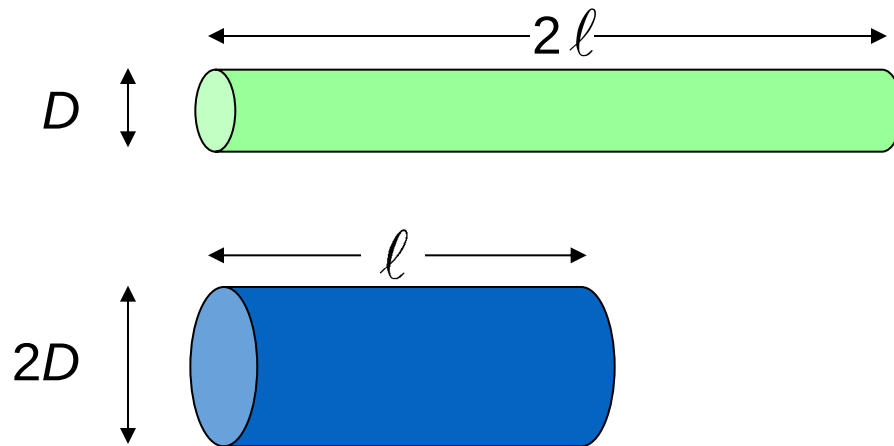
power, radiant flux	watt	W	J/s	$\text{m}^2 \cdot \text{kg} \cdot \text{s}^{-3}$
electric charge, quantity of electricity	coulomb	C	-	$\text{s} \cdot \text{A}$
electric potential difference, electromotive force	volt	V	W/A	$\text{m}^2 \cdot \text{kg} \cdot \text{s}^{-3} \cdot \text{A}^{-1}$
capacitance	farad	F	C/V	$\text{m}^{-2} \cdot \text{kg}^{-1} \cdot \text{s}^4 \cdot \text{A}^2$
electric resistance	ohm	Ω	V/A	$\text{m}^2 \cdot \text{kg} \cdot \text{s}^{-3} \cdot \text{A}^{-2}$
electric conductance	siemens	S	A/V	$\text{m}^{-2} \cdot \text{kg}^{-1} \cdot \text{s}^3 \cdot \text{A}^2$
magnetic flux	weber	Wb	V·s	$\text{m}^2 \cdot \text{kg} \cdot \text{s}^{-2} \cdot \text{A}^{-1}$
magnetic flux density	tesla	T	Wb/m ²	$\text{kg} \cdot \text{s}^{-2} \cdot \text{A}^{-1}$
inductance	henry	H	Wb/A	$\text{m}^2 \cdot \text{kg} \cdot \text{s}^{-2} \cdot \text{A}^{-2}$



Electrical Properties of Materials

Electrical Conduction

Which will have the greater resistance?



$$R_1 = \frac{2\rho l}{\pi \left(\frac{D}{2}\right)^2} = \frac{8\rho l}{\pi D^2}$$

$$R_2 = \frac{\rho l}{\pi \left(\frac{2D}{2}\right)^2} = \frac{\rho l}{\pi D^2} = \frac{R_1}{8}$$

- Analogous to flow of water in a pipe
- Resistance depends on sample geometry and size

What about resistivity?



Electrical Properties of Materials

Electrical Conduction

Further definitions

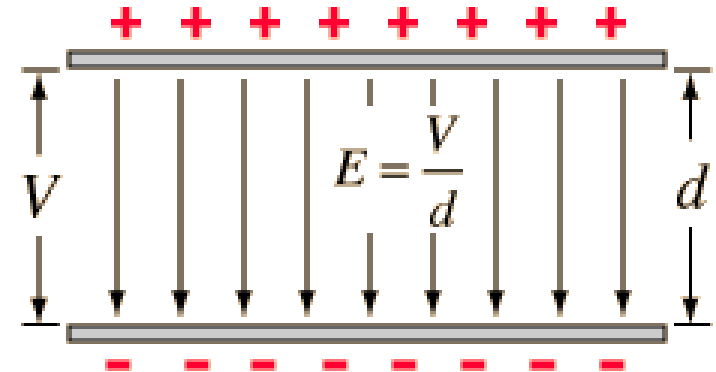
$$J = \sigma \varepsilon \quad \Leftarrow \text{another way to state Ohm's law}$$

$$J \equiv \text{current density} = \frac{\text{current}}{\text{surface area}} = \frac{I}{A} \quad \text{like a flux}$$

$$\varepsilon \equiv \text{electric field potential} = V/\ell$$

$$J = \sigma (V/\ell)$$

Electron flux conductivity voltage gradient



ε is the electric field intensity, or the voltage difference between two points divided by the distance separating them

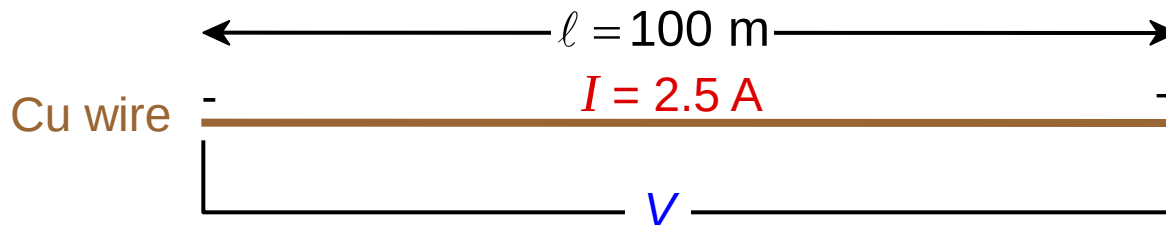
$$\text{Units: } \frac{\text{N}}{\text{C}} \text{ m} = \frac{\text{N m}}{\text{C}} = \frac{\text{Joule}}{\text{C}} = \text{Volts}$$



Electrical Properties of Materials

Electrical Conduction

What is the minimum diameter (D) of the wire so that $V < 1.5$ V?
conductivity of copper is 6.07×10^7 (Ohm-m) $^{-1}$?



$$R = \frac{\ell}{A\sigma} = \frac{V}{I}$$

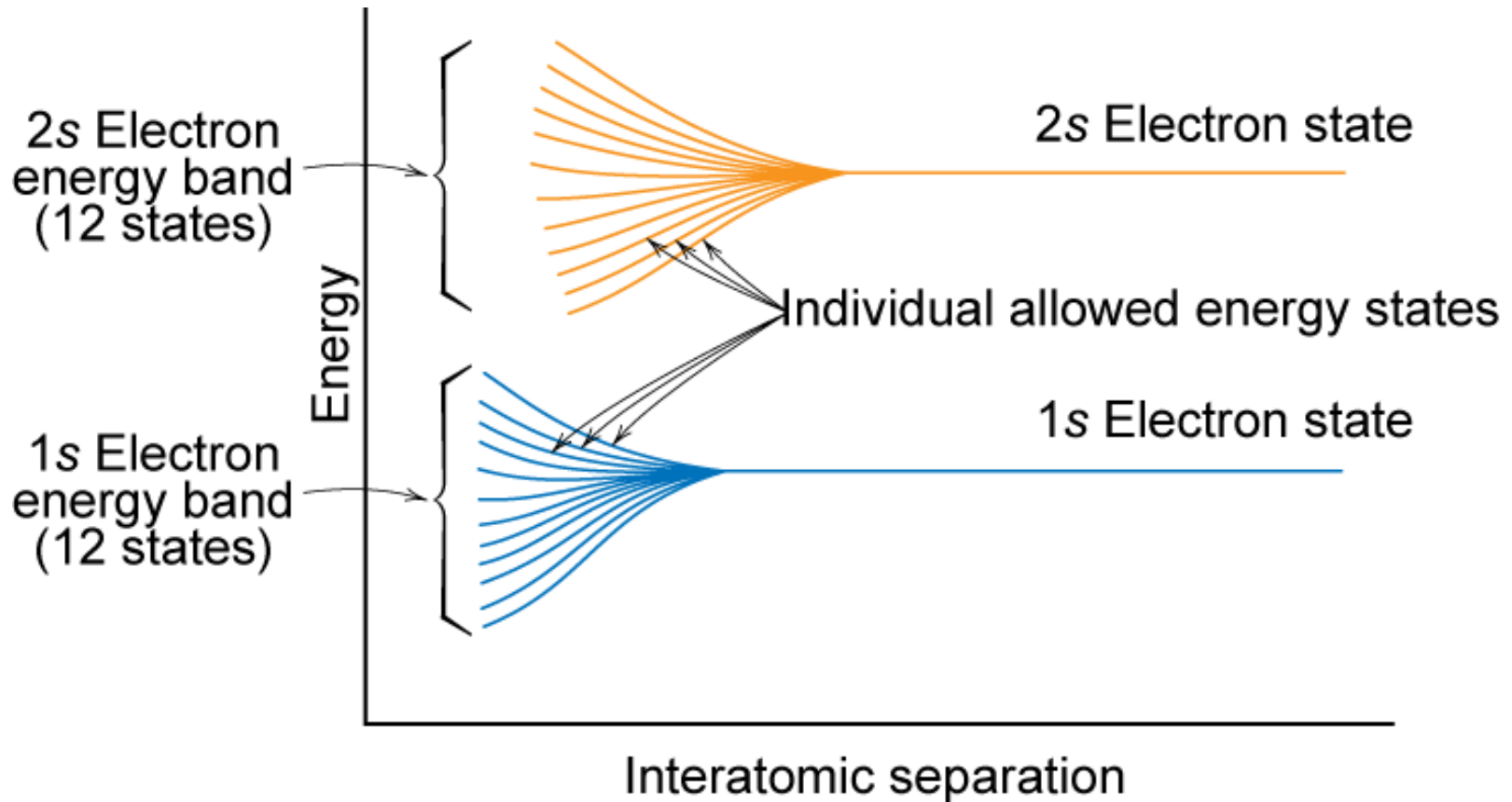
Diagram illustrating the calculation of the minimum diameter D of the wire. The formula for resistance R is shown as $R = \frac{\ell}{A\sigma}$, where ℓ is the length (100 m), A is the cross-sectional area ($\frac{\pi D^2}{4}$), and σ is the conductivity (6.07×10^7 (Ohm-m) $^{-1}$). The resistance is also expressed as $R = \frac{V}{I}$, where V is the voltage (< 1.5 V) and I is the current (2.5 A).

Solve to get $D > 1.87$ mm

Electrical Properties of Materials



Electron Energy Band Structures

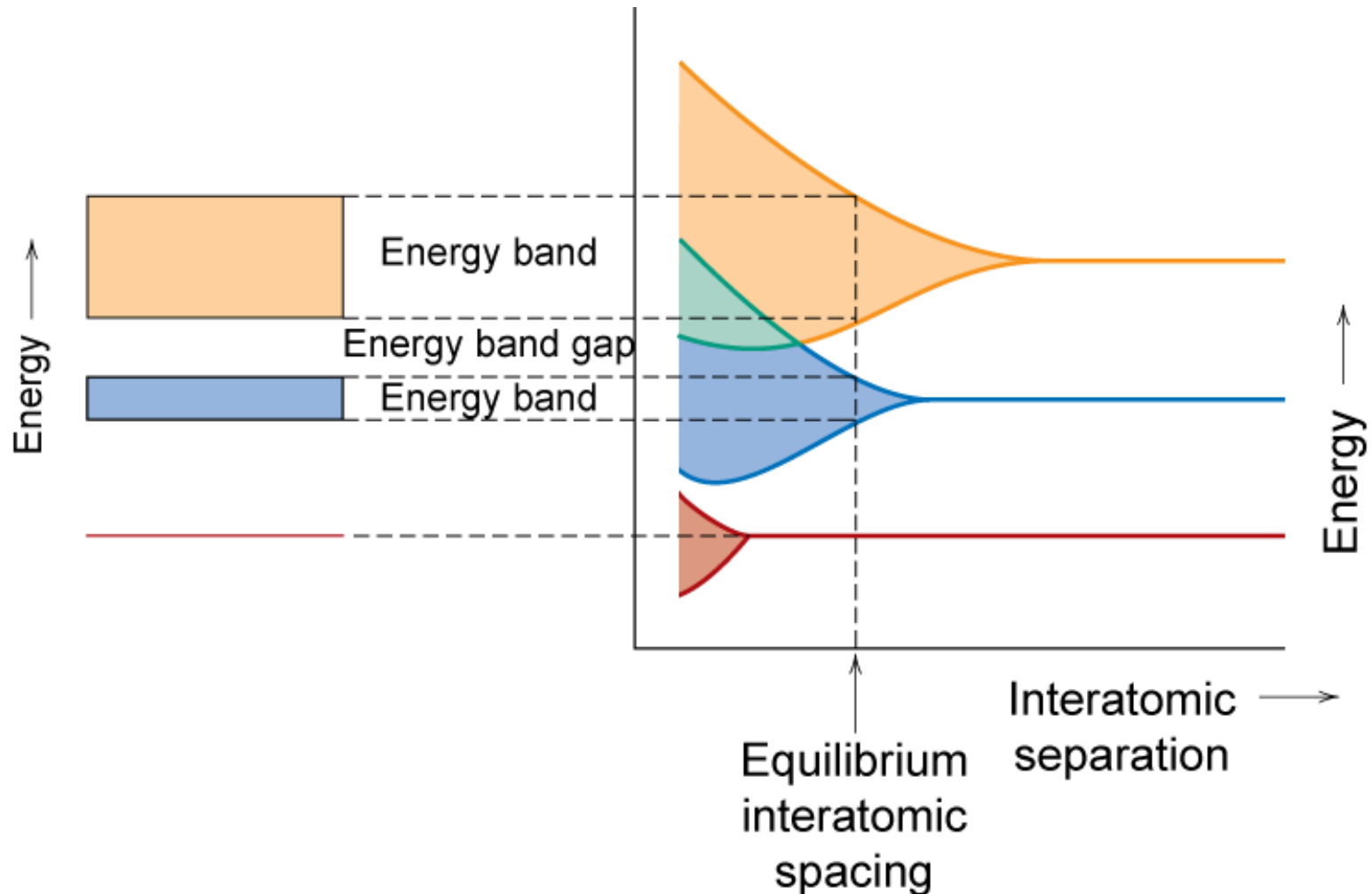


- As the atoms come within close proximity of one another, electrons are acted upon, or *perturbed*, by the electrons and nuclei of adjacent atoms.
- Each distinct atomic state may split into a series of closely spaced electron states in the solid to form what is termed an **electron energy band**.



Electrical Properties of Materials

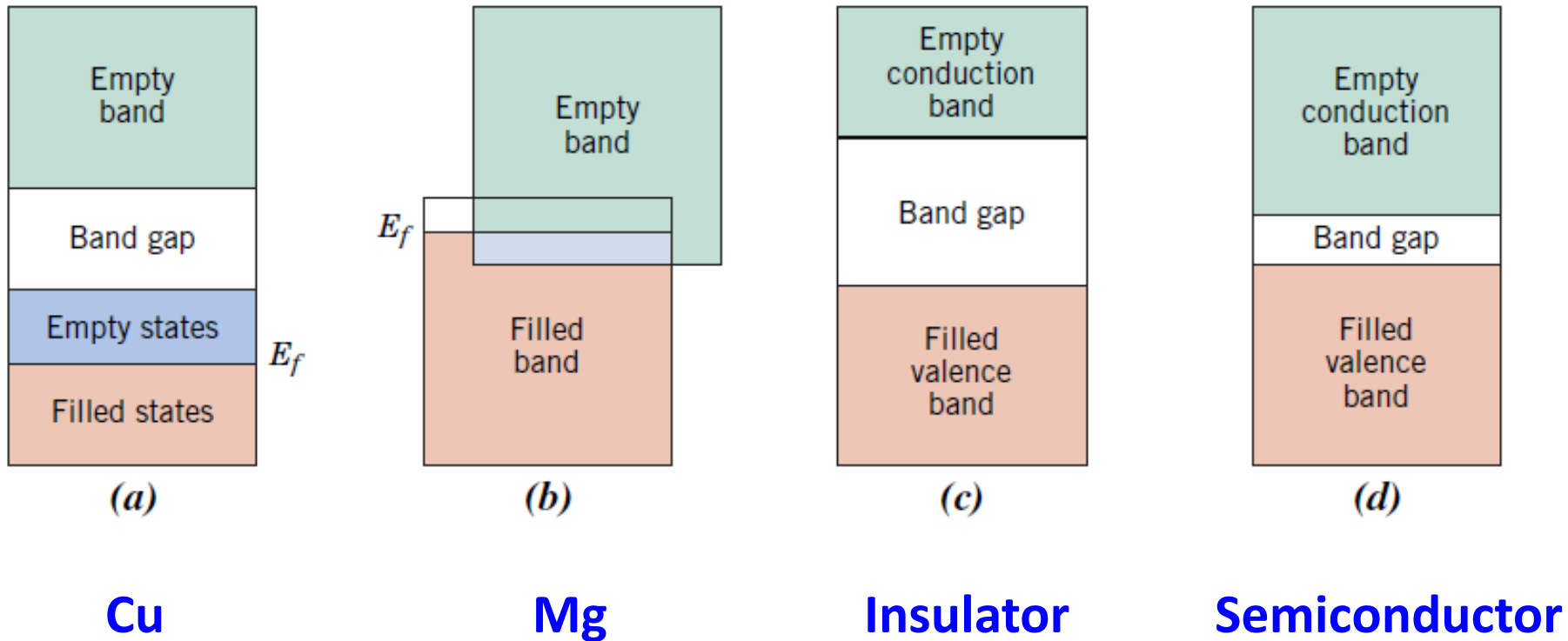
Electron Energy Band Structures





Electrical Properties of Materials

Electron Energy Band Structures

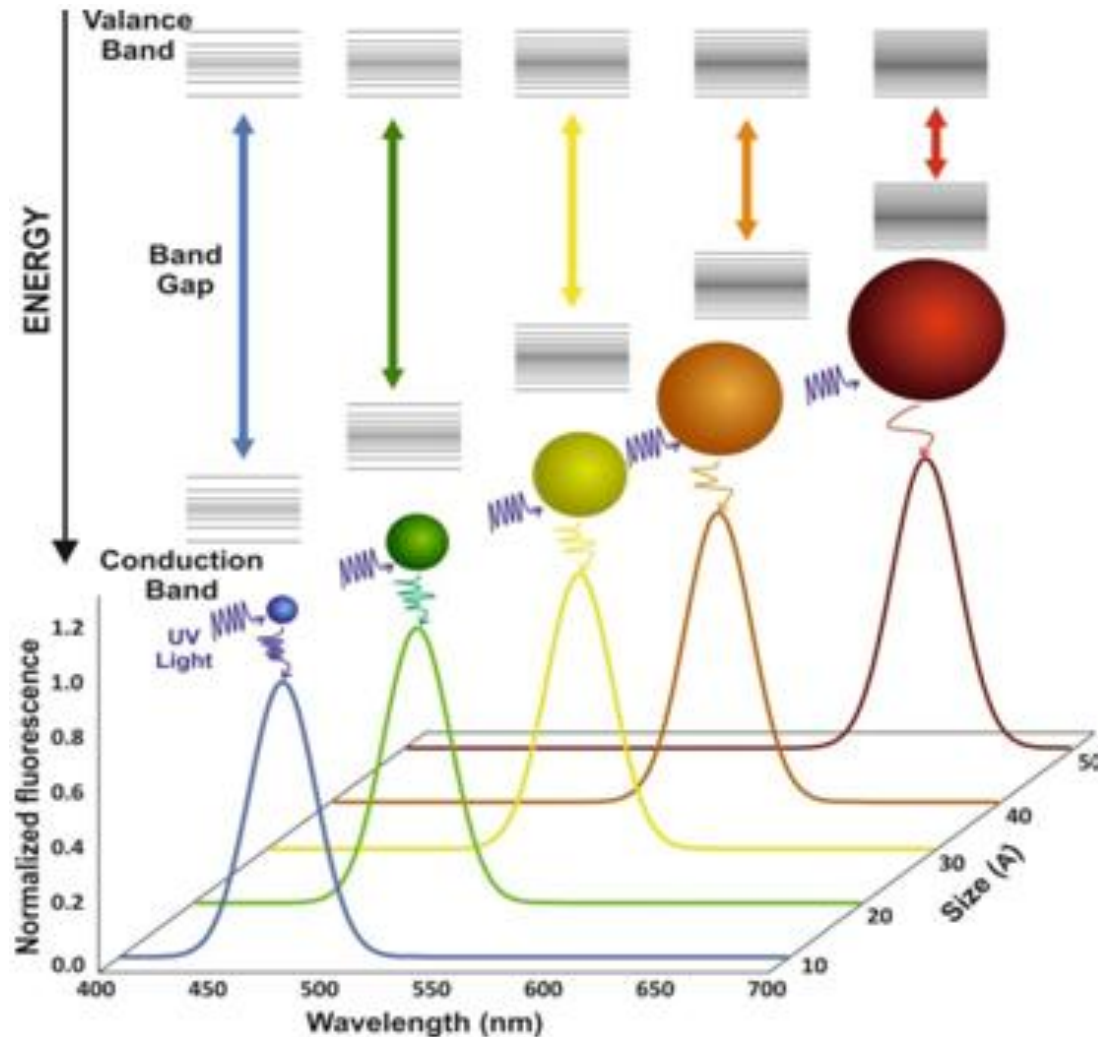


Electrical Properties of Materials

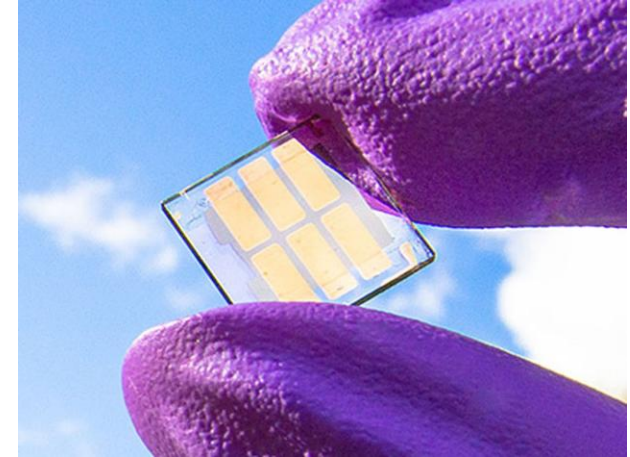


Effect of crystallite size on band gap (E_g)

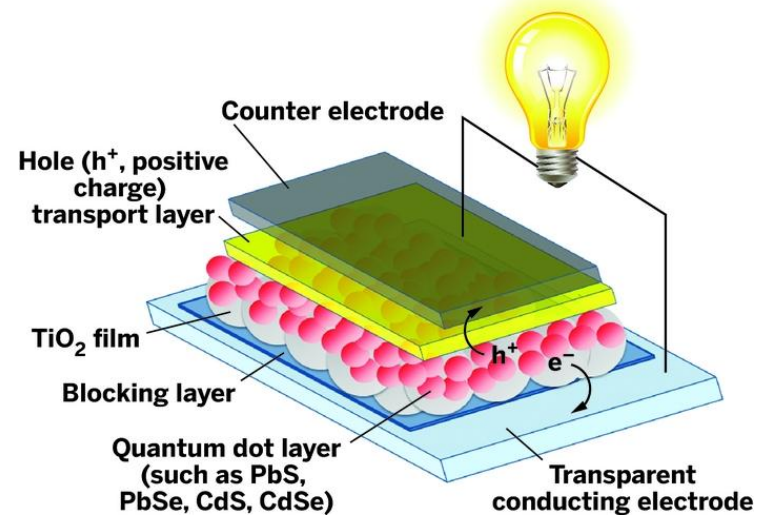
Quantum Dot: semiconductor nanocrystal (2 – 10 nm)



Quantum Dot solar cells



<https://cen.acs.org/articles/91/i38/Powerful-Dot-Solar-Energy.html>

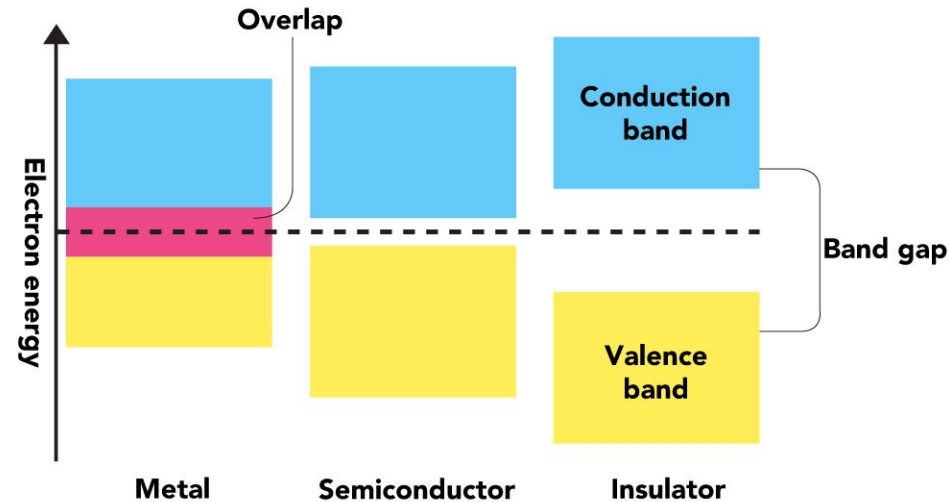
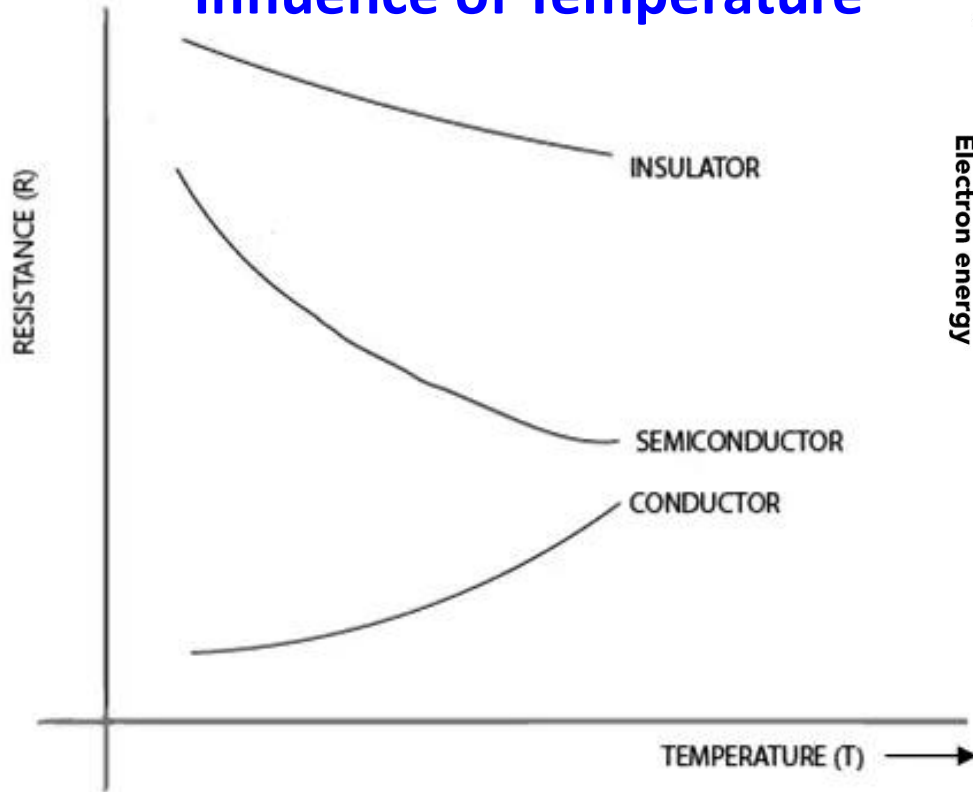




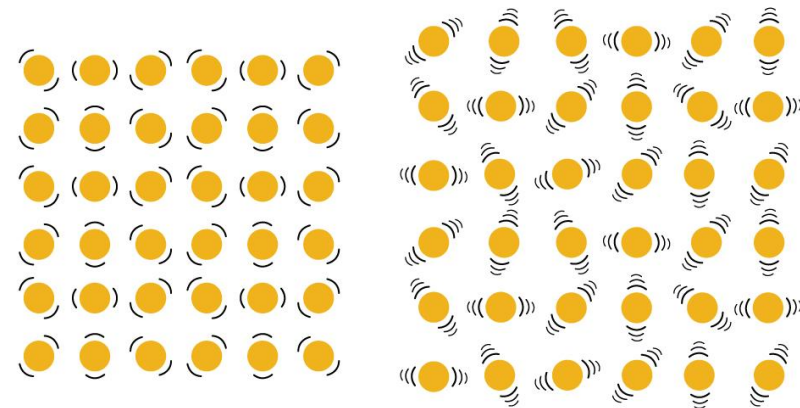
Electrical Properties of Materials

Electrical Conduction

Influence of Temperature



Why does conductivity decrease for conductor and increase for insulator and semiconductor?



Before heat

After heat

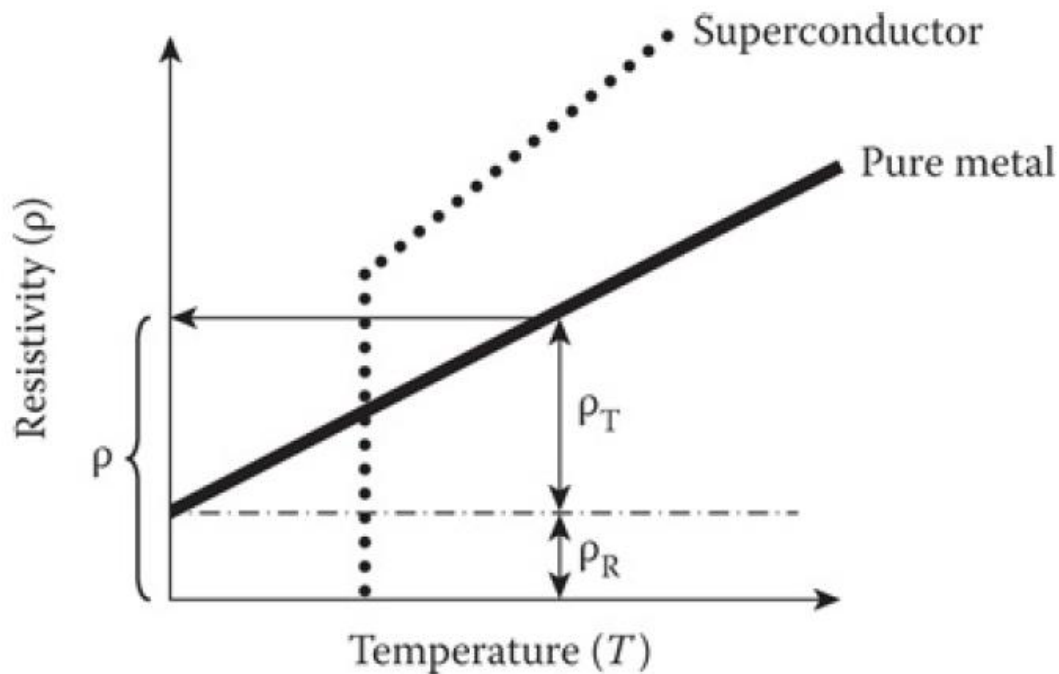


Electrical Properties of Materials

Electrical Conduction

Temperature dependence of resistivity

$$\rho = \rho_T + \rho_R$$



ρ_R : resistivity due to microstructural defects, impurities, alloying additions

ρ_T : temperature-dependent resistivity, related to scattering of conduction electrons from phonons



Electrical Properties of Materials

Electrical Conduction

Thickness dependence of resistivity

$$\rho = \rho_T + \rho_R + \rho_{\text{thickness}}$$

- Possible if the thickness (t) is too small i.e., t is equivalent to **mean free path length (λ)** of conduction electrons.

$$\lambda = v_{\text{avg}} \times \tau$$

- **λ :** Average distance that an electron travels between two collision events
- **Example:** Bulk Cu (large piece or wire) has $\lambda_{\text{Cu}} = 40 \text{ nm}$
 - If $t_{\text{Cu}} \approx \lambda_{\text{Cu}} \dots$ the **ρ** will increase?
 - Scattering of conduction electrons from the surface
 - Resistivity due to contaminations (gaseous, oxide) from the surface



Electrical Properties of Materials

Electrical Conduction

Thickness dependence of resistivity

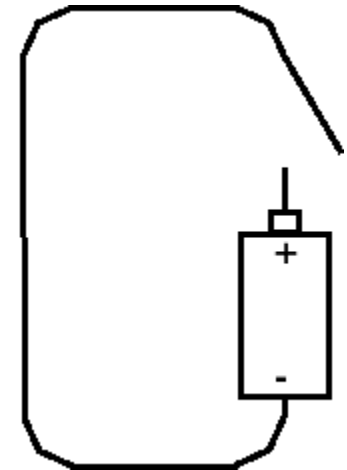
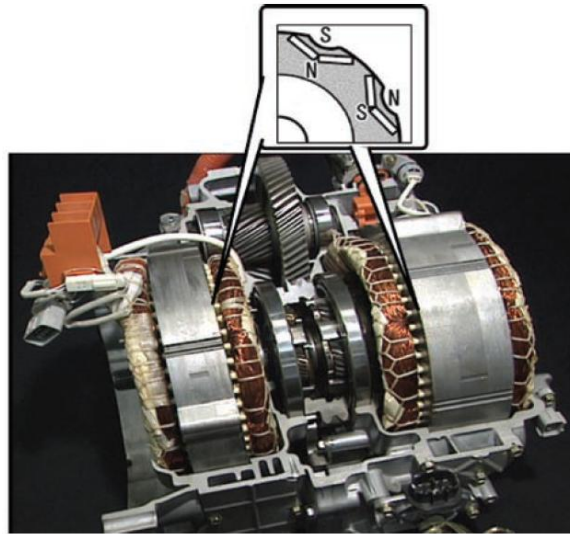
$$\rho = \rho_T + \rho_R + \rho_{\text{thickness}}$$

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Magnetic Properties of Materials



Hans Oersted

- Compass needle deflected perpendicular to a current carrying wire
- Current carrying wire behaves as a bar magnet

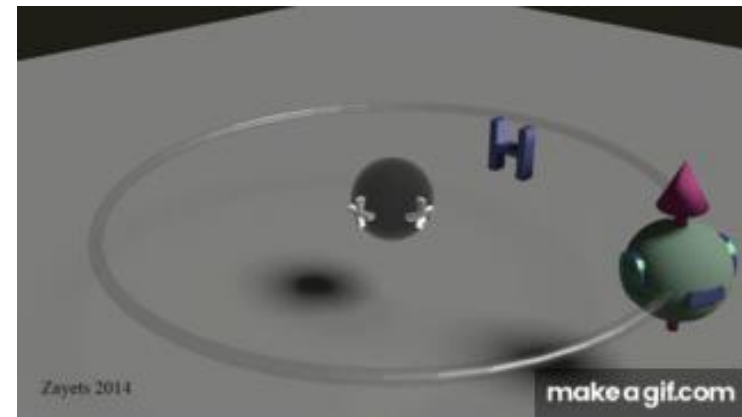
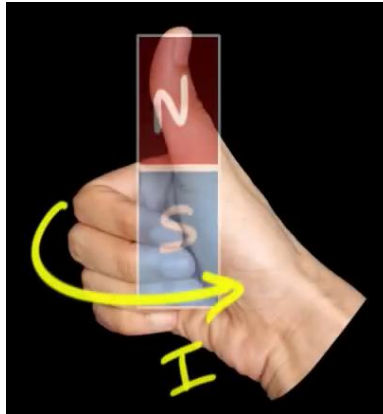
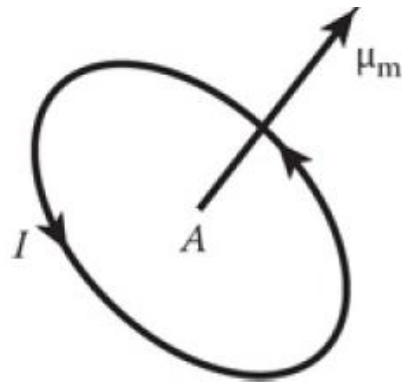




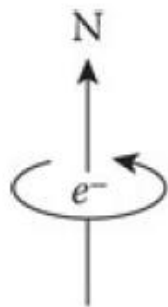
Magnetic Properties of Materials

Magnetic Moment

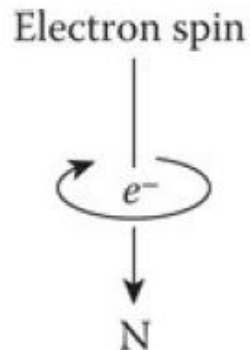
Tendency of the object to align with the magnetic field



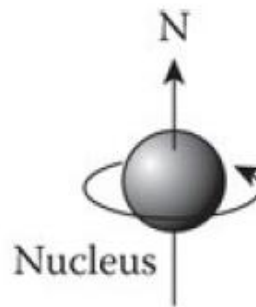
Origin of Magnetism



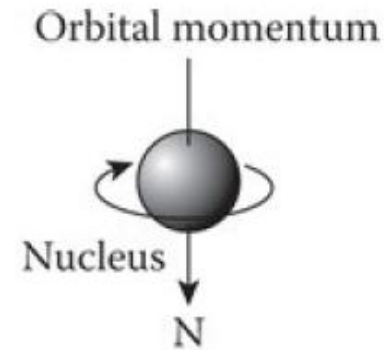
or



(a)



or



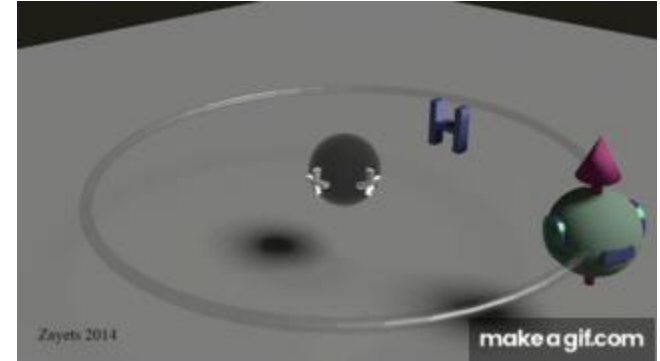
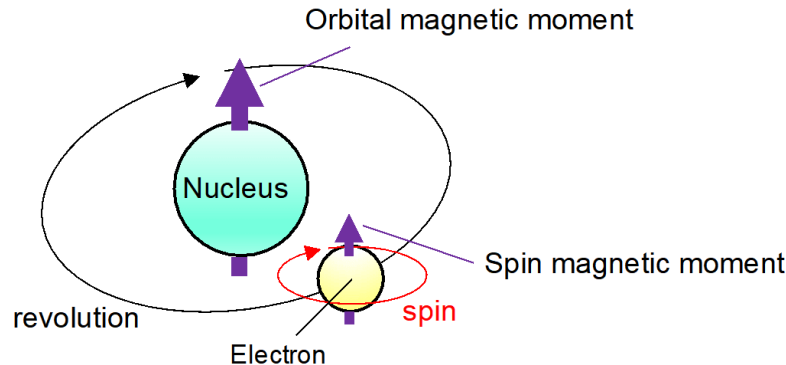
(b)



Magnetic Properties of Materials

Origin of Magnetism

Angular Magnetic Moment

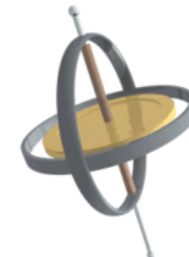


- Charged particle rotating in an orbit creates an angular magnetic moment:
(**L**: angular momentum, **m**: mass, **q**: charge)
- **Bohr**: angular momentum of electrons is quantized ... basic unit of magnetic moment ... known as **Bohr's magnetron**:
- **Gyromagnetic ratio**: links the angular momentum to magnetic moment

$$\mu = \frac{q \times L}{2 \times m}$$

$$\mu_B = \frac{h}{2\pi} \frac{q_e}{2m_e} = \hbar \frac{q_e}{2m_e}$$

$$\gamma = \frac{q}{2m}$$





Magnetic Properties of Materials

Origin of Magnetism

- **Bohr's magnetron (μ_B)** ... Basic unit that helps us define the total magnetic moment of ions, atoms or molecules
- Sum of all the magnetic dipole moments in a material is referred to as **Magnetization (M)** ... Or the total magnetic dipole moment per unit volume **Unit:** A/m or A.m²/cm³
- **Nuclear spin** ... nuclear magnetic moment, small value ($\sim 10^{-3}$ Bohr's magnetron) ... thus 'M' does not take into account nuclear spin
- Free atoms have magnetic moment ... **what about a solid?**
 - Net magnetic moment is zero ... different atoms interact
 - Most materials are non-magnetic ... in the absence of an external magnetic field

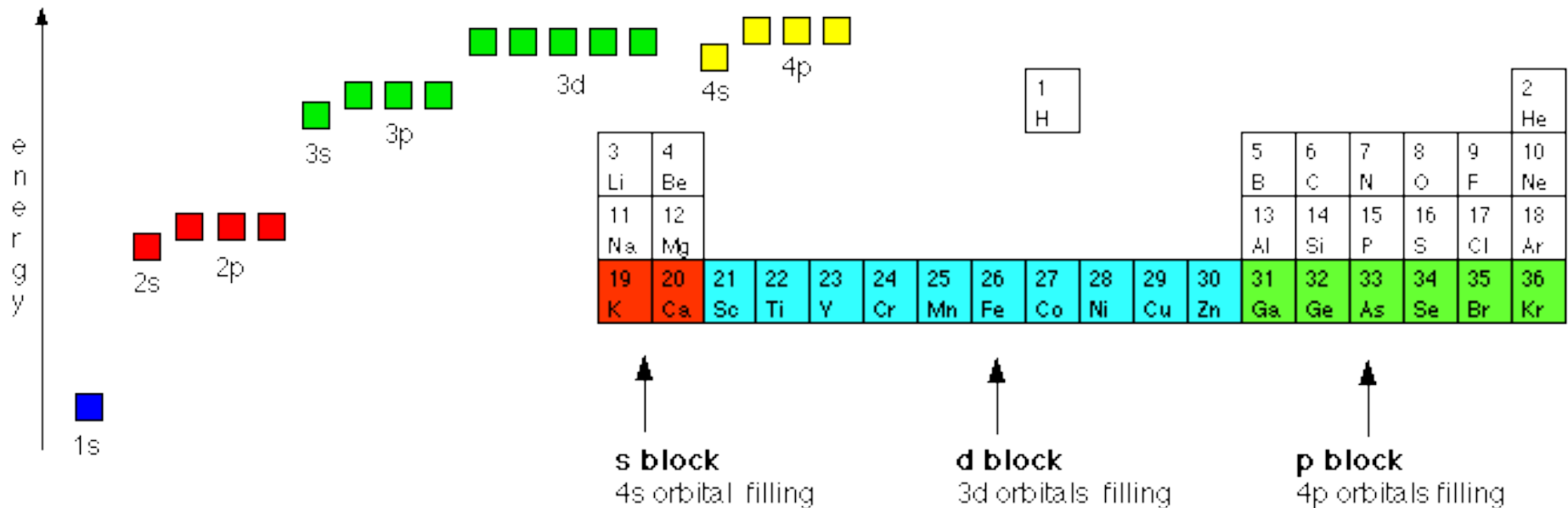


Magnetic Properties of Materials

Origin of Magnetism

- **Transition metals** (partially filled 3, 4, 5d sub-shell), Lanthanides (partially filled 4f), Actinides (partially filled 5f)
- **Magnetic spin dominates the orbital response** Some materials exhibit net magnetization
- **Examples:** Mn (Z=25) ... free atom has moment of $5\mu_B$, Fe (Z=26) free atom has moment of $4\mu_B$

Electronic configuration



Magnetic Properties of Materials



Element	Z	Electron configuration	Noble gas configuration	Electron in box diagram					
Scandium	21	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^1$	[Ar] $4s^2 3d^1$	$\uparrow\downarrow$	$\uparrow$				
Titanium	22	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^2$	[Ar] $4s^2 3d^2$	$\uparrow\downarrow$	$\uparrow$	$\uparrow$			
Vanadium	23	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^3$	[Ar] $4s^2 3d^3$	$\uparrow\downarrow$	$\uparrow$	$\uparrow$	$\uparrow$		
Chromium	24	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$	[Ar] $4s^1 3d^5$	$\uparrow$	$\uparrow$	$\uparrow$	$\uparrow$	$\uparrow$	$\uparrow$
Manganese	25	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^5$	[Ar] $4s^2 3d^5$	$\uparrow\downarrow$	$\uparrow$	$\uparrow$	$\uparrow$	$\uparrow$	$\uparrow$
Iron	26	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$	[Ar] $4s^2 3d^6$	$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow$	$\uparrow$	$\uparrow$	$\uparrow$
Cobalt	27	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^7$	[Ar] $4s^2 3d^7$	$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow$	$\uparrow$	$\uparrow$
Nickel	28	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^8$	[Ar] $4s^2 3d^8$	$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow$	$\uparrow$
Copper	29	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^{10}$	[Ar] $4s^1 3d^{10}$	$\uparrow$	$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$
Zinc	30	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10}$	[Ar] $4s^2 3d^{10}$	$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$

Cr^{3+}

Zn^{2+}

Sc^{3+}

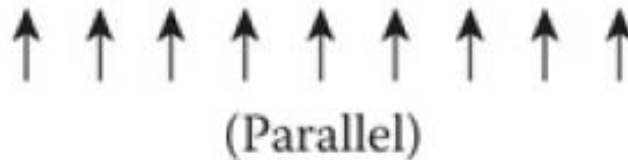
Magnetic Properties of Materials

Classification of Magnetic Materials

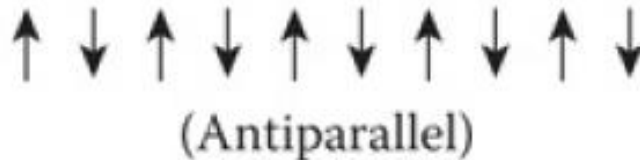
Paramagnet



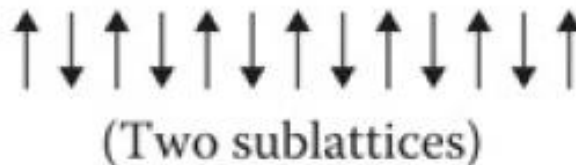
Ferromagnet



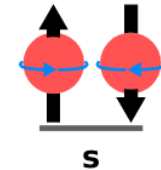
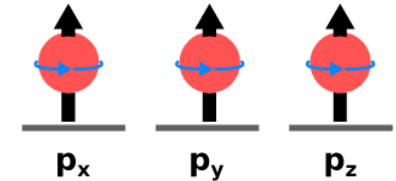
Antiferromagnet



Ferrimagnet

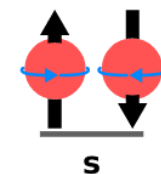
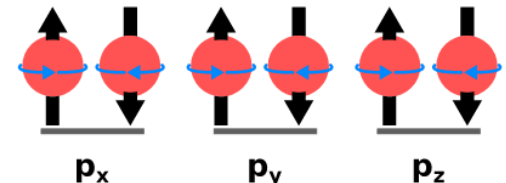


Paramagnetism



Unpaired Electrons

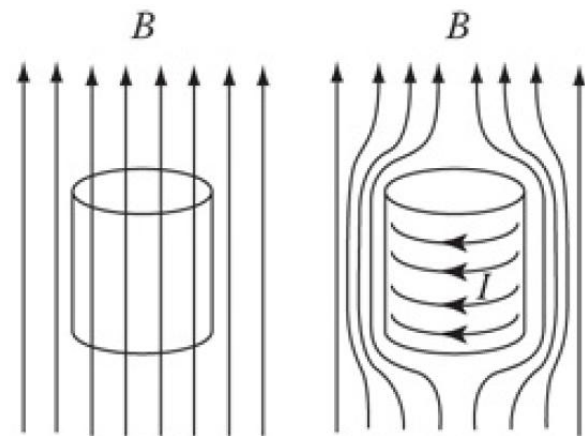
Diamagnetism



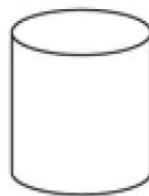
Paired Electrons¹²⁹

Magnetic Properties of Materials

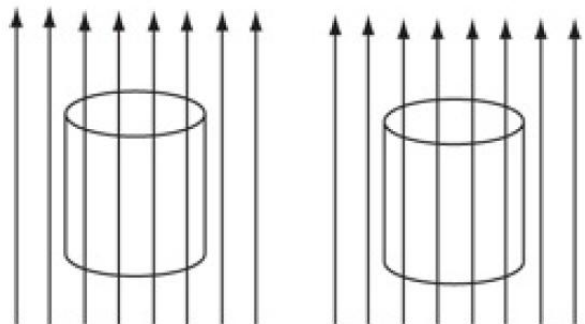
Classification of Magnetic Materials



B off

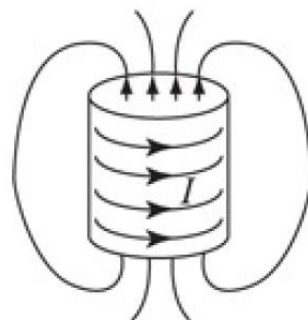


Superconductor



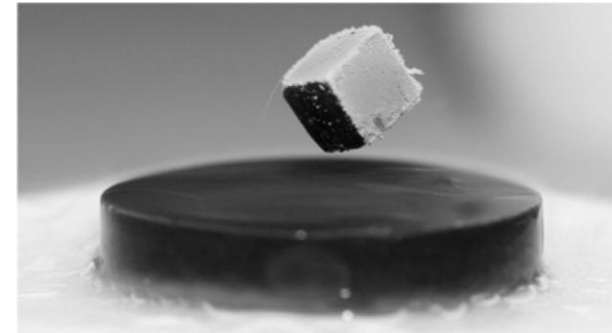
$T > T_c$

$T < T_c$



$T < T_c$

Perfect conductor



Superconductors are perfect diamagnets

Magnetic Properties of Materials

Classification of Magnetic Materials

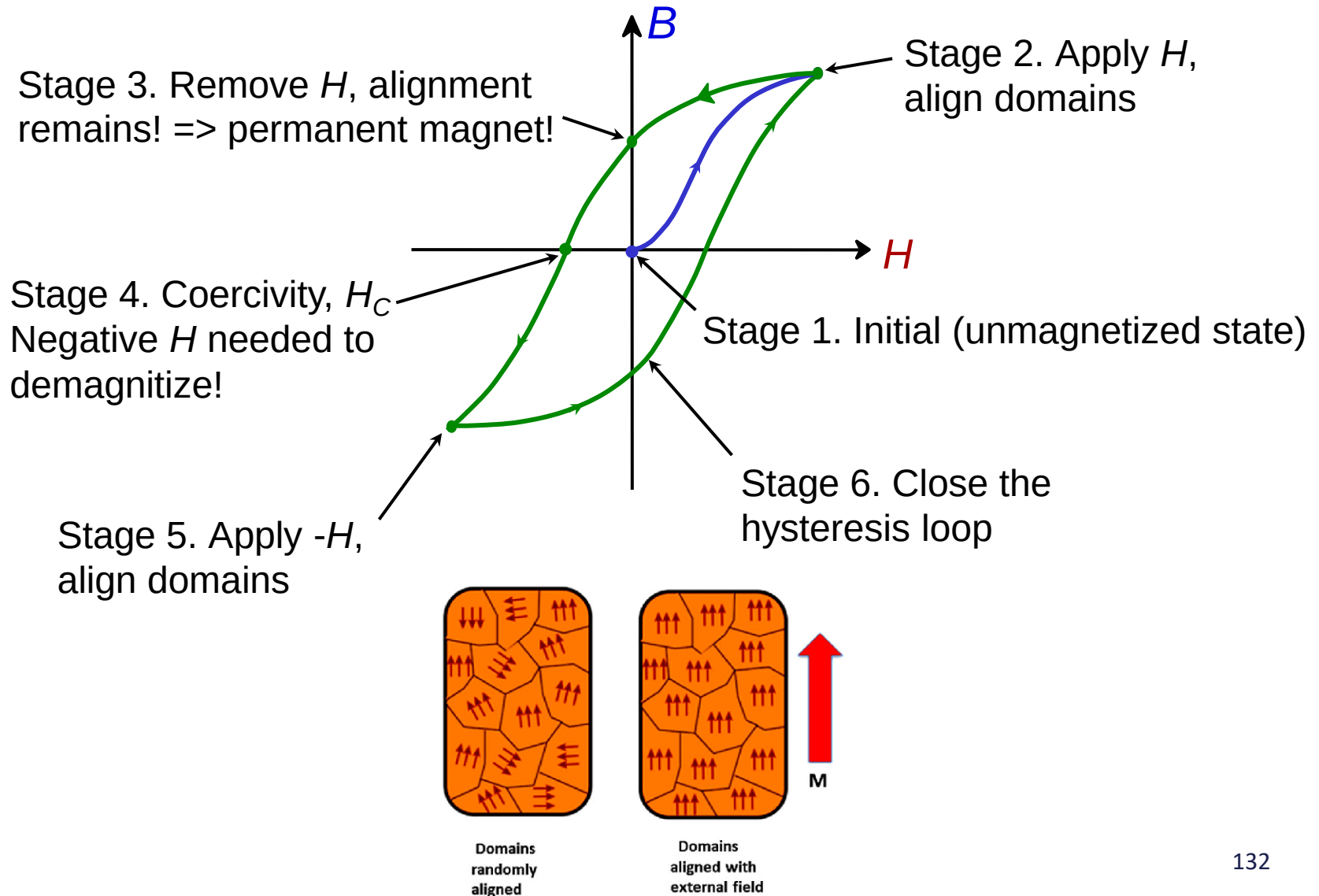
Types of Magnetic Behavior (in order of decrease strength):
everything related to magnetics is due to electron spin....

type	spin alignment	spin in simplified plot	examples
ferromagnetic	all spins align parallel to one another: spontaneous magnetization- $M = a + b$		Fe, Co, Ni, Gd, Dy, SmCo_5 , $\text{Sm}_2\text{Co}_{17}$, $\text{Nd}_2\text{Fe}_{14}\text{B}$
ferrimagnetic	most spins parallel to one another, some spins antiparallel: spontaneous magnetization- $M = a - b > 0$		magnetite (Fe_3O_4), yttrium iron garnet (YIG), GdCo_5
antiferromagnetic	periodic parallel-antiparallel spin distribution: $M = a - b = 0$		chromium, FeMn, NiO
paramagnetic	spins tend to align parallel to an external magnetic field: $M = 0 @ H=0$, $M > 0 @ H > 0$		oxygen, sodium, aluminum, calcium, uranium
diamagnetic	spins tend to align antiparallel to an external magnetic field $M = 0 @ H=0$, $M < 0 @ H > 0$		superconductors, nitrogen, copper, silver, gold, water, organic compounds



Magnetic Materials

Hysteresis and Permanent Magnetization





Magnetic Materials

Hysteresis and Permanent Magnetization

